# Causal survival analysis

## Estimation of the Average Treatment Effect (ATE) in Causal Survival Analysis : Practical Recommendations

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**Abstract**

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# 1 Introduction

Survival analysis is a key research area with applications in biostatistics, engineering, and social sciences. It deals with time-to-event data and faces the challenge of censoring, where only partial information about an event is known. Removing censored data introduces bias, while treating it as complete is equally problematic. This so-called censoring can be most effectively handled using survival analysis techniques, which depend on the specific characteristics of the censoring. The most common type of censoring is right-censoring, where the event has not occurred by the last observation, suggesting it happens later. Left-censoring occurs when the event predates the study, while interval censoring happens when the event occurs between two observations. The censoring mechanism also plays a crucial role: independent censoring (or non-informative censoring) assumes that censoring is unrelated to covariates, a requirement for estimators like Kaplan-Meier and Cox models. In contrast, informative censoring depends on the survival time and covariates, introducing bias if not accounted for, making standard techniques invalid. Causal inference evaluates the effect of treatments or interventions on outcomes, a widely used approach in health, economics, and social sciences. In a binary treatment setting ($A = 1$: treated, $A = 0$: untreated), outcomes $Y^{a=1}$ and $Y^{a=0}$ represent potential outcomes under different treatments. The individual treatment effect (ITE), defined as $Y^{a=1} - Y^{a=0}$, is central to causal inference. However, only one potential outcome can be observed per individual. The Average Treatment Effect (ATE) addresses this limitation by averaging ITEs. Randomized controlled trials (RCTs) are the gold standard for causal effect estimation as they balance covariates between treated and untreated groups. However, RCTs often lack generalizability due to restrictive inclusion criteria. Consequently, observational studies using real-world data (RWD) are increasingly used. Unlike RCTs, observational studies must address confounding bias, often handled using propensity scores or regression-based methods, under strong assumptions like unconfoundedness.

This article explores causal survival analysis, which combines causal inference and survival analysis to assess the effect of a treatment on time-to-event outcomes in the presence of censoring. Specifically, we focus on estimating the ATE for time-to-event data with static treatment assignment, baseline covariates, and right-censoring. In Section 2, we define the causal treatment effect for survival outcomes, emphasizing the easily interpretable Restricted Mean Survival Time (RMST). In Section 3, we discuss identifiability assumptions to address the challenges of censoring and treatment assignment, followed by the presentation of corresponding estimators in the RCT context. Similarly, Section 4 extends these estimators to observational studies.

Despite extensive literature on causal survival estimators, few software packages exist. We provide implementations for each estimator discussed in Section 3 and Section 4. Finally, in Section 7, we evaluate these estimators through simulations representing RCTs and observational studies under different conditions: independent or conditionally independent censoring, correct or incorrect model specifications, and positivity violations. We conclude with practical recommendations on estimator selection based on criteria like convergence, complexity, and computational efficiency.

# 2 Context and Notations

## 2.1 Notations

Let's consider a sample of $n$ i.i.d observations that are described by:

- $X_i$: the baseline covariates, $X \in \mathbb{R}^p$.
- $A_i$: the binary treatment, $A \in \{0, 1\}$.
- $C_i$: the time to censoring, $C \in \mathbb{R}^+$.
- $T_i(0)$: the survival time to the event of interest had the patient received control $A_i = 0$.
- $T_i(1)$: the survival time to the event of interest had the patient received treatment $A_i = 1$.
- $T_i = A_i T_i(1) + (1 - A_i) T_i(0)$, $T \in \mathbb{R}^+$: the observed outcome (see identifiability assumption Equation 1).
- $T_i \wedge \tau = min(T_i, \tau)$,: the truncated observed outcome at $\tau$.
- $\Delta_i = I\{T_i \leq C_i\}$: the status of censoring, where $I\{\cdot\}$ is the indicator.
- $\Delta_i^\tau = I\{T_i \wedge \tau \leq C_i\}$: the status of censoring truncated at $\tau$ (introduced later)
- $\tilde{T}_i = T_i \wedge C_i = \min(T_i, C_i)$: the observed time. When an observation is censured, then its observed time is equal to the censoring time. The censoring time is type II censoring (right censoring).
- $S(t) = P(T \geq t)$: the survival curve, which represents a key function in survival analysis. It denotes the probability that an individual will survive beyond a given time $t$.

The observed data can be summarized as a quadruplet $(X_i, A_i, \Delta_i, \tilde{T}_i)$ represented in Table 1.

Table 1: Example of survival data with covariates, treatment, the censoring time, the status of censoring and the potential outcomes and observed outcomes.

| ID | Covariates | | | Treatment | Censoring | Status | Outcomes | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ID | $X_1$ | $X_2$ | $X_3$ | A | C | $\Delta$ | T(0) | T(1) | T | $\tilde{T}$ |
| 1 | 1 | 1.5 | 4 | 1 | ? | 1 | ? | 200 | 200 | 200 |
| 2 | 5 | 1 | 2 | 0 | ? | 1 | 100 | ? | 100 | 100 |
| 3 | 9 | 0.5 | 3 | 1 | 200 | 0 | ? | ? | ? | 200 |

## 2.2 Definition of treatment effect

In causal inference, the primary goal is to estimate the individual causal effect of the treatment denoted as $\theta_i = T_i(1) - T_i(0)$ (D. B. Rubin 1974; Hernán and Robins 2010). However, this quantity cannot be observed because at most one outcome can be observed per sample (see Table 1). Furthermore, censoring may also mask outcomes (Turkson, ayiah-mensah, and Nimoh 2021).

Despite these challenges, certain identifiability assumptions, described in Section 2.3, enable us for estimating the average treatment effect (Díaz et al. 2019; Ozenne et al. 2020) (ATE) which is defined as follows:

**Definition 2.1** (Causal effect: Average treatment effect in survival analysis (ATE))**.**

$$\theta = \mathbb{E}\left[y(T(1)) - y(T(0))\right]$$

- with $y(T) = T \wedge \tau = \min(T, \tau)$ with $\tau$ a fixed time horizon; then, $E(y(T))$ becomes the restricted mean survival time (RMST) at time $\tau$ (Chen and Tsiatis 2001).
- with $y(T) = I\{T > \tau\}$ the indicator of survival with $\tau$ a fixed time horizon.

In this article, we only focus on restricted mean survival time ($y(T) = T \wedge \tau = \min(T, \tau)$) as the estimand of interest.

There is a direct relationship between RMST and the survival curve. As a result, the expression $E(T \wedge \tau)$ can be also expressed as:

$$E(T \wedge \tau) = E\left(\int_0^{T \wedge \tau} 1dt\right) = E\left(\int_0^{\tau} I\{T > t\}dt\right) = \int_0^{\tau} E(I\{T > t\})dt = \int_0^{\tau} S(t)dt$$

With this expression, we can interpret $\theta_{RMST}$ as the mean difference between the survival function of treated and control until a fixed time horizon $\tau$. RMST can be interpreted as the average survival time from baseline to a pre-specified time $\tau$. Thus, a difference in RMST ($\theta_{RMST}$) value of 10 days with $\tau = 200$ means that on average the treatment increases the survival time by 10 days at 200 days.

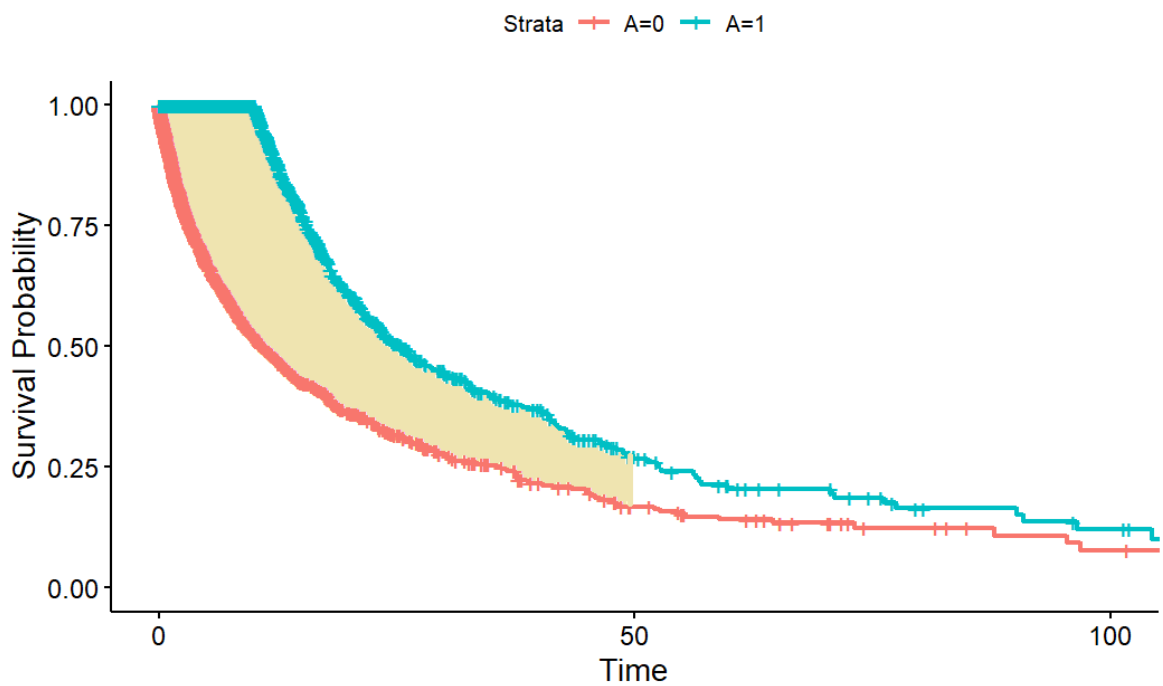



Figure 1: Plot of Kaplan-Meier survival curve for treated and control. The $\theta_{RMST}$ at $\tau = 50$ is represented in yellow

The difference in Restricted Mean Survival Time is a time-dependent measure that can be easily understood with the help of Figure 1. The average survival time will naturally vary based on the value of $\tau$.

A very naive (and biased) estimator of the average treatment effect could be to compute the difference of the mean of the uncensored survival time between the treated and the untreated group:

```
# Naive estimator:
# Warning, this estimator does not take into account censoring
# This estimator is in all context biased
Naive <- function(data, tau) {
  # Remove censored observations
  data<- data[data$status == 1, ]
  # Compute the restricted survival time
  data$T_obs_tau <- ifelse(data$T_obs >= tau, tau, data$T_obs)

  # Compute the difference of the restricted mean survival time of treated
  # and control
  mean_naive <- mean(data$T_obs_tau[data$A == 1]) -
    mean(data$T_obs_tau[data$A == 0])

  return(mean_naive)
}
```

This estimator eliminates censored observations, which represents a considerable loss of information. A reliable estimator of $\theta_{RMST}$ must include censored observations but also be expressed with observed quantities. To do this, certain identifiability assumptions are necessary to allow the ATE to be identified in different contexts.

## 2.3 General identifiability assumptions

A general identifiability assumption, already well-known in the field of causal inference, needs to be introduced:

**Assumption 1: Consistency and no-interference Assumption (Stable Unit Treatment Value Assumption: SUTVA)**

$$T = AT(1) + (1 - A)T(0) \tag{1}$$

This Equation 1 means that unit $i$'s potential outcomes do not depend on the treatments of other units. This is known as the no-interference assumption. Basically, this assumption can be violated in cases of infectious diseases. Additionally, Equation 1 means that there are no other versions of the treatment (Ding 2023).

As introduced in Section 1, it is crucial to consider the mechanism of censoring to identify appropriate estimators of $\theta_{RMST}$.

### 2.3.1 Censoring mechanism

This article focuses only on right censoring with two mechanisms of censoring considered. The first one is independent censoring:

**Assumption 2: Independent/ Non informative censoring**

$$C \perp\!\!\!\perp T(0), T(1), X, A \tag{2}$$

Under Equation 2, subjects censored at time $t$ are representative of all subjects who remain at risk at time $t$. It is as if the censored subjects were randomly selected from all subjects.

The case of dependent censoring considers that censoring is conditionally independent on covariates.

**Assumption 3: Conditionally independent censoring**

$$C \perp\!\!\!\perp T(0), T(1) | X_c, A \tag{3}$$

with $X_c$ is the set of covariables which influence the censoring mechanism.

Under Equation 3, within subgroups represented by $X_c = x$, subjects censored at time $t$ are representative of all subjects in their subgroup who remain at risk at time $t$. It is as if the censored subjects were randomly selected inside each subgroup. This assumption is very similar to the assumption of uncounfoundedness in causal inference. This assumption implies that there are no unmeasured confounders for censoring.

Another assumption for identifiability of $\theta_{RMST}$ is required in case of conditionally independent censoring: we need to assume that all subjects have a positive probability to remain uncensored at their failure time.

**Assumption 4: Positivity / Overlap for censoring**

$$0 < P(C > t \mid X_c = x, A = a) < 1, \forall t \leq \tau. \tag{4}$$

This assumption ensures that, for any time $t$, the probability of censoring is neither 0 nor 1, allowing for balance in the censoring mechanism across subgroups. If violated, it means that results are either only observed before or after time $t$, which can limit the analysis. In practice, adjusting the threshold time $\tau$ can help meet this assumption. For example, in a 5-year clinical study, if patients leave due to severe side effects or worsening health, censoring becomes dependent. In such cases, the likelihood of remaining uncensored for severely ill patients at 5 years is zero. To address this, $\tau$ can be adjusted so that participants have a chance to remain uncensored up to a revised threshold time. Despite its strength, this assumption can be verified by modeling the conditional probability of censoring using methods like the Cox model or survival forest, and checking for non-zero probabilities across covariate subgroups. We will next consider the identifiability assumptions for randomized clinical trials, starting with this simpler but more robust design.

# 3 Causal survival analysis with a Randomized Control Trial

Randomized clinical trials (RCTs) are the gold standard for establishing the effect of a treatment on an outcome, because treatment allocation is under control, which ensures (asymptotically) the balance of covariates between treated and controls, and thus avoids problems of confounding between covariables and treatment. The core assumption in a RCT is the random assignment of the treatment (D. B. Rubin 1974).

**Assumption 5: Random treatment assignment**

$$A \perp\!\!\!\perp (T(0), T(1), C, X) \tag{5}$$

Equation 5 implies that the treatment is given at random and is independent of both the potential outcomes and the covariates. It is like flipping a coin to decide the treatment assignment. In that case, as the covariates are balanced between treated and control (the treatment assignment does not depend on covariates), the causal effect is direct.

In the context of no censoring, causal effect is straightforward and can be measure with one simple estimator (such as OLS estimator). But in our case, censoring is present and the censoring mechanism has to be considered. In the next section, we will introduce identifiability of $\theta_{RMST}$ and the corresponding estimator in the simple case: independent censoring.

## 3.1 Independent censoring

The Figure 2 illustrates a simple causal graph when the study is randomized with independent censoring.

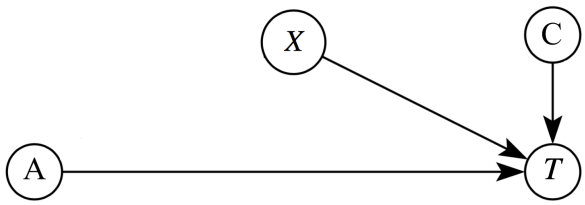



Figure 2: Illustration of a simple causal graph in RCT survival data with independent censoring ($A$ is the treatment, $X$ the precision variables, $T$ is the time to event outcome and $C$ is the time of censoring)

### 3.1.1 Identifiability

Under Equation 5 (random treatment assignment) and Equation 2 (independent censoring), the difference in RMST, $\theta_{RMST}$, can be identified as follows:

$$
\begin{aligned}
\theta_{RMST} &= \mathbb{E}[T(1) \wedge \tau - T(0) \wedge \tau] \\
&= \int_0^\tau \mathbb{E}[I\{T(1) > t\} - I\{T(0) > t\}]dt && \text{(By definition)} \\
&= \int_0^\tau \mathbb{E}[I\{T(1) > t\}] - \mathbb{E}[I\{T(0) > t\}]dt && \text{(By linearity of expectation)} \\
&= \int_0^\tau \mathbb{E}[I\{T(1) > t|A = 1\}] - \mathbb{E}[I\{T(0) > t|A = 0\}]dt && \text{(Random treatment assignment As. 5)} \\
&= \int_0^\tau \mathbb{E}[I\{T > t|A = 1\}] - \mathbb{E}[I\{T > t|A = 0\}]dt && \text{(By consistency As.1)} \\
&= \int_0^\tau \mathbb{P}(T > t|A = 1) - \mathbb{P}(T > t|A = 0)dt
\end{aligned}
\tag{6}
$$

**Voir si c'est cohérent de pousser un peu plus les équations pour que l'utilisation de Kaplan-Meier soit evident**

$\mathbb{P}(T > t|A = a)$ ($S(t, A = a)$) is the survival function and can be estimated using usual estimator of survival function such as Kaplan meier estimator, cox model, etc.

### 3.1.2 Estimation with Unadjusted Kaplan-meier

The straightforward estimator of the Equation 6 is the difference of Unadjusted Kaplan Meier estimator for the treated and control group:

**Definition 3.1** (Unadjusted Kaplan meier estimator)**.**

$$\begin{aligned}\hat{S}_{KM}(t \mid a) &= \prod_{j=1, t_j<=t} \left(1 - \frac{\sum_i I\left\{T_i = t_j, C_i \geq t_j, A_i = a\right\}}{\sum_k I\left\{T_k \geq t_j, C_k \geq t_j, A_k = a\right\}}\right) \\ &= \prod_{j=1, t_j<=t} \left(1 - \frac{\sum_i I\left\{\tilde{T}_i = t_j, \Delta_i = 1, A_i = a\right\}}{\sum_k I\left\{\tilde{T}_k \geq t_j, A_i = a\right\}}\right)\end{aligned} \tag{7}$$

The corresponding $\theta_{RMST}$ is obtained in integrating from 0 to $\tau$ the difference between unadjusted Kaplan Meier estimator of the treated and controls (Equation 6):

$$\hat{\theta}_{RMST}(\tau) = \int_0^\tau \left(\hat{S}_1(t) - \hat{S}_0(t)\right) dt$$

#### 3.1.2.1 Properties of Unadjusted Kaplan Meier estimator

Unadjusted Kaplan Meier for the survival function is a nonparametric MLE (maximum likelihood estimator). Its asymptotic properties has been derived firstly by Breslow and Crowley (1974). An other simple way of deriving the asymptotic properties is to consider Kaplan-Meier estimator as a Martingale process (*Survival and Event History Analysis: A Process Point of View* 2008). Kaplan-Meier is proved to be:

- Uniformly consistent: $\sup_{s\in[0,t]} |\hat{S}(s) - S(s)| \xrightarrow{\mathbb{P}} 0$.
- Asymptotically normal for a fixed $t$: $\sqrt{n}\left(\hat{S}(t) - S(t)\right) \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}\left(0, V^2(t)\right)$ with $V^2(t) = -S^2(t_0)\int_0^t \frac{S(du)}{S^2(u)G(u)}$.

To derive this asymptotic distribution, we need to assume that the proportion of the sample at risk at time $t$ becomes stable as the sample size increases (*Survival and Event History Analysis: A Process Point of View* 2008) and that the censoring is not informative (independent).

Under the same conditions, the Restricted Mean Survival Time ($F(t) = \int_0^\tau S(t)dt$) derived by an unadjusted Kaplan-Meier is proved to be:

- Almost surely consistent estimator of the true RMST (Per Kragh Andersen et al. 1993)
- Asymptotically normal with a variance that may be estimated by $\hat{V}(t) = \sum_{t_j \leq t} \frac{\left(\hat{F}(t) - \hat{F}(t_j)\right)^2}{Y(t_j)^2}$ (*Survival and Event History Analysis: A Process Point of View* 2008 ; Per Kragh Andersen et al. 1993).

#### 3.1.2.2 Implementation

For better interpretability and transparency, we implement an unadjusted Kaplan-Meier estimator and apply the rectangle integration method (detailed in Section 9) between 0 and $\tau$ for both survival curves (control and treated) to obtain $\theta_{RMST}$ (as shown in Figure 1). Alternatively, existing packages can be used for these calculations. For instance, the Kaplan-Meier curve can be generated with the survfit() function from the survival package (Therneau 2001), and the restricted mean survival time

can be derived using the rmean() function through the same rectangle integration method, followed by calculating the difference. Both approaches are equivalent and yield the same results.

The following code includes several functions :

- Integral_rectangles computes the integral of a given function by the method of rectangle integration
- Kaplan_meier_handmade enables to compute the Kaplan-Meier estimator for both treated and control group.
- RMST_1 computes $\theta_{RMST}$ by using the two previous function.

```
# Function to calculate the integral of a decreasing function using
# the rectangle method
# x corresponds to the x coordinate of the function to integrate
# y corresponds to the y coordinate
integral_rectangles <- function(x, y) {
  # Check if the lengths of x and y are the same
  if (length(x) != length(y)) {
    stop("Lengths of x and y must be the same")
  }

  # Calculate the width of each rectangle
  dx <- diff(x)

  # Initialize the sum
  integral_sum <- 0

  # Iterate through each rectangle and sum up the areas
  for (i in 1:(length(x) - 1)) {
    # Calculate the height of the current rectangle
    height <- min(y[i], y[i + 1])

    # Multiply the height by the width and add it to the sum
    integral_sum <- integral_sum + height * dx[i]
  }
  mean <- integral_sum + x[1]
  # Return the final integral sum
  return(mean)
}

# Kaplan-Meier estimator handmade implementation
# The database 'data' must be in the same form as that shown in
# notation (Table 1) and with the same variable name (status, T_obs)
Kaplan_meier_handmade <- function(data,
                                  status = data$status,
                                  T_obs = data$T_obs) {
  # Sort unique observed times
  Y.grid <- sort(unique(T_obs))

  # Initialize vectors for number of events, number at risk, and survival
```

```r
  # probability
  d <- rep(NA, length(Y.grid))  # Number of events at time Y.grid[i]
  n <- rep(NA, length(Y.grid))  # Number at risk just before time Y.grid[i]
  S <- rep(NA, length(Y.grid))  # Survival probability at time Y.grid[i]

  # Loop over each unique observed time
  for (i in 1:length(Y.grid)) {
    d[i] <- sum(T_obs == Y.grid[i] & status == 1, na.rm = TRUE)  # Count events
    n[i] <- sum(T_obs >= Y.grid[i])  # Count at risk

    # Calculate survival probability
    S[i] <- cumprod(1 - d / n)[i]
  }

  # Create a data frame with the results
  df <- data.frame(d = d, n = n, S = S, T = Y.grid)

  return(df)
}

# Function to calculate RMST (Restricted Mean Survival Time):
# Method 1: Handmade KM with no truncation

# Two possibilities of computing RMST :
# - in using directly S_A1 and S_A0 (survival function of treated and control)
# - in using the dataframe and the function computes the survival functions
RMST_1 <- function(data = NULL, A1 = 1, A0 = 0, tau, S_A1 = NULL, S_A0 = NULL) {
  if (is.null(S_A1) & is.null(S_A0)) {
    # Subset data for treatment groups
    data1 <- data[data$A == A1,]
    data0 <- data[data$A == A0,]

    # Calculate Kaplan-Meier survival estimates
    S_A1 <- Kaplan_meier_handmade(data1, status = data1$status,
                                  T_obs = data1$T_obs)
    S_A0 <- Kaplan_meier_handmade(data0, status = data0$status,
                                  T_obs = data0$T_obs)

    # Restrict observations to those less than or equal to tau
    Y.grid1 <- data1$T_obs[data1$T_obs <= tau]
    Y.grid0 <- data0$T_obs[data0$T_obs <= tau]
  } else {
    # Restrict observations to those less than or equal to tau
    Y.grid1 <- S_A1$T[S_A1$T <= tau]
    Y.grid0 <- S_A0$T[S_A0$T <= tau]
  }

  # Filter survival estimates to restricted observations
  S_A1 <- S_A1 %>%
```

```
    dplyr::filter(T %in% Y.grid1)
  S_A0 <- S_A0 %>%
    dplyr::filter(T %in% Y.grid0)

  # Check if there is any event at tau for S_A1
  if (!any(S_A1$T == tau)) {
    new_row <- tibble(T = tau, S = S_A1$S[nrow(S_A1)])
    S_A1 <- dplyr::bind_rows(S_A1, new_row)
  }

  # Check if there is any event at tau for S_A0
  if (!any(S_A0$T == tau)) {
    new_row <- tibble(T = tau, S = S_A0$S[nrow(S_A0)])
    S_A0 <- dplyr::bind_rows(S_A0, new_row)
  }

  # Calculate integrals from 0 to tau of survival probabilities
  intA1 <- integral_rectangles(S_A1$T, S_A1$S)
  intA0 <- integral_rectangles(S_A0$T, S_A0$S)
  RMST1 <- intA1 - intA0

  return(list(RMST=RMST1, intA1=intA1,intA0=intA0))
}
```

It is crucial to adapt the integration method to the type of function being integrated. Since the Kaplan-Meier estimator is a step function, the rectangle method is the most suitable. However, when the survival function comes from a parametric model (e.g., Weibull or Exponential), a more precise method like the trapezoidal method is recommended (details in Section 9) to calculate $\theta_{RMST}$.

When censoring becomes conditionally independent on covariates, the previously mentioned estimators are insufficient, as they assume independent censoring. The next section introduces strategies to address this issue.

### 3.2 Conditional independent censoring

The Figure 3 illustrates a simple causal graph when the study is randomized with conditionally independent censoring.

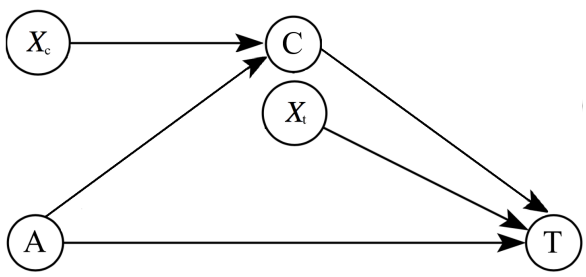



Figure 3: Illustration of a simple causal graph in RCT survival data with dependent censoring ($A$ is the treatment, $X_t$ the precision variables of the outcome, $T$ is the time to event outcome, $X_c$ the covariates which impact the censoring and $C$ is the time of censoring)

Under Equation 5 (random treatment assignment) and Equation 3 (conditional independent censoring), a strategy to make the identification of the $\theta_{RMST}$ possible can be to use a censoring unbiased transformation. A notable advantage of this transformation is that it enables the use of causal inference estimators in the same way as with fully observed data.

### 3.2.1 Notion of censoring unbiased transformation

The concept introduced by Fan and Gijbels (1994a) for local polynomial modeling in regression addresses censoring by transforming the data to create an unbiased fully observed population. This is achieved by ensuring $E(T^*|X, A) = E(T \wedge \tau|X, A)$. Transformations such as the Buckley-James and inverse probability of censoring (IPC) transformations are used for this purpose. The IPC transformation is examined first, followed by the Buckley-James transformation. We will define these transformations, present the identifiability formula for $\theta_{RMST}$ using the IPC transformation, and discuss estimation strategies for both transformations.

#### 3.2.1.1 The inverse probability of censoring transformation (IPC transformation)

The inverse probability censoring weighting approach has been introduced in survival analysis by Koul, Susarla, and Ryzin (1981) to overcome bias due to conditionally independent censoring.

The IPC transformation is given by :

$$T^*(\tau) = \frac{\widetilde{T} \wedge \tau * \Delta^\tau}{S_c(\widetilde{T} \wedge \tau|X, A)}$$

where $S_c(T \wedge \tau|X, A = a)$ is the survival function of remain uncensored truncated at $\tau$ given the covariate $X$ in the treatment arm $A = a$ and $\Delta^\tau = I\{T \wedge \tau < C\}$ is the status of the individual truncated at $\tau$.

IPC transformation verify $E(T^*|X, A) = E(T \wedge \tau|X, A)$ (proof in Section 9).

Considering Equation 19, this transformation does not consider censored observations by weighting only the uncensored observations. It exits other censoring unbiased transformation such as Buckley James that consider also the censored observations.

#### 3.2.1.2 Buckley James unbiased censoring transformation

This transformation has been introduced by Buckley and James (1979) (BJ). This is the earliest unbiased censoring transformation:

$$\begin{aligned} T^*(O, \tau) &= \Delta^\tau * (\widetilde{T} \wedge \tau) + (1 - \Delta^\tau) * \mathbb{E}[T \wedge \tau|X, A, T \wedge \tau > \widetilde{T} \wedge \tau] \\ &= \Delta^\tau * (\widetilde{T} \wedge \tau) + (1 - \Delta^\tau) * Q_S(C|X, A) \end{aligned}$$

with $Q_S(t|x, a) = E[T \wedge \tau|X = x, A = a, T \wedge \tau > t] = \frac{1}{S(T \wedge \tau|X=x, A=a)} \int_t^{+\infty} t.d(1 - S(T \wedge \tau|X = x, A = a))$ (proof in Section 9) with $F$ the function of the cumulative probability ($F(t) = \mathbb{P}(t \leq T)$) and $\bar{F}$ the complementary cumulative probability ($S(t) = 1 - F(t)$).

This transformation uses the uncensored observed values of the time-to-event $\widetilde{T} \wedge \tau$ directly in the formula (as uncensored observations are fully complete) while the censored values of the time-to-event outcome $\widetilde{T} \wedge \tau$ are extrapolated from an estimator $Q_S(t|x, a)$ which corresponds to the expected remaining survival time given the covariates and treatment for censored observation.

Exactly than the inverse probability of censoring transformation, the Buckley James transformation verify $E(T^*|X, A) = E(T \wedge \tau|X, A)$ (proof in Section 9).

Visually, these two transformations will create fully complete population as below:

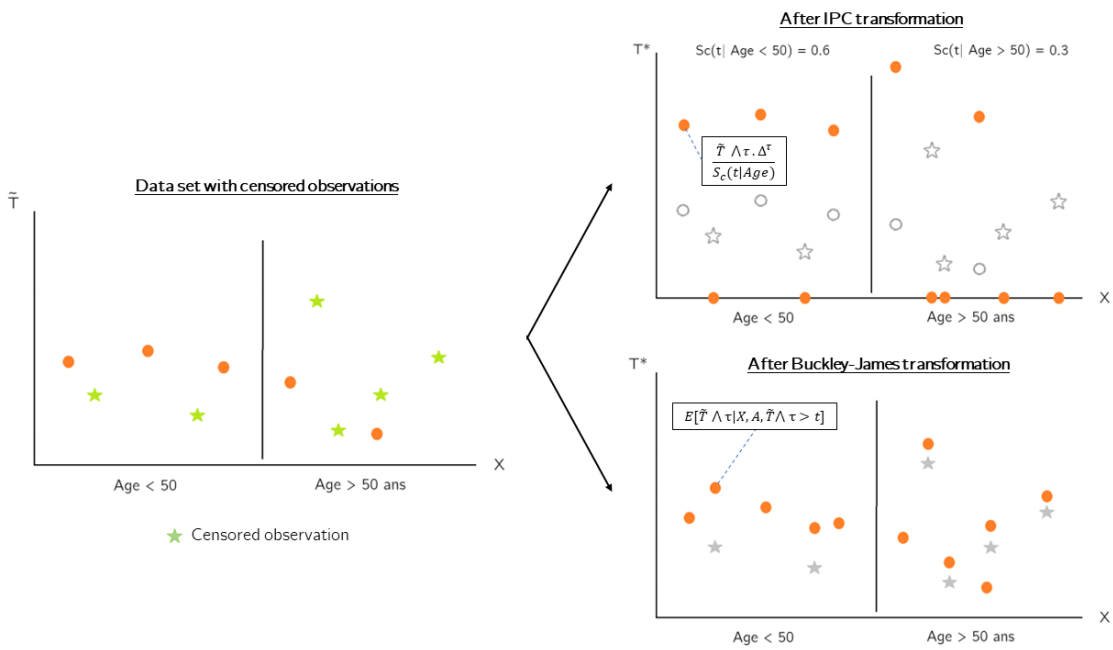



Figure 4: Illustration on Inverse Probability of Censoring and Buckley-James transformation

Unfortunately, the two previous transformation depends on nuisance parameters:

- the censoring distribution $S_c(t|x, a)$ for the IPC weighting transformation.
- the conditional survival distribution $Q_S(t|x, a)$ for the Buckley-James transformation.

If this nuisance parameters are not well estimated, the estimator using the transformation will be biased (Fan and Gijbels 1994b).

### 3.2.2 Identifiability using IPC transformation

Thus, under Equation 5 (random treatment assignment) and Equation 3 (conditionally independent censoring), the $\theta_{RMST}$ can be identified as follows:

$$
\begin{aligned}
\theta_{\text{RMST}} &= \mathbb{E}\left[T(1) \wedge \tau - T(0) \wedge \tau\right] \quad (1) \\
&= \mathbb{E}\left[T(1) \wedge \tau\right] - \mathbb{E}\left[T(0) \wedge \tau\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[T(1) \wedge \tau \mid A = 1, X\right]\right] - \mathbb{E}\left[\mathbb{E}\left[T(0) \wedge \tau \mid A = 0, X\right]\right] \quad (2) \\
&\text{(Law of total probability and Ignorability)} \\
&= \mathbb{E}\left[\mathbb{E}\left[T^*(1) \mid A = 1, X\right]\right] - \mathbb{E}\left[\mathbb{E}\left[T^*(0) \mid A = 0, X\right]\right] \quad (3) \\
&\text{(IPC transformation)} \\
&= \mathbb{E}\left[T^*(1) \mid A = 1\right] - \mathbb{E}\left[T^*(0) \mid A = 0\right] \quad (4) \\
&\text{(Law of total probability)} \\
&= \mathbb{E}\left[\frac{\widetilde{T}(1) \wedge \tau \cdot \Delta^\tau}{S_c(T \wedge \tau \mid A = 1, X)} \mid A = 1\right] - \mathbb{E}\left[\frac{\widetilde{T} \wedge \tau \cdot \Delta^\tau}{S_c(T \wedge \tau \mid A = 0, X)} \mid A = 0\right] \quad (5) \\
&= \int_0^\tau \mathbb{E}\left[\frac{1\{T(1) \geq t\} \cdot \Delta^\tau}{S_c(T \wedge \tau \mid A = 1, X)} \mid A = 1\right] - \mathbb{E}\left[\frac{1\{T(0) \geq t\} \cdot \Delta^\tau}{S_c(T \wedge \tau \mid A = 0, X)} \mid A = 0\right] dt \quad (6)
\end{aligned}
\tag{8}
$$

The IPC transformation used in this identifiability equation has been detailed in the Section 3.2.1.1.

#### 3.2.2.1 Estimation with IPCW Kaplan Meier

Under the assumptions of random treatment assignment (Equation 5), conditional censoring (Equation 3), and positivity for censoring (Equation 4), the unadjusted Kaplan-Meier (KM) estimator in Definition 3.1 can misestimate survival probabilities due to conditionally independent censoring (Willems et al. 2018). To correct for this, the adjusted IPCW (inverse probability of censoring weighting) Kaplan-Meier estimator (James M. Robins and Rotnitzky 1992; James M. Robins and Finkelstein 2000) is used. This estimator applies the weight $\frac{\Delta^\tau}{S_c(T \wedge \tau | A=1, X)}$ to observations in the KM estimator:

**Definition 3.2** (IPCW adjusted Kaplan Meier estimator)**.**

$$
\hat{S}_{IPCW-KM}(t \mid A = a) = \prod_{j=1, t_j <= t} \left(1 - \frac{\sum_i \hat{w}_i(t_j, X_i) * I\left\{T_i = t_j, C_i \geq t_j, A_i = a\right\}}{\sum_k \hat{w}_k(t_j, X_k) * I\left\{T_k \geq t_j, C_k \geq t_j, A_k = a\right\}}\right)
$$

- $\hat{w}_i(t, X_i) = \frac{\Delta_i^\tau}{\hat{S}_c(t|X_i, A_i)}$ is the inverse of the probability of remain uncensored given the covariates $X_i$ (IPC transformation) .
- $\Delta^\tau = I\{T \wedge \tau < C\} = I\{\widetilde{T} \geq \tau\} + I\{\widetilde{T} \leq \tau\}.\Delta$ (proof in Section 9).
- $\hat{S}_c(t|X_i, A_i)$ is based on the fit of parametric, semi-parametric (for example a Cox model) or even non-parametric model (such as survival forest) for censoring with $X_i$ and $A_i$ the covariates.

This estimator assigns higher weights to uncensored subjects within the same covariate group, correcting for conditionally independent censoring and reducing selection bias (Howe et al. 2016).

To estimate $\theta_{RMST}$, integrate the difference between the adjusted Kaplan-Meier estimates for treated and control groups from 0 to $\tau$:

$$
\hat{\theta}_{RMST} = \int_0^\tau \hat{S}_{IPCW-KM}(t, A = 1) - \hat{S}_{IPCW-KM}(t, A = 0) dt \tag{9}
$$

##### 3.2.2.1.1 Properties of IPCW Kaplan Meier

James M. Robins et al. (1993) **(fameuse publi introuvable mais très cité)** shows that our IPCW estimate $\hat{S}_{IPCW-KM}(t, a)$ is guaranteed to be consistent and asymptotically normal under Equation 3 and when the model of conditional censoring is correct (J. Robins, Rotnitzky, and Bonetti 2004). Also, James M. Robins and Finkelstein (2000) shows that IPCW estimate is asymptotically more efficient than the standard Kaplan-Meier estimator for failure in treatment arm a whenever the latter estimator is consistent (i.e., whenever censoring is independent).

#### 3.2.2.1.2 Implementation

The following code includes several functions :

- Adjusted.KM allows to compute the adjusted IPCW survival curve for treated and control (stratification on treatment) from given weights, times and events.
- Estimate_survival_function enables to compute the probability of remain uncensored over the time, $S_c$ in using cox model or survival forest (survival_forest function from grf (Tibshirani et al. 2017)). It allows cross-fitting for survival forest (n.folds>1).
- IPCW_Kaplan_meier computes IPCW KM estimator by using the previous functions.

```
# Kaplan-Meier adjusted
# Times of event
# Failures:  1 if event, 0 if censored
# Variable:  1 if treated, 0 if control
# Weights:  weight of the individual
adjusted.KM <- function(times, failures, variable, weights = NULL) {
  # Sanity checks
  if (sum(times < 0) > 0) {
    stop("Error: times must be positive")
  }
  if (!is.null(weights) && sum(weights < 0) > 0) {
    stop("Error: weights must be superior to 0")
  }
  if (sum(failures != 0 & failures != 1) > 0) {
    stop("Error: failures must be a vector of 0 or 1")
  }
  # If 'weights' is NULL, initialize 'w' with ones of the same length as 'times',
  # otherwise use 'weights'
  w <- if (is.null(weights)) rep(1, length(times)) else weights

  # Create a DataFrame 'data' with columns t (times), f (failures),
  # v (stratification variable: often treatment variable), and w (weights)
  data <- data.frame(t = times, f = failures, v = variable, w = w)

  # Remove rows from the DataFrame where the stratification variable is NA
  data <- data[!is.na(data$v),]

  # Initialize an empty DataFrame to store the Kaplan-Meier results
  table_KM <- data.frame(times = NULL, n.risk = NULL, n.event = NULL,
                         survival = NULL, variable = NULL)

  # Loop over each unique value of the stratification variable
```

```r
  for (i in unique(variable)) {
    # Subset the data for the current stratification variable value
    d <- data[data$v == i,]

    # Create a sorted vector of unique event times, including time 0 and the
    # maximum time
    tj <- c(0, sort(unique(d$t[d$f == 1])), max(d$t))

    # Calculate the number of events at each time point
    dj <- sapply(tj, function(x) {
      sum(d$w[d$t == x & d$f == 1])
    })

    # Calculate the number of individuals at risk at each time point
    nj <- sapply(tj, function(x) {
      sum(d$w[d$t >= x])
    })

    # Compute the cumulative product for the survival probabilities
    st <- cumprod((nj - dj) / nj)

    # Append the results to the Kaplan-Meier table
    table_KM <- rbind(table_KM, data.frame(T = tj, n = nj, d = dj,
                                           S = st, variable = i))
  }
  return(table_KM)
}


# Estimate survival function with covariates for each individual at each time Y.grid
# Type of model can be cox or survival forest (n.fold must be completed in this case)

# This function is used also to compute S_c with status = censor.status
estimate_survival_function <- function(data, X.names,
                                       Y.grid,
                                       type_of_model = "cox",
                                       T_obs = "T_obs",
                                       status = "status",
                                       n.folds = NULL) {
  if (type_of_model == "cox") {
    # Formula for cox model (single learner: A as a covariate,
    # T-learner: Stratified fit on A)
    # Here only T-learner
    outcome <- paste0('Surv(', T_obs, ',', status, ')')
    formula <- as.formula(paste(outcome, paste(c(X.names),
                                               collapse = " + "), sep = " ~ "))
    data.1 <- data%>%
      filter(A==1)
```

```
    data.0 <- data%>%
      filter(A==0)

    # Cox model fitting stratified on A=1
    fitS1 <- suppressWarnings(coxph(formula, data = data.1, x = TRUE))
    # Suppress NA in coefficient
    fitS1$coefficients[is.na(fitS1$coefficients)] <- 0

    # Cox model fitting on A=0
    fitS0 <- suppressWarnings(coxph(formula, data = data.0, x = TRUE))
    # Suppress NA in coefficient
    fitS0$coefficients[is.na(fitS0$coefficients)] <- 0

    fit.pred1 <- predictCox(fitS1, newdata = data, times = Y.grid,
                            type = "survival")
    fit.pred0 <- predictCox(fitS0, newdata = data, times = Y.grid,
                            type = "survival")
    S_hat1 <- fit.pred1$survival
    S_hat0 <- fit.pred0$survival

  } else { # Survival forest
    # Initialization
    n <- nrow(data)
    fit.pred1 <- matrix(NA, nrow = n, ncol = length(Y.grid))
    fit.pred0 <- matrix(NA, nrow = n, ncol = length(Y.grid))

    if (n.folds > 1) {# Cross-fitting
      # Split the dataset into n-folds
      indices <- split(seq(n), sort(seq(n) %% n.folds))

      # For each index in each split
      for (idx in indices) {
        # Fit survival forest on observations removed from idx (training set) and A=1
        # A is not included in covariates (T-learner)
        forest.grf1 <- survival_forest(X = as.matrix(data[-idx & data[,"A"] == 1,
                                                          X.names]),
                                       Y = data[-idx & data[,"A"] == 1, T_obs],
                                       D = data[-idx & data[,"A"] == 1, status],
                                       failure.times = Y.grid)
        # Fit survival forest on observations removed from idx (training set) and A=0
        # A is not included in covariates (T-learner)
        forest.grf0 <- survival_forest(X =  as.matrix(data[-idx & data[,"A"] == 0,
                                                           X.names]),
                                       Y = data[-idx & data[,"A"] == 0, T_obs],
                                       D = data[-idx & data[,"A"] == 0, status],
                                       failure.times = Y.grid)
        # Prediction on idx to avoid overfitting
        fit.pred1[idx,] <- predict(forest.grf1,  as.matrix(data[idx, X.names]),
                                   failure.times = Y.grid)$predictions
```

```
        fit.pred0[idx,] <- predict(forest.grf0,  as.matrix(data[idx, X.names]),
                                   failure.times = Y.grid)$predictions
      }
    } else {# No cross-fitting
      # Fit survival forest on all observations with A=1
      # A is not included in covariates (T-learner)
      forest.grf1 <- survival_forest(X =  as.matrix(data[data[,"A"] == 1,
                                                         X.names]),
                                     Y = data[data[,"A"] == 1, T_obs],
                                     D = data[data[,"A"] == 1, status],
                                     failure.times = Y.grid)

      # Fit survival forest on all observations with A=0
      # A is not included in covariates (T-learner)
      forest.grf0 <- survival_forest(X =  as.matrix(data[data[,"A"] == 0,
                                                         X.names]),
                                     Y = data[data[,"A"] == 0, T_obs],
                                     D = data[data[,"A"] == 0, status],
                                     failure.times = Y.grid)

      # Prediction on all observations
      fit.pred1 <- predict(forest.grf1,  as.matrix(data[, X.names]),
                           failure.times = Y.grid)$predictions
      fit.pred0 <- predict(forest.grf0,  as.matrix(data[, X.names]),
                           failure.times = Y.grid)$predictions
    }

    S_hat1 <- fit.pred1
    S_hat0 <- fit.pred0
  }

  # Associate the corresponding Survival curve to the observation
  S_hat <- S_hat1 * data$A + (1 - data$A) * S_hat0

  return(list('S_hat' = S_hat, "S_hat1" = S_hat1, "S_hat0" = S_hat0, "T" = Y.grid))
}

# IPCW Kaplan-Meier estimator with restricted tau
IPCW_Kaplan_meier <- function(data, tau,
                              X.names.censoring,
                              nuisance_censoring = "cox",
                              n.folds = NULL) {

  # Compute of truncated T_obs, status and censored status
  data$T_obs_tau <- ifelse(data$T_obs >= tau, tau, data$T_obs)
  data$censor.status_tau <- 1 - as.numeric((data$T_obs >= tau) |
                                           (data$T_obs < tau & data$status == 1))
  data$status_tau <- as.numeric((data$T_obs >= tau) |
```

```
                                    (data$T_obs < tau & data$status == 1))
  Y.grid <- sort(unique(data$T_obs_tau))

  # Estimate probability of remaining uncensored based on nuisance model
  S_C_hat <- estimate_survival_function(data = data, X.names = X.names.censoring,
                                        Y.grid = Y.grid, T_obs = "T_obs_tau",
                                        status = "censor.status_tau",
                                        type_of_model = nuisance_censoring,
                                        n.folds = n.folds)

  # Select the probability of censoring for each observe T_obs_tau from the all
  # curve
  data$S_C <- S_C_hat$S_hat[cbind(1:nrow(data), match(data$T_obs_tau, Y.grid))]

  # Compute IPC weights
  data$weights <- data$status_tau / data$S_C

  # Compute the adjusted IPCW Kaplan Meier
  S <- adjusted.KM(times = data$T_obs, failures = data$status,
                   variable = data$A, weights = data$weights)

  # Compute difference in RMST
  RMST <- RMST_1(S_A1 = S[S$variable == 1,], S_A0 = S[S$variable == 0,], tau = tau)

  return(list(RMST = RMST$RMST,
              intA1 = RMST$intA1,
              intA0 = RMST$intA0,
              weights = data$weights))
}
```

It exists other solutions to identify $\theta_{RMST}$. One of them use directly the results using pseudo-observations from the unbiased censoring transformation in Section 3.2.1.1 or Section 3.2.1.2 without using survival function. In the following section, we will introduce the identifiability formulae using Buckley-James transformation known as the best restoration transformation. It will give us another estimator described in Section 3.2.3.1 to estimate $\theta_{RMST}$.

### 3.2.3 Identifiability using BJ transformation

Under Equation 5 (random treatment assignment) and Equation 3 (conditionally independent censoring), another identifiability formula can be derived from line 4 in Equation 8 in using Buckley-James transformation (Section 3.2.1.2):

$$
\begin{aligned}
\theta_{\text{RMST}} &= \mathbb{E}\left[T(1) \wedge \tau - T(0) \wedge \tau\right] \\
&= \mathbb{E}\left[T^*(1) \mid A = 1\right] - \mathbb{E}\left[T^*(0) \mid A = 0\right] \\
&= \mathbb{E}\left[\Delta^\tau * (\widetilde{T}(1) \wedge \tau) + (1 - \Delta^\tau) * Q_S(C|X, A) \mid A = 1\right] - \\
&\qquad \mathbb{E}\left[\Delta^\tau * (\widetilde{T}(0) \wedge \tau) + (1 - \Delta^\tau) * Q_S(C|X, A) \mid A = 0\right] \\
&\text{(BJ transformation)} \\
&= \mathbb{E}\left[\Delta^\tau * (\widetilde{T} \wedge \tau) + (1 - \Delta^\tau) * Q_S(C|X, A) \mid A = 1\right] - \\
&\qquad \mathbb{E}\left[\Delta^\tau * (\widetilde{T} \wedge \tau) + (1 - \Delta^\tau) * Q_S(C|X, A) \mid A = 0\right] \\
&\text{(By consistency)}
\end{aligned}
\tag{10}
$$

The straightforward corresponding estimator from Equation 10 is described below.

#### 3.2.3.1 Estimation of BJ estimator

Based on the identifiability formula Equation 10, it is possible to implement BJ estimator directly without using survival function:

**Definition 3.3** (BJ estimator)**.**

$$
\begin{aligned}
\theta_{RMST} = \frac{1}{n_1} * \sum_{i=1}^{n_1} \left[\Delta_i^\tau * (\widetilde{T}_i \wedge \tau) + (1 - \Delta_i^\tau) * \hat{Q}_S(C_i|X, A) \mid A = 1\right] - \\
\frac{1}{n_0} * \sum_{j=1}^{n_0} \left[\Delta_j^\tau * (\widetilde{T}_j \wedge \tau) + (1 - \Delta_j^\tau) * \hat{Q}_S(C_j|X, A) \mid A = 0\right]
\end{aligned}
$$

with $n_1$ corresponds to the number of observations in the treated group, $n_0$ corresponds to the number of observations in the control group and $\hat{Q}_S(\widetilde{T} \wedge \tau \mid X, A) = \frac{1}{\hat{S}(\widetilde{T} \wedge \tau|X,A)} \int_{\widetilde{T} \wedge \tau}^{+\infty} \widetilde{T} \wedge \tau . d\hat{F}(\widetilde{T} \wedge \tau \mid X, A)$ the estimation function of the remaining survival function ($E[\widetilde{T} \wedge \tau \mid X, A, \widetilde{T} \wedge \tau > t]$)

This estimator is the illustration that $\theta_{RMST}$ can be computed directly without integrate the restricted survival function. It does the average of the RMST using fully complete pseudo-observation from Buckley-James transformation of each group. This estimator behaves as if there is no censoring.

##### 3.2.3.1.1 Properties of Buckley-James estimator

BJ transformation is considered as the best predictor of the original response in the sens that $\mathbb{E}(T - T^*(O))^2 \leq \mathbb{E}(T - T^*)^2$ among all censoring unbiased transformation. It can be regarded as the best restoration (Fan and Gijbels 1994b). But, as presented in Section 3.2.1.2, the properties of BJ estimator is dependent on the model specification of $Q_s$. If the model specification is correct, then BJ transformation is considered as the best predictor of the original response among all censoring unbiased transformation.

##### 3.2.3.1.2 Implementation

The following code includes several functions :

- Q_t_hat computes $Q_s(t|x, a)$ for each timepoint and individuals which uses the previously implemented function estimate_survival_function.
- Q_Y compute specifically $Q_s(T_i \wedge \tau|X_i, A_i)$.

- BJ computes the $\theta_{RMST}$ by implementing the Buckley-James estimator and in using the previous functions.

The previous estimators are suitable for RCT settings but not for more complex contexts such as observational studies. The next section will introduce more sophisticated estimators to measure the Average Treatment Effect (ATE) in the context of observational studies.

# 4 Causal survival analysis with an observational study

In contrast to randomized controlled trials (RCTs), observational data, such as data from registries, electronic health records, and national health data systems, are collected without controlled interventions on treatment allocation.

In this setting, treated and control groups can be unbalanced due to the non-randomized design, meaning that the treatment allocation is not controlled. Consequently, the causal effect of the treatment is confounded by variables $X_t$, which affect both the time-to-event outcome $T$ and the treatment allocation $A$.

The assumption of randomized treatment assignment, as presented in Equation 5 (see Section 3), is not satisfied in observational studies. To enable the identifiability of the causal estimand, additional assumptions regarding treatment allocation are needed. These assumptions are standard in causal inference methods with observational data and can be extended to identify $\theta_{RMST}$:

**Assumption 5: Conditional exchangeability / Uncounfoundedness**

$$A \perp\!\!\!\perp (T(0), T(1)) | X_t \tag{11}$$

with $X_t$ the set of covariates that are related both to treatment's assignment and outcomes.

Under Assumption Equation 11, the treatment assignment is randomly assigned conditionally on the covariates $X$. It is as if the treatment for all subjects were randomly selected inside each subgroup. Exactly than Equation 3, this assumption assumes that there are no unmeasured confounders as unobserved confounders make it impossible to separate correlation and causality.

**Assumption 6: Positivity / Overlap for treatment**

$$1 > P(A = a \mid X_t = x) > 0 \tag{12}$$

The Equation 12 assumption requires adequate overlap in covariate distributions between treatment groups, meaning every observation must have a non-zero probability of being treated.

In addition to confounding bias, censoring bias must also be addressed, as discussed in Section 3. The censoring mechanism assumptions from Section 2.3.1 remain applicable. The next section will present the identifiability formula, corresponding estimator, and implementation when censoring is independent in observational studies. Following this, we will cover identifiability formulas when censoring is conditionally dependent on $X_c$, with their estimators and implementations.

In the following section, we will consider that the variables $X_t = X_c = X$.

## 4.1 Independent censoring

The figure below (referenced as Figure 5) illustrates a simple causal graph in observational survival data with independent censoring, showing that the causal effect is not easily identifiable without additional assumptions:

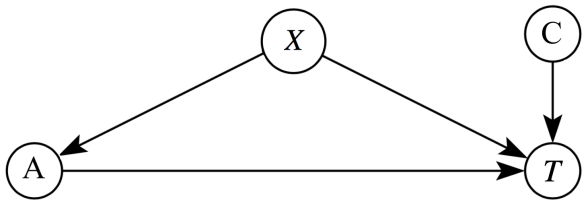



Figure 5: Illustration of a simple causal graph in observational survival data with independent censoring (A is the treatment, $X_t$ the confounding variable and $T$ is the time to event outcome and $C$ is the censoring)

#### 4.1.1 Identifiability

Under Equation 11 (Uncounfoundedness) and Equation 2 (independent censoring), the RMST can be identified as follows:

$$
\begin{aligned}
\theta &= \mathbb{E}[T(1) \wedge \tau - T(0) \wedge \tau] \\
&= \int_0^\tau \mathbb{E}[I\{T(1) > t\}] - \mathbb{E}[I\{T(0) > t\}]dt \\
&\text{(By linearity)} \\
&= \int_0^\tau \mathbb{E}\left[\mathbb{E}[I\{T(1) > t\}|X]\right] - \mathbb{E}\left[\mathbb{E}[I\{T(0) > t\}|X]\right] \\
&\text{(Law of total probability and Ignorability)} \\
&= \int_0^\tau \mathbb{E}\left[\frac{\mathbb{E}[I\{T(1) > t|X\}] * \mathbb{E}[A|X\}]}{e(X)}\right] - \mathbb{E}\left[\frac{\mathbb{E}[I\{T(0) > t|X\}] * \mathbb{E}[1 - A|X\}]}{1 - e(X)}\right] dt \\
&\text{(In color, the terms are equal)} \\
&= \int_0^\tau \mathbb{E}\left[\frac{\mathbb{E}[I\{T(1) > t\} * A|X\}}{e(X)}\right] - \mathbb{E}\left[\frac{\mathbb{E}[I\{T(0) > t\} * (1 - A)|X]]}{1 - e(X)}\right] dt \\
&\text{(By unconfoundedness)} \\
&= \int_0^\tau \mathbb{E}\left[\frac{I\{T(1) > t\} * A}{e(X)}\right] - \mathbb{E}\left[\frac{\mathbb{E}[I\{T(0) > t\} * (1 - A)}{1 - e(X)}\right] dt \\
&\text{(Law of total probability)} \\
&= \int_0^\tau \mathbb{E}\left[\frac{I\{T(1) > t|A = 1\} * A}{e(X)}\right] - \mathbb{E}\left[\frac{I\{T(0) > t|A = 0\} * (1 - A)}{1 - e(X)}\right] dt \\
&= \int_0^\tau \mathbb{E}\left[\frac{I\{T > t|A = 1\} * A}{e(X)}\right] - \mathbb{E}\left[\frac{I\{T > t|A = 0\} * (1 - A)}{1 - e(X)}\right] dt \\
&\text{(By consistency)}
\end{aligned}
\tag{13}
$$

This inverse probability treatment weighting approach is similar to the method for IPW estimator used in causal inference to correct the bias due to some confounding variables. The use of propensity score in causal inference has been introduced by Rosenbaum and Rubin (1983) and extended to survival analysis by Xie and Liu (2005).

##### 4.1.1.1 Estimation with IPTW Kaplan Meier

Under Equation 11 (Uncounfoundedness) and Equation 2 (independent censoring), an adjusted Kaplan Meier estimator is easily derived from the identifiability Equation 13. This estimator includes a weighting term to take into account that the treated and control groups are unbalanced. This

weighted estimator is called the inverse probability of treatment weighted Kaplan Meier estimator (IPTW-KM) (Xie and Liu 2005):

**Definition 4.1** (IPTW Kaplan Meier estimator)**.**

$$\hat{S}_{IPTW-KM}(t \mid A = a) = \prod_{j=1, t_j <= t} \left(1 - \frac{\sum_i \hat{w}_i * I\left\{T_i = t_j, C_i \geq t_j, A_i = a\right\}}{\sum_k \hat{w}_k * I\left\{T_k \geq t_j, C_k \geq t_j, A_k = a\right\}}\right) \tag{14}$$

with:

- $\hat{w}_i = \frac{A_i}{\hat{e}(X_i)} + \frac{1-A_i}{1-\hat{e}(X_i)}$ the inverse of the propensity score.
- $e(X_i) = P(A_i = 1|X_i)$ is the propensity score.

In the exact same way than before, the corresponding $\theta_{RMST}$ is obtained in integrating from 0 to $\tau$ the difference between IPTW adjusted Kaplan Meier estimator of the treated and controls.

#### 4.1.1.1.1 Properties of IPTW Kaplan Meier estimator

Under Equation 11 and Equation 12, and that $\sup_{X_i} |\hat{e}(X_i) - e(X_i)| \xrightarrow{P} 0$, IPTW adjusted Kaplan-Meier is a consistent estimate of the survival function :

$$\hat{S}_{IPTW}(t) \xrightarrow{P} S(t) \quad \forall t \leq \tau$$

The propensity score can be estimated given $X, A$ in using parametric or non-parametric model, then the corresponding variance of IPTW Kaplan-Meier estimator is estimated by (Xie and Liu 2005):

$$\text{Var}\left[\hat{S}(t)\right] = \left(\hat{S}(t)\right)^2 \sum_{j:t_j \leq t} \frac{1 - \hat{s_j}}{\hat{M}_j \hat{s_j}}$$

with

$$\hat{M}_j = \begin{cases} \frac{\left(\sum_{i:T_i \geq t_j} \frac{1}{\hat{e}_i}\right)^2}{\sum_{i:T_i \geq t_j} \left(\frac{1}{\hat{e}_i}\right)^2} & \text{when } A = 1, \\ \frac{\left(\sum_{i:T_i \geq t_j} \frac{1}{1-\hat{e}_i}\right)^2}{\sum_{i:T_i \geq t_j} \left(\frac{1}{1-\hat{e}_i}\right)^2} & \text{when } A = 0. \end{cases}$$

#### 4.1.1.1.2 Implementation

The following code includes several functions :

- Estimate_propensity_score enables to compute the probability of being treated over the time $e(X)$ in using logistic regression or probability forest (probability_forest function from grf (Tibshirani et al. 2017)). It allows cross-fitting for probability forest (n.folds>1).
- IPTW_Kaplan_meier computes the presented estimator and regroups the previous function and adjusted.KM presented in Section 3.2.2.1.2.

```
# Function to estimate propensity score
estimate_propensity_score <- function(data, treatment_covariates,
                                      type_of_model = "glm", n.folds = NULL) {
  # Generalized Linear Model (GLM)
  if (type_of_model == "glm") {
```

```r
    outcome <- 'A'
    f <- as.formula(paste(outcome, paste(c(treatment_covariates),
                                         collapse = " + "), sep = " ~ "))
    fitA <- glm(f, data = data, family = binomial(link = "logit"))
    e_hat <- predict(fitA, newdata = data, type = "response")
  }

  # Probability Forest (only for continuous variables,
  # categorical variables need one-hot encoding)
  if (type_of_model == "probability forest" && !is.null(n.folds)) {
    # Initialization
    n <- nrow(data)
    e_hat <- rep(NA, n)
    A <- data$A

    # Cross-fitting to avoid overfitting
    if (n.folds > 1) {
      # Split the dataset into n folds
      indices <- split(seq(n), sort(seq(n) %% n.folds))

      # Learn and predict for each fold
      for (idx in indices) {

        # Learn on all data except idx
        propensity_model <- probability_forest(
          as.matrix(data[-idx, treatment_covariates]),
          as.factor(A[-idx]))

        # Predict on idx
        e_hat[idx] <- predict(
          propensity_model,
          newdata = as.matrix(data[idx, treatment_covariates]))$predictions[, 2]
      }
    }
    # No cross-fitting
    else if (n.folds == 0 | n.folds == 1) {

      propensity_model <- probability_forest(
        as.matrix(data[, treatment_covariates]),
        as.factor(A))

      e_hat <- predict(
        propensity_model,
        newdata = as.matrix(data[, treatment_covariates]))$predictions[, 2]
    }
  }
  return(e_hat)
}
```

```r
# Function to calculate IPTW Kaplan-Meier
IPTW_Kaplan_meier <- function(data, tau, X.names.propensity,
                              nuisance_propensity = "glm", n.folds = NULL) {
  # Estimate propensity scores
  data$e_hat <- estimate_propensity_score(
    data,
    treatment_covariates = X.names.propensity,
    type_of_model = nuisance_propensity,
    n.folds = n.folds)

  # Truncate observed times at tau
  data$T_obs_tau <- pmin(data$T_obs, tau)

  # Define censoring status at tau
  data$status_tau <- as.numeric((data$T_obs >= tau) |
                                (data$T_obs < tau & data$status == 1))

  # Calculate weights
  data$weights <- (data$A) * (1 / data$e_hat) + (1 - data$A) / (1 - data$e_hat)

  # Adjusted Kaplan-Meier estimator
  S <- adjusted.KM(
    times = data$T_obs,
    failures = data$status,
    variable = data$A,
    weights = data$weights)

  # Calculate RMST from the adjusted survival curves
  RMST <- RMST_1(S_A1 = S[S$variable == 1,],
                 S_A0 = S[S$variable == 0,],
                 tau = tau)

  return(list("intA0" = RMST$intA0, "intA1" = RMST$intA1, "RMST" = RMST$RMST))
}
```

The next section will present several identifiability formulae when conditional independent censoring is met in an observational study.

## 4.2 Conditional independent censoring

The figure below (referenced as Figure 6) illustrates a simple causal graph in observational survival data with conditionally independent censoring:

Under Equation 11 (uncounfoundedness) and Equation 3 (conditional independent censoring), the causal effect is affected both by confounding variables (confounding bias) and by conditional censoring. In this context, $\theta_{RMST}$ can be identified using the conditional survival function without the need for weighting corrections. This approach results in the following identifiability formula and plug-in estimator.

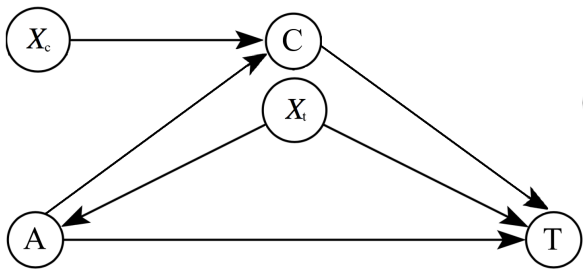



Figure 6: Illustration of a simple causal graph in observational survival data with independent censoring (A is the treatment, $X_t$ the confounding variable and $T$ is the time to event outcome and $C$ is the censoring)

#### 4.2.1 Identifiability of $\theta_{RMST}$ by using conditional survival function

Under Equation 11 (uncounfoundedness), the $\theta_{RMST}$ can be identified very easily as follow:

$$\begin{aligned}
\theta &= \mathbb{E}\left[T(1) \wedge \tau - T(0) \wedge \tau\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[T(1) \wedge \tau - T(0) \wedge \tau \mid X\right]\right] \\
&\text{(Law of total probability)} \\
&= \mathbb{E}\left[\mathbb{E}\left[T(1) \wedge \tau \mid X, A=1\right] - \mathbb{E}\left[T(0) \wedge \tau \mid X=X, A=0\right]\right] \\
&\text{(Uncounfoundedness and Positivity of treatment)} \\
&= \mathbb{E}\left[\mathbb{E}\left[T \wedge \tau \mid X, A=1\right] - \mathbb{E}\left[T \wedge \tau \mid X, A=0\right]\right] \\
&\text{(Consistency)}
\end{aligned} \tag{15}$$

This identifiability formula stands out from the others in this rather complex context because of its simplicity. The plug-in estimator derived from this formula is the g-formula. In the identification formula, it seems that this corresponding estimator does not need the Equation 3 and Equation 4 to be verified (conditional independent censoring and positivity for censoring). But these assumptions are implicit and necessary to allow an estimation of the conditional restricted mean survival.

##### 4.2.1.1 Estimation with G-formula estimator

Introduced by J. Robins (1986), the G-formula addresses confounding bias in causal inference and can be applied to survival data. Unlike IPCW, it models the conditional mean of the outcome and marginalizes over the empirical covariate distributions for each treatment arm. The plug-in estimator is:

**Definition 4.2** (G-formula plug-in estimator)**.**

$$\widehat{\theta}_{\text{g-formula}}(\tau) = \frac{1}{n} \sum_{i=1}^{n} \left(\hat{F}\left(X_i, 1\right) - \hat{F}\left(X_i, 0\right)\right)$$

with $\hat{F}(x, a) \triangleq \mathbb{E}[T \wedge \tau \mid X = x, A = a] = \int_0^\tau S_a(t, x) dt$ the estimation of the conditional restricted mean survival time.

An illustration of G-formula computation is displayed below:

$S_a(t, x)$ can be estimated through parametric (e.g., Weibull), semi-parametric (e.g., Cox), or non-parametric models (e.g., survival random forests). Two methods can be used:

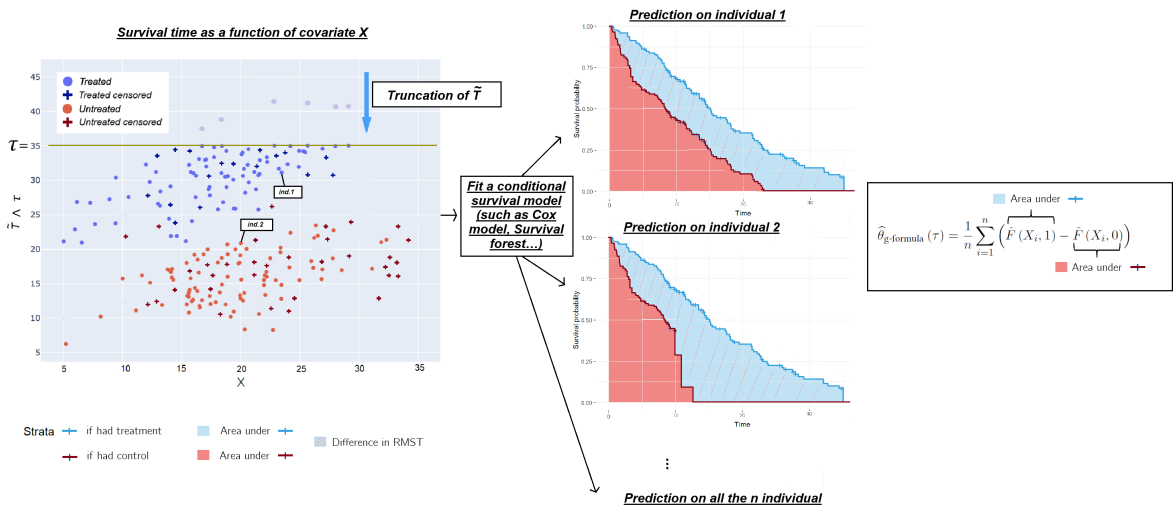



Figure 7: Illustration of the G-formula Process for calculating $\theta_{RMST}$

- S-learner: Fit a single model on both covariates and treatment, predicting for both treated and control groups.
- T-learner: Fit separate models for treated and control groups, then predict as if all observations had received each treatment.

The choice between the two methods depends on the model used. For instance, if you are using a Cox model and the proportional hazards assumption is violated for the treatment variable, the S-learner may not be appropriate, and the T-learner would be a better option.

#### 4.2.1.1.1 Properties of G-formula plug-in estimator

Chen and Tsiatis (2001) proove the asymptotic normality of the G-estimator assuming a proportional hazard relationship. The article also derive an expression of the asymptotic variance of G-formula estimator.

Moreover, G-formula plug-in estimator is simple to implement and also very stable.

#### 4.2.1.1.2 Implementation

The following code implements several function :

- Expected_survival computes the integral by the trapezoidal rule (described in Section 9) of a given survival function.
- g_formula_T_learner computes the presented G-formula plug in estimator in using two-learners survival forest or cox regression. It allows cross-fitting for survival forest (n.folds>1).
- g_formula_S_learner computes the presented G-formula plug in estimator in using a single learner survival forest or cox regression. It allows also cross-fitting for survival forest (n.folds>1).

```
# Compute the area under the survival curve for each individual using the
# Trapezoidal rule.
# S.hat: predicted survival function for each individual.
expected_survival <- function(S.hat, Y.grid) {
```

```
  # Y.grid: vector of time at which to evaluate the survival estimates
  # (same as S.hat).

  # Calculate the distance between each time point.
  grid.diff <- diff(c(0, Y.grid, max(Y.grid)))

  # Compute the area under each survival curve.
  area <- c(base::cbind(1, S.hat) %*% grid.diff)

  return(area)
}

# Function to estimate the g-formula Two-learner.
g_formula_T_learner <- function(data,
                                X.names.outcome,
                                tau,
                                nuisance_survival = "cox",
                                n.folds = NULL) {
  # Compute min(T_obs,tau)
  data$T_obs_tau <- ifelse(data$T_obs >= tau, tau, data$T_obs)

  # Y.grid is the grid of time points where we want to estimate the
  # survival function.
  Y.grid <- sort(unique(data$T_obs_tau))

  # Subset data for A == 0.
  data0 <- data %>% filter(A == 0)

  # Subset data for A == 1.
  data1 <- data %>% filter(A == 1)

  # Cox
  if (nuisance_survival == "cox") {
    outcome <- 'Surv(T_obs, status)'

    # cph do not support notation I(X2) but X2 directly (contrary to coxph)
    X.names.outcome <- gsub("I\\((X[0-9]+\\2)\\)", "\\1", X.names.outcome)


    # Learn Cox regression on two datasets: A|X.
    f <- as.formula(paste(outcome, paste(c(X.names.outcome), collapse = " + "),
                          sep = " ~ "))

    # Fit the two models on the covariates of time Y.grid.
    fitS0 <- cph(f, data = data0, y = TRUE, x = TRUE, times = Y.grid)
    fitS1 <- cph(f, data = data1, y = TRUE, x = TRUE, times = Y.grid)

    # Predict survival probabilities for each individual at each Y.grid.
    fit.pred1 <- predictCox(fitS1, newdata = data, times = Y.grid,
```

```
                              type = "survival")
  fit.pred0 <- predictCox(fitS0, newdata = data, times = Y.grid,
                              type = "survival")

  # Survival probabilities for each individual at each Y.grid.
  S_hat1 <- fit.pred1$survival
  S_hat0 <- fit.pred0$survival
} else {
  # Survival forest.
  # Initialize objects
  n <- nrow(data)
  fit.pred1 <- matrix(NA, nrow = n, ncol = length(Y.grid))
  fit.pred0 <- matrix(NA, nrow = n, ncol = length(Y.grid))

  if (n.folds > 1) {
    # Split the dataset into n-folds.
    indices <- split(seq(n), sort(seq(n) %% n.folds))

    # For all index in each split.
    for (idx in indices) {
      # Fit survival forest on all observations with A=1 except idx
      forest.grf1 <- survival_forest(
        X = as.matrix(data[-idx & data[, "A"] == 1, X.names.outcome]),
        Y = data[-idx & data[, "A"] == 1, "T_obs"],
        D = data[-idx & data[, "A"] == 1, "status"],
        failure.times = Y.grid
      )
      # Fit survival forest on all observations with A=0 except idx
      forest.grf0 <- survival_forest(
        X = as.matrix(data[-idx & data[, "A"] == 0, X.names.outcome]),
        Y = data[-idx & data[, "A"] == 0, "T_obs"],
        D = data[-idx & data[, "A"] == 0, "status"],
        failure.times = Y.grid
      )
      # Predict on idx
      fit.pred1[idx, ] <- predict(
        forest.grf1, as.matrix(data[idx, X.names.outcome]),
        failure.times = Y.grid)$predictions

      fit.pred0[idx, ] <- predict(
        forest.grf0, as.matrix(data[idx, X.names.outcome]),
        failure.times = Y.grid)$predictions
    }
  } else if (n.folds == 0 | n.folds == 1) {
    # If no cross-fitting
    # Fit survival forest on all observation with A=1
    forest.grf1 <- survival_forest(
      X = as.matrix(data[data[, "A"] == 1, X.names.outcome]),
      Y = data[data[, "A"] == 1, "T_obs"],
```

```
        D = data[data[, "A"] == 1, "status"],
        failure.times = Y.grid
      )
      # Fit survival forest on all observation with A=0
      forest.grf0 <- survival_forest(
        X = as.matrix(data[data[, "A"] == 0, X.names.outcome]),
        Y = data[data[, "A"] == 0, "T_obs"],
        D = data[data[, "A"] == 0, "status"],
        failure.times = Y.grid
      )
      # Predict on all observations
      fit.pred1 <- predict(forest.grf1, as.matrix(data[, X.names.outcome]),
                           failure.times = Y.grid)$predictions
      fit.pred0 <- predict(forest.grf0, as.matrix(data[, X.names.outcome]),
                           failure.times = Y.grid)$predictions
    }

    S_hat1 <- fit.pred1
    S_hat0 <- fit.pred0
  }

  # Compute the area under each survival curve until max(Y.grid) = tau.
  E_hat1 <- expected_survival(S_hat1, Y.grid)
  E_hat0 <- expected_survival(S_hat0, Y.grid)

  # Calculate the mean difference.
  theta_g_formula <- mean(E_hat1 - E_hat0)

  return(theta_g_formula)
}

# Function to estimate the g-formula Single-learner.
g_formula_S_learner <- function(data,
                                X.names.outcome,
                                tau,
                                nuisance_survival = "cox",
                                n.folds = NULL) {
  # Compute min(T_obs,tau)
  data$T_obs_tau <- ifelse(data$T_obs >= tau, tau, data$T_obs)

  # Y.grid is the grid of time points where we want to estimate the
  # survival function.
  Y.grid <- sort(unique(data$T_obs_tau))

  # Set A=0 for all data
  data0 <- data
  data0$A <- 0

  # Set A=1 for all data
```

```r
data1 <- data
data1$A <- 1

# Cox
if (nuisance_survival == "cox") {
  outcome <- 'Surv(T_obs, status)'

  # cph do not support notation I(X2) but X2 directly (contrary to coxph)
  X.names.outcome <- gsub("I\\((X[0-9]+\\2)\\)", "\\1", X.names.outcome)

  # Learn Cox regression on one datasets and add A as covariate
  f <- as.formula(paste(outcome, paste(c(X.names.outcome,"A"),
                                       collapse = " + "),
                        sep = " ~ "))

  # Fit the two models on the covariates of time Y.grid.
  fitS <- cph(f, data = data, y = TRUE, x = TRUE, times = Y.grid)

  # Predict survival probabilities for each individual at each Y.grid.
  fit.pred1 <- predictCox(fitS, newdata = data1, times = Y.grid,
                          type = "survival")
  fit.pred0 <- predictCox(fitS, newdata = data0, times = Y.grid,
                          type = "survival")

  # Survival probabilities for each individual at each Y.grid.
  S_hat1 <- fit.pred1$survival
  S_hat0 <- fit.pred0$survival
} else {
  # Survival forest.
  # Initialize objects
  n <- nrow(data)
  fit.pred1 <- matrix(NA, nrow = n, ncol = length(Y.grid))
  fit.pred0 <- matrix(NA, nrow = n, ncol = length(Y.grid))

  if (n.folds > 1) {
    # Split the dataset into n-folds.
    indices <- split(seq(n), sort(seq(n) %% n.folds))

    # For all index in each split.
    for (idx in indices) {
      # Fit survival forest on all observations except idx (add A as covariate)
      forest.grf <- survival_forest(
        X = as.matrix(data[-idx, c(X.names.outcome,"A")]),
        Y = data[-idx, "T_obs"],
        D = data[-idx, "status"],
        failure.times = Y.grid
      )
      # Predict on idx
      fit.pred1[idx, ] <- predict(
```

```
          forest.grf, as.matrix(data1[idx, c(X.names.outcome,"A")]),
          failure.times = Y.grid)$predictions

        fit.pred0[idx, ] <- predict(
          forest.grf, as.matrix(data0[idx, c(X.names.outcome,"A")]),
          failure.times = Y.grid)$predictions
      }
    } else if (n.folds == 0 | n.folds == 1) {
      # If no cross-fitting
      # Fit survival forest on all observation (add A as covariate)
      forest.grf <- survival_forest(
        X = as.matrix(data[, c(X.names.outcome,"A")]),
        Y = data[, "T_obs"],
        D = data[, "status"],
        failure.times = Y.grid
      )

      # Predict on all observations
      fit.pred1 <- predict(
        forest.grf, as.matrix(data1[, c(X.names.outcome,"A")]),
        failure.times = Y.grid)$predictions
      fit.pred0 <- predict(
        forest.grf, as.matrix(data0[, c(X.names.outcome,"A")]),
        failure.times = Y.grid)$predictions
    }

    S_hat1 <- fit.pred1
    S_hat0 <- fit.pred0
  }

  # Compute the area under each survival curve until max(Y.grid) = tau.
  E_hat1 <- expected_survival(S_hat1, Y.grid)
  E_hat0 <- expected_survival(S_hat0, Y.grid)

  # Calculate the mean difference.
  theta_g_formula <- mean(E_hat1 - E_hat0)

  return(theta_g_formula)
}
```

It exists other type of estimators valid in this context. The following section will present the identifiability formulae, corresponding estimators and implementations of another estimator with double correction weighting.

### 4.2.2 Identifiability of $\theta_{RMST}$ by double weighting with IPC transformation

Under Equation 11 (uncounfoundedness) and Equation 3 (conditional independent censoring), $\theta_{RMST}$ can be also identified as follow:

$$
\begin{aligned}
\theta_{RMST} &= \mathbb{E}[T(1) \wedge \tau - T(0) \wedge \tau] \\
&= \int_0^\tau \mathbb{E}[I\{T(1) > t\}] - \mathbb{E}[I\{T(0) > t\}]dt \\
&\text{(By linearity)} \\
&= \int_0^\tau \mathbb{E}\left[\mathbb{E}[I\{T(1) > t\}|X, A]\right] - \mathbb{E}\left[\mathbb{E}[I\{T(0) > t\}|X, A]\right] \\
&\text{(Law of total probability)} \\
&= \int_0^\tau \mathbb{E}\left[1(\{T(1) \geq t\}) * \frac{\Delta^\tau}{S_C(T \wedge \tau \mid A, X)} * \frac{A}{e(X)}|A = 1\right] - \\
&\mathbb{E}\left[1(\{T(0) \geq t\}) * \frac{\Delta^\tau}{S_C(T \wedge \tau \mid A, X)} * \frac{1 - A}{1 - e(X)}|A = 0\right] dt \\
&\text{(By def of IPCW and IPTW)} \\
&= \int_0^\tau \mathbb{E}\left[1(\{T \geq t\}) * \frac{\Delta^\tau}{S_C(T \wedge \tau \mid A, X)} * \frac{A}{e(X)}|A = 1\right] - \\
&\mathbb{E}\left[1(\{T \geq t\}) * \frac{\Delta^\tau}{S_C(T \wedge \tau \mid A, X)} * \frac{1 - A}{1 - e(X)}|A = 0\right] dt \\
&\text{(By consistency)}
\end{aligned}
\tag{16}
$$

The identifiability formula Equation 16 uses the two previous weighting correction: IPCW presented in Section 3.2 and IPTW in Section 4.1.

#### 4.2.2.1 Estimation with IPCW-IPTW Kaplan Meier

The IPTW-IPCW Kaplan Meier estimator can be used to estimate the causal treatment effect according to the identifiability formula Equation 16:

**Definition 4.3** (IPTW-IPCW Kaplan Meier estimator)**.**

$$
\hat{S}_{IPTW-IPCW}(t \mid A = a) = \prod_{j=1, t_j <= t} \left(1 - \frac{\sum_i \hat{w}_i(t, X_i) * I\left\{T_i = t_j, C_i \geq t_j, A_i = a\right\}}{\sum_i \hat{w}_i(t, X_i) * I\left\{T_i \geq t_j, C_i \geq t_j, A_i = a\right\}}\right) \tag{17}
$$

with $\hat{w}_i(t, X_i) = \frac{\Delta_i^\tau}{\hat{S}_C(\widetilde{T} \wedge \tau | A_i, X_i)} * (\frac{A_i}{\hat{e}(X_i)} + \frac{1 - A_i}{1 - \hat{e}(X_i)})$ the corresponding weight including the inverse of the estimated propensity score ($\hat{e}(X)$) and the inverse estimated probability of remain uncensored given the covariates ($\hat{S}_c(t|X)$) and the censoring status at $\tau$ ($\Delta^\tau$) already presented in Section 3.2 and Section 4.1.

This IPTW-IPCW Kaplan Meier estimator enables a balance between treatment and control groups and between censored and uncensored individuals. The corresponding $\theta_{RMST}$ is the integral of the difference between the survival curve with $A = 1$ and the $A = 0$.

##### 4.2.2.1.1 Properties of IPTW-IPCW Kaplan Meier estimator

Under the regularity conditions $\hat{e}(x) = e^*(x)$ for almost all $x$ and $\hat{S}_C\left(\widetilde{T} \wedge \tau \mid a, x\right) = S_C^*\left(\widetilde{T} \wedge \tau \mid a, x\right)$ for all $t \in [0, \tau]$), Schaubel and Wei (2011) shows that estimated $\hat{\theta}_{RMST}$ in using double weighting method converge almost surely and uniformly to $\theta_{RMST}$ and that $\hat{\theta}_{RMST}$

- Uniformly consistent: $\sup_{s \in [0,\tau]} |\hat{\theta}_{RMST}(s) - \theta_{RMST}(s)| \xrightarrow{\mathbb{P}} 0$.

- Asymptotically normal for a fixed $t$: $\sqrt{n}\left(\hat{\theta}_{RMST}(t) - \theta_{RMST}(t)\right) \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}\left(0, V^2(t)\right)$ with $V^2$ the covariance function expressed in Schaubel and Wei (2011).

Like the IPTW Kaplan-Meier (in Section 4.1.1.1) or the IPCW Kaplan-Meier (in Section 3.2.2.1), this estimator is even more subject to extreme weights since it is weighted by two inverse probabilities. This makes it very sensitive to positivity for censoring and treatment.

#### 4.2.2.1.2 Implementation

The following code implements the IPTW-IPCW Kaplan Meier function, which uses the previously implemented functions:

- estimate_survival_function (detailed and implemented in Section 3.2.2.1.2) which computes the probability of remaining uncensored over time $S_c$ using either a Cox model or a survival forest.
- estimate_propensity_score (detailed and implemented in Section 4.1.1.1.2) which computes the propensity score of each observation using a logistic model or a probability forest.
- adjusted.KM (detailed and implemented in Section 3.2.2.1.2) which computes an adjusted Kaplan Meier estimator.
- RMST_1 (detailed and implemented in Section 3.1.2.2) which computes RMST.

```
IPTW_IPCW_Kaplan_meier <- function(data,
                                   X.names.propensity,
                                   X.names.censoring,
                                   tau,
                                   nuisance_propensity = "glm",
                                   nuisance_censoring = "cox",
                                   n.folds = NULL) {
  # Censoring time to tau if observed time exceeds tau
  data$T_obs_tau <- ifelse(data$T_obs >= tau, tau, data$T_obs)

  # Create censoring status for tau
  data$censor.status_tau <- 1 - as.numeric((data$T_obs >= tau) |
                                           (data$T_obs < tau & data$status == 1))

  # Create status at tau
  data$status_tau <- as.numeric((data$T_obs >= tau) |
                                (data$T_obs < tau & data$status == 1))

  # Grid of unique observed times truncated at tau
  Y.grid <- sort(unique(data$T_obs_tau))

  # Estimate propensity scores
  data$e_hat <- estimate_propensity_score(data,
                                          treatment_covariates = X.names.propensity,
                                          type_of_model = nuisance_propensity,
                                          n.folds = n.folds)

  # Estimate survival function for censoring
```

```
  S_C_hat <- estimate_survival_function(data, X.names = X.names.censoring,
                                        Y.grid = Y.grid, T_obs = "T_obs_tau",
                                        status = "censor.status_tau",
                                        type_of_model = nuisance_censoring,
                                        n.folds = n.folds)

  # Get estimated survival probabilities for censoring
  data$S_C <- S_C_hat$S_hat[cbind(1:nrow(data), match(data$T_obs_tau, Y.grid))]

  # Calculate weights
  data$weights <- data$status_tau / data$S_C *
                  (data$A * (1 / data$e_hat) +
                     (1 - data$A) * (1 / (1 - data$e_hat)))

  # Compute adjusted Kaplan-Meier estimator
  S <- adjusted.KM(times = data$T_obs_tau,
                   failures = data$status_tau,
                   variable = data$A,
                   weights = data$weights)

  # Compute Restricted Mean Survival Time (RMST)
  RMST <- RMST_1(S_A1 = S[S$variable == 1, ],
                 S_A0 = S[S$variable == 0, ],
                 tau = tau)

  # Return RMST and ATE for treated and not treated groups
  return(list(RMST = RMST$RMST, ATE_treated = RMST$intA1,
              ATE_not_treated = RMST$intA0))
}
```

As in Section 3.2.3, it exists other solutions to identify $\theta_{RMST}$. One of them apply directly the inverse propensity weighting on the pseudo-observation without using survival function. It leads to the following identifiability formula.

### 4.2.3 Identifiability of $\theta_{RMST}$ by double weighting with BJ transformation

Under Equation 11 (uncounfoundedness), Equation 3 (conditional independent censoring) and Equation 4 (positivity for censoring) the $\theta_{RMST}$ can be identified as follow:

$$
\begin{aligned}
\theta &= \mathbb{E}[T(1) \wedge \tau - T(0) \wedge \tau] \quad (1) \\
&= \mathbb{E}[T(1) \wedge \tau] - \mathbb{E}[T(0) \wedge \tau] \quad (2) \\
&\text{(By linearity)} \\
&= \mathbb{E}\left[\mathbb{E}[T(1) \wedge \tau \mid X]\right] - \mathbb{E}[T(0) \wedge \tau \mid X] \quad (3) \\
&\text{(By the total probability law)} \\
&= \mathbb{E}\left[\mathbb{E}[(T \wedge \tau) \mid A, X]\left(\frac{A}{e(X)} - \frac{1-A}{1-e(X)}\right)\right] \quad (4) \\
&\text{(def of IPTW)} \\
&= \mathbb{E}\left[\mathbb{E}[T^* \mid A, X]\left(\frac{A}{e(X)} - \frac{1-A}{1-e(X)}\right)\right] \quad (5) \\
&= \mathbb{E}\left[(\Delta^\tau * (\widetilde{T} \wedge \tau) + (1-\Delta^\tau) * Q_S(C|X,A)) * \left(\frac{A}{e(X)} - \frac{1-A}{1-e(X)}\right)\right] \quad (6) \\
&\text{(by Buckley James transformation)}
\end{aligned}
\tag{18}
$$

The transition from line 5 to line 6 has been explained by the Buckley-James transformation explained in Section 3.2.1.2. The next section will start with the estimation based on the identifiability formula in Equation 18.

#### 4.2.3.1 Estimation with IPTW-BJ estimator

Based on the identifiability of Equation 18, the IPTW-BJ estimator is defined as follow:

**Definition 4.4** (IPTW-BJ estimator)**.**

$$
\hat{\theta}_{\text{IPTW-BJ}}(\tau) = \frac{1}{n}\sum_{i=1}^{n}\left(\Delta^\tau \widetilde{T} \wedge \tau + (1-\Delta^\tau)\hat{Q}_S(\widetilde{T} \wedge \tau | X, A)\right)\left(\frac{A_i}{\hat{e}\left(X_i\right)} - \frac{1-A_i}{1-\hat{e}\left(X_i\right)}\right).
$$

Exactly than simple BJ estimator (introduced in Section 3.2.3.1), this estimator is easier to implement than the weighting survival function.

##### 4.2.3.1.1 Properties of IPTW-BJ estimator

**Pas de publi** (peut-être Per K. Andersen, Syriopoulou, and Parner (2017))

##### 4.2.3.1.2 Implementation of IPTW-BJ estimator

The following code implements the IPTW-BJ function, which uses the previously implemented functions:

- estimate_survival_function (detailed and implemented in Section 3.2.2.1.2) which computes the probability of remaining uncensored over time $S_c$ using either a Cox model or a survival forest.
- estimate_propensity_score (detailed and implemented in Section 4.1.1.1.2) which computes the propensity score of each observation using a logistic model or a probability forest.

```
# In using the min
IPTW_BJ <- function(data,
                    X.names.propensity,
                    X.names.outcome,
                    tau,
```

```
                    nuisance_propensity = "glm",
                    nuisance = "cox",
                    n.folds = NULL) {
  # Minimum of T_obs and tau
  data$T_obs_tau <- ifelse(data$T_obs >= tau, tau, data$T_obs)

  # Grid of unique observed times truncated at tau
  Y.grid <- sort(unique(data$T_obs_tau))

  # Indicator for min(T, tau) < C
  data$status_tau <- as.numeric((data$T_obs >= tau) |
                                (data$T_obs < tau & data$status == 1))


  # Estimate propensity scores
  data$e_hat <- estimate_propensity_score(data,
                                          treatment_covariates = X.names.propensity,
                                          type_of_model = nuisance_propensity,
                                          n.folds = n.folds)


  # Estimation of Q_s
  Q_t <- Q_t_hat(data, tau, X.names.outcome, nuisance, n.folds)
  data$Q_y <-  Q_Y(data,tau,Q_t)

  # BJ transformation
  data$Y <-  data$status_tau * data$T_obs_tau +
                           (1 - data$status_tau) * data$Q_y

  # IPTW on BJ transformation
  data$RST <- data$Y * (data$A/data$e_hat-(1-data$A)/(1-data$e_hat))

  RMST <- mean(data$RST)

  # Return RMST and other relevant metrics
  return(RMST)
}
```

As seen in the previous properties, the estimators with inverse weighting such as IPTW-IPCW or BJ-IPTW are subject to instability and sensible to mis-specification of nuisance models. Regression based estimators such as G-formula are also subject to mis-specification of the outcome model. A possibility to overcome these issues is to use doubly robust corrections that is by property more robust to mis-specification. The next section will discuss the identifiability of $\theta_{RMST}$ using doubly robust methods. First, by introducing the concept of the influence function, which enables the construction of the augmented version of IPTW. Then, the properties of this AIPTW estimator will be disclosed. Next, we will present the augmented version of IPCW, another unbiased censoring transformation. Finally, we will describe the corresponding estimator with double augmented weighting corrections for treatment and censoring and explain its theoretical properties.

#### 4.2.4 Identifiability of $\theta_{RMST}$ by augmented corrections

##### 4.2.4.1 Augmented weighting for treatment from semi-parametric approach

The semi-parametric approach yields $\sqrt{n}$-consistent estimators with valid confidence intervals by deriving the efficient influence function (Hines et al. 2022). The influence function measures the sensitivity of an estimator to small perturbations in the sample distribution. Suppose we have a distribution $P$, perturbed distribution $\bar{P}$, and "mixture" distribution $P_\epsilon = P + \epsilon(\bar{P} - P)$ with $\epsilon \in [0; 1]$ (Fisher and Kennedy 2021). For the unknown true distribution $P_0$, the goal is to estimate $\psi(P_0)$.

For the plug-in estimator $\psi(P) = \mathbb{E}_P[\mathbb{E}_P(T \wedge \tau | X, A = 1)]$, the influence function which satisfies Equation 22 (in Section 9) is expressed as (detail in Kennedy (2023)):

$$\varphi(\psi) = \frac{A}{e(X)}(T \wedge \tau - F(X, A = 1)) + F(X, A = 1) - \psi$$

Thus, based on Equation 22 in Section 9, the corresponding bias-corrected estimator is:

$$\psi(P) = \mathbb{P}_n \left[ \hat{F}(X, A = 1) + \frac{A.(T \wedge \tau - \hat{F}(X, A = 1))}{\hat{e}(X)} \right]$$

where $\mathbb{P}_n$ means the corresponding sample average, $\hat{F}(X)$ is the estimated conditional survival and $\hat{e}(X)$ is the estimated propensity score.

When we come back to our estimand of interest $\theta_{RMST}$, $\psi(P) = \mathbb{E}_P[\mathbb{E}_P(T \wedge \tau | X, A = 1) - \mathbb{E}_P(T \wedge \tau | X, A = 0)]$, the efficient influence function is then:

$$\begin{aligned} \varphi(\psi) = F(X, A = 1) - F(X, A = 0) + \\ \frac{A}{e(X)}(T \wedge \tau - F(X, A = 1)) - \frac{1 - A}{1 - e(X)}(T \wedge \tau - F(X, A = 0)) \end{aligned}$$

Finally, the bias-corrected estimator can be defined as:

$$\begin{aligned} \hat{\theta}_{AIPTW} &= \mathbb{E}[\mathbb{E}(T \wedge \tau | X, A = 1)] - \mathbb{E}[\mathbb{E}(T \wedge \tau | X, A = 0)] \\ &= \mathbb{E}\left[ \underbrace{\hat{F}(X, A = 1)}_{1} + \underbrace{\frac{A.(T \wedge \tau - \hat{F}(X, A = 1))}{\hat{e}(X)}}_{2} \right] - \\ &\quad \mathbb{E}\left[ \underbrace{\hat{F}(X, A = 0)}_{1} + \underbrace{\frac{(1 - A).(T \wedge \tau - \hat{F}(X, A = 0))}{1 - \hat{e}(X)}}_{2} \right] \end{aligned}$$

This augmented inverse probability of treatment weighting (AIPTW) estimator does direct conditional survival (1) and IPW to conditional survival residuals (2).

Alternatively, AIPTW can also be defined as:

$$
\begin{aligned}
\hat{\theta}_{AIPTW} &= \mathbb{E}[\mathbb{E}(T \wedge \tau | X, A = 1)] - \mathbb{E}[\mathbb{E}(T \wedge \tau | X, A = 0)] \\
&= \mathbb{E}\left[\underbrace{\frac{A.T \wedge \tau}{\hat{e}(X)}}_{1} + \underbrace{\hat{F}(X, A = 1) * \frac{\hat{e}(X) - A}{\hat{e}(X)}}_{2}\right] - \\
&\quad \mathbb{E}\left[\underbrace{\frac{(1 - A) * T \wedge \tau}{1 - \hat{e}(X)}}_{1} + \underbrace{\hat{F}(X, A = 0) * \frac{(1 - \hat{e}(X)) - A}{1 - \hat{e}(X)}}_{2}\right]
\end{aligned}
$$

In that case, AIPTW does direct IPTW adjustment (1) and conditional survival adjustment on IPTW residuals (2).

This estimator is an adaptation of the AIPW estimator in causal inference (James M. Robins and Zhao 1994, 1995; Chernozhukov et al. 2016).

#### 4.2.4.1.1 Properties of AIPTW estimator

Semi-parametric estimators, like AIPTW, are doubly robust, meaning that AIPTW remains consistent if either the treatment model $\hat{F}(X, A) = E[T \wedge \tau | X]$ or the outcome model $\hat{e}(X)$ is correctly specified. This robustness can be explained by the decomposition of $\hat{\psi} - \psi$

$$
\begin{aligned}
\widehat{\psi} - \psi &= \psi(\widehat{\mathbb{P}}) + \mathbb{P}_n\{\varphi(Z; \widehat{\mathbb{P}})\} - \psi(\mathbb{P}) \\
&= (\mathbb{P}_n - \mathbb{P})\{\varphi(Z; \widehat{\mathbb{P}})\} + R_2(\widehat{\mathbb{P}}, \mathbb{P}) \\
&= \underbrace{(\mathbb{P}_n - \mathbb{P})\{\varphi(Z; \mathbb{P})\}}_{\mathrm{S}^*} + \underbrace{(\mathbb{P}_n - \mathbb{P})\{\varphi(Z; \widehat{\mathbb{P}}) - \varphi(Z; \mathbb{P})\}}_{\mathrm{T}_1} + \underbrace{R_2(\widehat{\mathbb{P}}, \mathbb{P})}_{\mathrm{T}_2}
\end{aligned}
$$

with $S^* = o_{\mathbb{P}}(1/\sqrt{n})$ by the central limit theorem. $T_1$, known as the empirical process, is of smallest order (sample average of a short term as $\varphi(Z; \widehat{\mathbb{P}})$ converge to $\varphi(Z; \mathbb{P})$) when working with not too complex estimators (such as Donsker estimator) and in using sample splitting. The double robustness properties can be explained by the remainder term $T_2$.

By Cauchy-Schwarz theorem, this remainder term $T_2$ can be lower and gives:

$$
\left|R_2(\widehat{\mathbb{P}}, \mathbb{P})\right| \leq \left(\frac{1}{\epsilon}\right) \int \left|e(x) - \widehat{e}(x)\right| \left\|F(x) - \widehat{F}(x)\right| d\mathbb{P}(x) \leq \left(\frac{1}{\epsilon}\right) \left\|\widehat{e} - e\right\| \|\widehat{F} - F\|
$$

If $\widehat{e}(x) \geq \epsilon > 0$ and that the combination product of $\|\widehat{e} - e\|$ and $\|\widehat{F} - F\|$ give $o_{\mathbb{P}}(\frac{1}{\sqrt{n}})$, thus, $\left|R_2(\widehat{\mathbb{P}}, \mathbb{P})\right| = o_{\mathbb{P}}(\frac{1}{\sqrt{n}})$. In other words, the residual term tends towards zero even if one of the estimators $widehatF(x)$ or $widehate(x)$ converges slowly while the other compensates by converging more quickly and vice-versa.

To conclude, AIPTW is root-$n$ consistent, asymptotically normal (even in non-parametric approach) and attains the nonparametric efficiency bound (no estimator can have smaller mean squared error than this estimator) (Kennedy 2023).

This estimator only considers correction for treatment in case of confounding effect. In addition, it can be applied on complete observations. In our context of Equation 11 (uncounfoundedness) and Equation 3 (conditional independent censoring), this estimator cannot be applied. This augmented

treatment weighting could be associated with transformations already mentioned above but an augmented transformation based on the semi-parametric approach also exists for censoring to overcome this problem and form an estimator that is also doubly robust for censoring.

This estimator is limited to complete observations. In the context of censoring, it becomes unsuitable as we refer to an incomplete data. This issue can be resolved with a previous introduced unbiased transformation for censoring. But, a semi-parametric approach would offer better solution, providing complete observation and a doubly robust estimator for censoring.

#### 4.2.4.2 Augmented weighting for conditionally independent censoring from semi-parametric approach

The augmented weighting to correct conditionally independent censoring is considered as an augmented censoring unbiased transformation (simple transformations were introduced in Section 3.2.1).

This augmented transformation from Fan and Gijbels (1994a) is inspired by the Buckley James transformation (Buckley and James 1979) and Inverse probability censoring transformation (Koul, Susarla, and Ryzin 1981) (detailed in Section 3.2.1.2 and Section 3.2.1.1) with an augmentation terms (Tsiatis 2006; Laan and Robins 2003).

This transformation can be expressed as (D. Rubin and Laan 2007):

$$\begin{aligned} T^{\star}(O) &= T_{S,S_c^{\star}}(O) \\ &= \frac{\widetilde{T} \wedge \tau \Delta^{\tau}}{S_c(\widetilde{T} \wedge \tau \mid X, A)} + \frac{Q_S(\widetilde{T} \wedge \tau \mid X, A)(1 - \Delta^{\tau})}{S_c(\widetilde{T} \wedge \tau \mid X, A)} \\ &\quad - \int_{-\infty}^{\widetilde{T} \wedge \tau} \frac{Q_S(\widetilde{T} \wedge \tau \mid X, A)}{S_c^2(\widetilde{T} \wedge \tau \mid X, A)} d(1 - S_c(\widetilde{T} \wedge \tau \mid X, A)) \end{aligned}$$

With $Q_S(\widetilde{T} \wedge \tau \mid X, A) = \frac{1}{S(\widetilde{T} \wedge \tau | X, A)} \int_{\widetilde{T} \wedge \tau}^{+\infty} \widetilde{T} \wedge \tau . dF(\widetilde{T} \wedge \tau \mid X, A)$ the estimation function of the remaining survival function ($E[\widetilde{T} \wedge \tau \mid X, A, \widetilde{T} \wedge \tau > t]$)

##### 4.2.4.2.1 Properties of AIPCW estimator

This estimator is root-$n$ consistent, asymptotically normal (even in non-parametric approach) and attains the nonparametric efficiency bound (details in Laan and Robins (2003)) if either $S_c(t|x, a)$ or $Q_S(t|x, a)$ is correctly estimated.

In order to obtain a complete correction for confounding effect and conditionally independent censoring, the both estimator can be combined.

#### 4.2.4.3 Double augmented weighting correction

The augmented inverse probability of treatment weighting (AIPTW) and the augmented inverse probability of censoring weighting (AIPCW) can be combined to form an augmented estimator of RMST with conditional censored data .

**Definition 4.5** (AIPTW- AIPCW estimator (IPTW on pseudo-observation and correction on IPTW

residuals)).

$$\hat{\theta}_{AIPTW-AIPCW} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{A_i}{\hat{e}\left(X_i\right)} - \frac{1-A_i}{1-\hat{e}\left(X_i\right)}\right)\hat{T}^*_{DR} + \hat{F}\left(X_i, A=1\right)\left(1-\frac{A_i}{\hat{e}\left(X_i\right)}\right) - \hat{F}\left(X_i, A=0\right)\left(1-\frac{1-A_i}{1-\hat{e}\left(X_i\right)}\right)$$

with

$\hat{T}^*_{DR} = \frac{\widetilde{T}_i \wedge \tau \cdot \Delta^\tau}{\hat{S_C}\left(\widetilde{T}_i \wedge \tau | X_i\right)} + \frac{Q_{\hat{S}}(\tilde{T}_i \wedge \tau | X, A)\cdot(1-\Delta^\tau)}{\hat{S_C}(\tilde{T}_i \wedge \tau | X_i)} - \int_0^{\widetilde{T}_i \wedge \tau} \frac{Q_{\hat{S}}(c|X_i, A_i)}{\hat{S_C}^2(c|X_i, A_i)} d(1-\hat{S_C}(c \mid X_i, A_i))$ which corresponds to the previous AIPCW estimator in Section 4.2.4.2.

Basically, the above AIPTW-AIPCW applies direct IPTW on the augmented unbiased pseudo-population (that would have been observed if there was no censoring) and regression correction on IPTW residuals.

This estimator can be also written as:

**Definition 4.6** (AIPTW- AIPCW estimator (Conditional survival and correction on survival residuals with pseudo-observation)).

$$\hat{\theta}_{AIPTW-AIPCW} = \frac{1}{n}\sum_{i=1}^{n}\hat{F}\left(X_i, A=1\right) - \hat{F}\left(X_i, A=0\right) + \left(\frac{A_i * (\hat{T}^*_{DR} - \hat{F}\left(X_i, A=1\right))}{\hat{e}\left(X_i\right)} - \frac{(1-A_i) * (\hat{T}^*_{DR} - \hat{F}\left(X_i, A=0\right))}{1-\hat{e}\left(X_i\right)}\right)$$

In this case, the estimator applies direct conditional survival and the residuals of the conditional survival (based on the augmented unbiased pseudo-population) is weighted by IPTW.

#### 4.2.4.3.1 Properties of AIPTW-AIPCW estimator

The estimator AIPTW-AIPCW involves nuisance models for the outcome $\hat{S}$ to estimate both $Q_S$ and $F$, for the treatment to estimate the propensity score $e$ and for the censoring to estimate $S_c$. As introduced in Section 4.2.4.1.1 and Section 4.2.4.2.1, this estimator achieves double robustness, root-$n$ consistency and is asymptotically normal and attains the nonparametric efficiency bound (Kennedy 2023). Concerning the double robustness, the estimator is expected to be consistent only if:

- $\hat{S}_c(t|a, x)$ and $\hat{e}(x)$ are consistent.
- $\hat{S}(t|a, x)$ is consistent.

In addition, this method has to incorporate cross-fitting (Zheng and Laan 2012; Chernozhukov et al. 2016) to provide an efficient (reduce possible overfitting) and unbiased method. Also, this estimator is also refereed as the locally efficient estimator written as the solution of the efficient influence curve equation.

In the exact same way than AIPTW and AIPCW, the AIPTW-AIPCW estimator enables the use of machine-learning estimation of nuisance functions while preserving the root-$n$ consistency of the AIPTW-AIPCW estimator (Chernozhukov et al. 2016).

#### 4.2.4.3.2 Implementation

The following code implements the AIPTW-AIPCW function, which uses other tool functions:

- Estimate_hazard_function which computes the instantaneous hazard function of the corresponding survival function necessary to compute the third term of AIPCW. This function compute the instantaneous hazard function as a forward difference of $-\log\left(\hat{S}(t \mid x)\right)$.
- Integrate which integrate from 0 to $T_i \wedge \tau$ for each individuals an integrand also to compute the third term of AIPCW.
- AIPCW which finally compute pseudo observation based on all previous functions, estimate_survival_function (detailed and implemented in Section 3.2.2.1.2), Q_t_hat function and Q_Y function (implemented in Section 3.2.3.1.2). The third term of the transformation is simplified by: $\int_0^{T_i \wedge \tau} \frac{\lambda_c(s|A_i,X_i)}{\hat{S}_C(s|A_i,X_i)} * Q_S(s \mid X_i, A_i) ds$ (detail in Section 9).

```r
# Tool functions
# Compute hazard function from survival function
estimate_hazard_function <- function(S_hat, Y.grid) {

  Y.grid[Y.grid==0]<-0.001

  # Calculate differences between successive elements in Y.grid
  Y.diff <- diff(c(0, Y.grid))

  # Get the number of columns in S_hat
  grid.length <- ncol(S_hat)

  # Compute -log of survival probabilities (cumulative hazard function),
  # Add 1 as the first value of survival function to ensure that lambda(0)=0
  log.surv.C <- -log(base::cbind(1, S_hat))

  # Calculate differences of -log survival probabilities to have
  # the instantaneous hazard function
  h_hat <- log.surv.C[, 2:(grid.length + 1)] - log.surv.C[, 1:grid.length]

  # Divide each column of h_hat by the corresponding element in Y.diff
  h_hat <- sweep(h_hat, 2, Y.diff, "/")

  # Return the estimated hazard function
  return(h_hat)
}

integrate <- function(integrand, Y.grid, times) {
  # Create a filter matrix to indicate which elements are within the time
  # interval
  filter <- sapply(1:length(Y.grid), function(i) {
    return(as.numeric(i <= findInterval(times, Y.grid)))
  })
```

```r
  # Apply the filter to the integrand
  integrand_filtered <- filter * integrand

  # Sum the rows of the filtered integrand to get the integrated values
  integrated_value <- rowSums(integrand_filtered)

  # Return the integrated values
  return(integrated_value)
}


# DR censoring transformation
AIPCW <-function(data,
                 tau,
                 X.names.censoring = c("X1","X2","X3","X4"),
                 X.names.outcome = c("X1","X2","X3","X4"),
                 nuisance_Qt = "cox",
                 nuisance_censoring = "cox",
                 n.folds = NULL,
                 h_C_hat = NULL,
                 method_aipw = 1) {

  # Truncate observed times at tau
  data$T_obs_tau <- pmin(data$T_obs, tau)

  # Define status at tau
  data$status_tau <-  as.numeric((data$T_obs > tau) |
                                 (data$T_obs <= tau &  data$status == 1 ))

  data$censor.status_tau <- 1- as.numeric(
    (data$T_obs > tau) | (data$T_obs <= tau &  data$status == 1 ))

  Y.grid <- sort(unique(data$T_obs_tau))

  # Estimate survival function for censoring
  S_C_hat <- estimate_survival_function(data = data,X.names.censoring,
                                        type_of_model = nuisance_censoring,
                                        n.folds = n.folds,
                                        Y.grid = Y.grid,
                                        T_obs = "T_obs_tau",
                                        status = "censor.status_tau")

  Y.index <- findInterval(data$T_obs_tau, Y.grid)

  data$S_C_hat_T_obs_tau <- S_C_hat$S_hat[cbind(seq_along(Y.index), Y.index)]


  if (is.null(h_C_hat)) {
      h_C_hat <- estimate_hazard_function(S_C_hat$S_hat,Y.grid)
```

```
  }

  # Compute Q.t.hat
  Q.t.hat <- Q_t_hat(data = data,
                     X.names = X.names.outcome,
                     tau = tau,
                     nuisance = nuisance_Qt,
                     n.folds = n.folds)

  # Compute Q.Y.hat
  data$Q.Y.hat <- Q_Y(data = data, tau, Q.t.hat)

  # Compute first term
  data$first_term <- (data$T_obs_tau * data$status_tau) /
    data$S_C_hat_T_obs_tau

  # Compute second term
  data$second_term <- (data$Q.Y.hat * (1 - data$status_tau)) /
    data$S_C_hat_T_obs_tau

  Y.diff <- diff(c(0, Y.grid))

  # Compute integrand for the third term
  integrand <- sweep( ( (h_C_hat) / S_C_hat$S_hat )* (Q.t.hat), 2, Y.diff, "*")

  # Compute third term
  data$third_term <- integrate(integrand, Y.grid, data$T_obs_tau)

  # Compute pseudo outcome
  pseudo_outcome <- data$first_term + data$second_term - data$third_term

  return(pseudo_outcome)
}
```

Then, we can compute the AIPTW-AIPCW estimator which uses all the previous functions and estimate_propensity_score (detailed and implemented in Section 4.1.1.1.2) which computes the propensity score of each observation using a logistic model or a probability forest:

```
AIPTW_AIPCW <- function(data,
                        tau,
                        X.names.propensity = c("X1", "X2", "X3", "X4"),
                        X.names.censoring = c("X1", "X2", "X3", "X4"),
                        X.names.outcome = c("X1", "X2", "X3", "X4"),
                        nuisance_propensity = "glm",
                        nuisance_regression = "cox",
                        nuisance_censoring = "cox",
                        nuisance_Qt = "cox",
                        n.folds = NULL) {
```

```r
# Estimate propensity scores
data$e_hat <- estimate_propensity_score(
  data = data,
  treatment_covariates = X.names.propensity,
  type_of_model = nuisance_propensity,
  n.folds = n.folds
)

# Prepare data for censoring model
data$T_obs_tau <- ifelse(data$T_obs >= tau, tau, data$T_obs)

data$censor.status_tau <- 1 - as.numeric((data$T_obs >= tau) |
                                         (data$T_obs < tau & data$status == 1))

data$status_tau <- as.numeric((data$T_obs >= tau) |
                              (data$T_obs < tau & data$status == 1))

# Create unique time grid
Y.grid <- sort(unique(data$T_obs_tau))

if (nuisance_regression == "cox") {
  # Survival formula for Cox model
  outcome <- 'Surv(T_obs, status)'

  # Split data by treatment group
  data.0 <- data %>% filter(A == 0)
  data.1 <- data %>% filter(A == 1)

  # cph do not support notation I(X2) but X2 directly (contrary to coxph)
  X.names.outcome <- gsub("I\\((X[0-9]+\\2)\\)", "\\1", X.names.outcome)

  # Construct formula
  f <- as.formula(paste(outcome, paste(X.names.outcome, collapse = " + "),
                        sep = " ~ "))

  # Fit Cox model to each subset
  fitS0 <- cph(f, data = data.0, y = TRUE, x = TRUE, times = Y.grid)
  fitS1 <- cph(f, data = data.1, y = TRUE, x = TRUE, times = Y.grid)

  # Predict survival on the time grid
  fit.pred1 <- predictCox(fitS1, newdata = data, times = Y.grid,
                          type = "survival")
  fit.pred0 <- predictCox(fitS0, newdata = data, times = Y.grid,
                          type = "survival")

  # Survival probabilities for each individual
  S_hat1 <- fit.pred1$survival
  S_hat0 <- fit.pred0$survival
```

```r
} else {
  # Initialize prediction matrices for survival forest
  n <- nrow(data)
  fit.pred1 <- matrix(NA, nrow = n, ncol = length(Y.grid))
  fit.pred0 <- matrix(NA, nrow = n, ncol = length(Y.grid))

  if (n.folds > 1) {
    # Split indices into n subsets
    indices <- split(seq(n), sort(seq(n) %% n.folds))

    for (idx in indices) {
      # Fit survival forest model to each subset
      forest.grf1 <- survival_forest(
        X = as.matrix(data[-idx & data[, "A"] == 1, X.names.outcome]),
        Y = data[-idx & data[, "A"] == 1, "T_obs_tau"],
        D = data[-idx & data[, "A"] == 1, "status_tau"],
        failure.times = Y.grid
      )

      forest.grf0 <- survival_forest(
        X = as.matrix(data[-idx & data[, "A"] == 0, X.names.outcome]),
        Y = data[-idx & data[, "A"] == 0, "T_obs_tau"],
        D = data[-idx & data[, "A"] == 0, "status_tau"],
        failure.times = Y.grid
      )

      # Predict survival probabilities
      fit.pred1[idx, ] <- predict(
        forest.grf1, as.matrix(data[idx, X.names.outcome]),
        failure.times = Y.grid)$predictions

      fit.pred0[idx, ] <- predict(
        forest.grf0, as.matrix(data[idx, X.names.outcome]),
        failure.times = Y.grid)$predictions
    }
  } else {
    # Fit survival forest model without subset splitting
    forest.grf1 <- survival_forest(
      X = as.matrix(data[data[, "A"] == 1, X.names.outcome]),
      Y = data[data[, "A"] == 1, "T_obs_tau"],
      D = data[data[, "A"] == 1, "status_tau"],
      failure.times = Y.grid
    )

    forest.grf0 <- survival_forest(
      X = as.matrix(data[data[, "A"] == 0, X.names.outcome]),
      Y = data[data[, "A"] == 0, "T_obs_tau"],
      D = data[data[, "A"] == 0, "status_tau"],
      failure.times = Y.grid
```

```r
    )

    # Predict survival probabilities
    fit.pred1 <- predict(
      forest.grf1, as.matrix(data[, X.names.outcome]),
      failure.times = Y.grid)$predictions

    fit.pred0 <- predict(
      forest.grf0, as.matrix(data[, X.names.outcome]),
      failure.times = Y.grid)$predictions
  }

  S_hat1 <- fit.pred1
  S_hat0 <- fit.pred0
}

# Compute area under the survival curve up to tau
data$E_hat1 <- expected_survival(S_hat1, Y.grid)
data$E_hat0 <- expected_survival(S_hat0, Y.grid)

# Compute IPW-weighted residuals
data$IPW_res <- data$E_hat1 * (1 - data$A / data$e_hat) -
  data$E_hat0 * (1 - (1 - data$A) / (1 - data$e_hat))

# Compute AIPCW weights
TDR <- AIPCW(
  data = data,
  tau = tau,
  X.names.censoring = X.names.censoring,
  X.names.outcome = X.names.outcome,
  nuisance_Qt = nuisance_Qt,
  nuisance_censoring = nuisance_censoring,
  n.folds = n.folds
)

data$TDR <- TDR

# Compute AIPCW-weighted residuals
data$AIPCW_w <- data$TDR * (data$A / data$e_hat -
                            (1 - data$A) / (1 - data$e_hat))

# Compute regression residuals
data$reg <- data$E_hat1 - data$E_hat0
data$reg_res <- data$A / data$e_hat * (data$TDR - data$E_hat1) -
  (1 - data$A) / (1 - data$e_hat) * (data$TDR - data$E_hat0)

# Compute estimators
# na.rm = TRUE to remove NA for the mean calculation
AIPTW_AIPCW_IPW_res <- mean(data$AIPCW_w + data$IPW_res, na.rm = TRUE)
```

```
  AIPTW_AIPCW_reg_res <- mean(data$reg + data$reg_res, na.rm = TRUE)

  return(list(AIPTW_AIPCW_reg_res = AIPTW_AIPCW_reg_res,
              AIPTW_AIPCW_IPW_res = AIPTW_AIPCW_IPW_res))
}
```

This function returns the two estimators introduced previously in Section 4.2.4.3.

In the previous section, we discussed methods for implementing various estimators. However, there are existing packages that can directly compute $\theta_{RMST}$ in certain contexts. The next section will introduce these packages and explain the specific conditions under which they are applicable.

# 5 Available packages to compute $\theta_{RMST}$

To date, there are very few packages that offer functions with a direct application of the methods presented above. The package selection is based on criteria:

- Calculation of ATE or CATE in a survival analysis framework similar to our case study (static treatment assignment and binary treatment, baseline covariates, right-censoring type)
- The package is not archived

The function that requires minimal user intervention is grf's (Tibshirani et al. 2017) causal_survival_forest or survRM2 ("survRM2: Comparing Restricted Mean Survival Time" 2015)'s rmst2. The other packages provide functions that demand significant user effort, such as calculating nuisance models, which account for the largest proportion of error.

## 5.1 SurvRM2 Packages

The following function is based on the package survRM2 ("survRM2: Comparing Restricted Mean Survival Time" 2015) which allows to compute the RMST under Equation 5 (random treatment assignment) and Equation 2 (independent censoring). It performs two-sample comparisons using the restricted mean survival time (RMST) as a summary measure of the unadjusted survival time distribution presented in Section 3.1.2. The arguments for this function are the observed time to event, status, arm (treatment in our case) and $\tau$.

```
library(survRM2)
RMST_survRM2 <- function(data, tau) {
  ATE_pack <- rmst2(data$T_obs, data$status, arm = data$A, tau = tau)
  RMST <- ATE_pack[[5]][1]
  return(RMST)
}
```

## 5.2 RISCA

The RISCA package (Foucher, Le Borgne, and Chatton 2019) enables the computation of $\theta_{RMST}$ through various methods: unadjusted survival functions (as discussed in Section 3.1.2), $S_{IPTW}$-adjusted survival functions (from Section 4.1.1.1), and g-computation, a maximum-likelihood estimator based on the g-formula (from Section 4.2.1.1) (Chatton et al. 2022). For adjusted IPTW survival functions, the weights must be calculated prior to using the relevant function.

- The rmst() function computes the RMST for a specified time horizon $\tau$ based on a survival function, which must be pre-calculated. To match Equation 7, the survival function is stratified by treatment group, and the RMST is computed for both treated and control groups, followed by calculating $\theta_{RMST}$ as the difference. The argument type in this function informs if the survival curve is a step-funtion ("s") or a continous function ("l").
- The ipw.survival() function estimates confounder-adjusted survival curves by weighting individual contributions based on the inverse probability of being in the treatment group (as in Equation 14). This function also supports other weights, such as IPTW-IPCW (as in Equation 17), provided the propensity scores are computed beforehand.
- The gc.survival() function estimates the marginal effect of treatment using G-computation for censored time-to-event data, with the Q-model specified by a Cox model. This function operates as a single learner.

```
# Function to estimate RMST using unadjusted method
RISCA_unadj <- function(data,
                        tau) {
  # Fit survival curves stratified by treatment group
  fit <- survfit(Surv(T_obs, status) ~ A, data = data)
  res <- summary(fit)

  # Calculate RMST for treatment group A=1
  RMST_A1 <- rmst(
    times = res$time[as.character(res$strata) == "A=1"],
    surv.rates = res$surv[as.character(res$strata) == "A=1"],
    max.time = tau,
    type = "s"
  )

  # Calculate RMST for treatment group A=0
  RMST_A0 <- rmst(
    times = res$time[as.character(res$strata) == "A=0"],
    surv.rates = res$surv[as.character(res$strata) == "A=0"],
    max.time = tau,
    type = "s"
  )

  # Compute ATE as the difference in RMST between groups
  ATE_RISCA_unadj <- RMST_A1 - RMST_A0
  return(ATE_RISCA_unadj)
}

# Function to estimate RMST using IPTW method
RISCA_iptw <- function(data,
                       tau,
                       X.names.propensity,
                       nuisance_propensity = "glm",
                       n.folds = NULL) {
```

```r
  # Estimate propensity scores
  e_hat <- estimate_propensity_score(
    data,
    treatment_covariates = X.names.propensity,
    type_of_model = nuisance_propensity,
    n.folds = n.folds
  )

  # Compute inverse probability weights
  weighted <- (data$A / e_hat) + ((1 - data$A) / (1 - e_hat))

  # Fit weighted survival curves
  IPW_pack <- ipw.survival(
    times = data$T_obs,
    failures = data$status,
    variable = data$A,
    weights = weighted
  )

  # Calculate RMST for treatment group A=1 using weighted survival curve
  RMST_RISCA_A1 <- rmst(
    times = IPW_pack$table.surv$times[IPW_pack$table.surv$variable == 1],
    surv.rates = IPW_pack$table.surv$survival[IPW_pack$table.surv$variable == 1],
    max.time = tau,
    type = "s"
  )

  # Calculate RMST for treatment group A=0 using weighted survival curve
  RMST_RISCA_A0 <- rmst(
    times = IPW_pack$table.surv$times[IPW_pack$table.surv$variable == 0],
    surv.rates = IPW_pack$table.surv$survival[IPW_pack$table.surv$variable == 0],
    max.time = tau,
    type = "s"
  )

  # Compute ATE as the difference in RMST between groups
  ATE_RISCA_IPW <- RMST_RISCA_A1 - RMST_RISCA_A0
  return(ATE_RISCA_IPW)
}

# Function to estimate RMST using G-formula method
RISCA_gf <- function(data,
                     tau,
                     X.names.outcome) {

  # Define the outcome formula for the Cox model
  outcome <- paste(c('Surv(', "T_obs", ',', "status", ')'), collapse = "")
  # Single learner : the treatment arm is a predictor
  formula <- as.formula(paste(outcome, paste(c(X.names.outcome, 'A'),
```

```
                                                    collapse = " + "), sep = " ~ "))

  # Fit the Cox proportional hazards model
  cox.cdt <- coxph(formula, data = data, x = TRUE)
  summary(cox.cdt)

  # Compute the marginal effect of the treatment (ATE) using the G-formula
  gc.ate <- gc.survival(
    object = cox.cdt,
    data = data,
    group = "A",
    times = "T_obs",
    failures = "status",
    max.time = tau,
    iterations = 100,
    effect = "ATE",
    n.cluster = 1
  )

  # Extract the ATE
  ATE_RISCA_gf <- gc.ate$delta[[1]]
  return(ATE_RISCA_gf)
}
```

## 5.3 Causal survival forest

In this section, the algorithm of causal survival forest will be introduced. It exists other way of using forest algorithm for observational study with conditionally independent censoring such as weighted IPCW causal forest but the causal survival estimator showed better performance (Cui et al. 2023).

### 5.3.1 Theory of causal survival forest

The causal survival forest (Cui et al. 2023) is an adaptation of the causal forest algorithm (Athey, Tibshirani, and Wager 2018) for time-to-event data with censoring. This estimator is primarily used to estimate heterogeneous treatment effects (CATE):

$$\theta(x) = E[y(T_i(1)) - y(T_i(0))|X_i = x]$$

Assumptions like consistency (Equation 1), conditional independent censoring (Equation 3), positivity of censoring (Equation 4), unconfoundedness (Equation 11), and positivity of the propensity score (Equation 12) are crucial for identifying and estimating CATE using causal survival forests.

Without censoring, the estimator adjusts for treatment effects by solving the localized equation:

$$\sum_{i=1}^{n} \alpha_i(x)\psi_\theta(X_i, T_i \wedge \tau, A_i, \hat{e}, \hat{F}) = 0$$

with $\psi_\theta(X_i, T_i \wedge \tau, A_i, \hat{e}, \hat{F}) = [A_i - \hat{e}(X_i)].[T_i \wedge \tau - \hat{F}(X_i) - \theta(A_i - \hat{e}(X_i)]$, $\theta$ is the conditional average treatment effect, $\alpha_{bi}(x) = \frac{\mathbf{1}(\{X_i \in L_b(x)\})}{|L_b(x)|}$, $\alpha_i(x) = \frac{1}{B}\sum_{b=1}^{B} \alpha_{bi}(x)$ with $L_b(x)$ the set of training observations falling in the same leaf as $x$ in the tree $b$. In other words, the weighting $\alpha$ is

used to express heterogeneity in $\theta_{RMST}$ by measuring the relevance of the i-th sample to fitting $\theta$ at $x$. The weights sum is equal to 1.

In the presence of censoring, an additional adjustment corrects for censoring bias, using the AIPCW transformation (Section 4.2.4.2):

$$\psi_\theta\left(X_i, T_i \wedge \tau, A_i, \Delta_i^\tau; \hat{e}, \hat{F}, \hat{\lambda}_a^C, \hat{S}_a^C, \hat{Q}_a\right) =$$
$$\left(\frac{\hat{Q}_{A_i}\left(T_i \wedge \tau \mid X_i\right) + \Delta_i^\tau\left[T_i \wedge \tau - \hat{Q}_{A_i}\left(T_i \wedge \tau \mid X_i\right)\right] - \hat{F}\left(X_i\right) - \theta\left(A_i - \hat{e}\left(X_i\right)\right)}{\hat{S}_{A_i}^C\left(T_i \wedge \tau \mid X_i\right)}\right.$$
$$\left. - \int_0^{T_i \wedge \tau} \frac{\hat{\lambda}_{A_i}^C\left(s \mid X_i\right)}{\hat{S}_{A_i}^C\left(s \mid X_i\right)}\left[\hat{Q}_{A_i}\left(s \mid X_i\right) - \hat{F}\left(X_i\right) - \theta\left(A_i - \hat{e}\left(X_i\right)\right)\right] ds\right)\left(A_i - \hat{e}\left(X_i\right)\right)$$

where $Q_a(t \mid x) = \mathbb{E}\left[T_i \wedge \tau \mid X_i = x, A_i = a, T_i \wedge \tau > t\right]$ is the conditional expectation of the survival time, $\hat{\lambda}_a^C(t \mid x)$

Exactly than illustrated before, the causal survival forest solves this localized equation:

$$\sum_{i=1}^{n} \alpha_i(x)\psi_\theta\left(X_i, T_i \wedge \tau, A_i, \Delta_i^\tau; \hat{e}, \hat{F}, \hat{\lambda}_a^C, \hat{S}_a^C, \hat{Q}_a\right) = 0$$

In the method from Athey and Wager (2019), splits in the forest favor maximizing heterogeneity in treatment effects, using a criterion:

$$\Delta\left(C_1, C_2\right) := \frac{n_{C_1} n_{C_2}}{n_P^2}\left(\hat{\theta}_{C_1}(\mathcal{J}) - \hat{\theta}_{C_2}(\mathcal{J})\right)^2$$

where $\hat{\theta}_{C_1}$ and $\hat{\theta}_{C_2}$ are solutions to the estimating equation computed in the children. and $n_P = |\{i \in \mathcal{J} : X_i \in P\}|$, the number of observations in the parent and $n_{C_j}$ for the number of observations in each child node.

In reality, an approximate criterion is used in practice for computational efficiency.

### 5.3.2 Properties of Causal survival forest

This estimator is Neyman-orthogonal in the sense discussed in Chernozhukov et al. (2016), and attains a $1/\sqrt{n}$ rate of convergence for $\tau$ under 4-th root rates for the nuisance components, provided we use cross-fitting and that assumptions detailed above hold (Cui et al. 2023; Kennedy 2023).

#### 5.3.2.1 Implementation

This causal survival forest is implemented in using grf package (Tibshirani et al. 2017):

```
# Function to estimate RMST using Causal Survival Random Forest (CSRF)
CSRF <- function(data, X.names, tau) {
  # Select and convert covariates to matrix
  X <- data %>%
    dplyr::select(all_of(X.names)) %>%
    as.matrix()

  # Select and convert observed times to matrix
  Y <- data %>%
```

```
    dplyr::select(T_obs) %>%
    as.matrix()

  # Select and convert treatment assignment to matrix
  W <- data %>%
    dplyr::select(A) %>%
    as.matrix()

  # Select and convert event status to matrix
  D <- data %>%
    dplyr::select(status) %>%
    as.matrix()

  # Set target and horizon for the causal forest
  target <- "RMST"
  horizon <- tau

  # Fit a causal survival forest
  cf <- causal_survival_forest(X = X, Y = Y, W = W, D = D, horizon = horizon)

  # Predict using the fitted forest
  cf.predict <- predict(cf)

  # Estimate the average treatment effect (ATE)
  ATE_csf <- average_treatment_effect(cf)

  # Return the estimated ATE
  return(ATE_csf[[1]])
}
```

# 6 Summary of the estimators

In this section, we use tables to summarise the notations and estimators presented and their consistency under mis-specification.

Here's a summary of the notation used in the previous sections:

Table 2: Notations reminder for all estimators

| Symbol | Description |
|---|---|
| $X$ | Covariates |
| $A$ | Treatment indicator ($A = 1$ for treatment, $A = 0$ for control) |
| $T$ | Time to event |
| $T(1), T(0)$ | Potential time to event respectively with and without treatment |
| $S_1, S_0$ | Potential survival curve $(S_a(t) = p(T(a) > t))_{a \in 0,1}$ of the potential time to event |
| $C$ | Censoring time |
| $\widetilde{T}$ | Observed time ($T \wedge C$) |
| $\Delta$ | Censoring indicator (or status) $\mathbb{I}(\{T \leq C\})$ |

| Symbol | Description |
|---|---|
| $\Delta^\tau$ | Censoring indicator of the restricted time (or restricted status) $\mathbb{I}(\{\widetilde{T} > \tau\}) + \mathbb{I}(\{\widetilde{T} \leq \tau\}) \cdot \Delta$ |
| $(t_1, t_2, \ldots, t_D)$ | $D$ ordered distinct times to event in the sample |
| $e(x)$ | Propensity score $\mathbb{E}[A\|X = x]$ |
| $F(a, x)$ | $\mathbb{E}[T \wedge \tau \mid X = x, A = a]$ |
| $S(t\|a, x)$ | Conditional survival function, $p[T \wedge \tau > t\|X = x, A = a]$ for $t \leq \tau$ |
| $S_C(t\|a, x)$ | conditional survival function of the censoring $p(C > t\|X, A)$ for $t \leq \tau$ |
| $Q_S(t\|x, a)$ | $\mathbb{E}[T \wedge \tau \mid X = x, A = a, T \wedge \tau > t]$ |

As illustrated previously, the estimators can be computed in two ways:

- Complete the censored observation by an unbiased censoring transformation, derive a consistent estimator of the RMST (context dependent) on the complete observation of the treated and untreated observation and compute the $\theta_{RMST}$ in doing the mean difference.
- Derive a consistent estimator of the survival mean (context dependent), do the difference of the integral of the treated and untreated survival curves.

As seen in the previous sections, the estimators do not all have the same nuisance parameters to estimate. The following table summarizes the estimators used in the simulation and the nuisance parameters used to estimate them:

Table 3: Nuisance parameter to compute for all estimators

| Estimator | Context of application | Outcome model | Censoring model | Treatment model |
|---|---|---|---|---|
| Unadjusted KM | RCT & Independent censoring | | | |
| ——————————————

models. The table below shows the consistency of the estimators under different mis-specification scenarios

Table 4: Consistency of estimator under model mis-specification. When all the nuisances models are mis-specified none of the estimators is consistent. ✓ indicates consistency of the estimator, ⊠ mean that the nuisance model is necessary for the estimator and that it is not consistent when the model is mis-specified and empty boxes indicate that the nuisance model not needed in the estimator thus mis-specfication has no impact.

| Estimator | Context of application | mis. outcome model | mis. censoring model | mis. treatment model | mis. outcome and censoring | mis. outcome and treatment | mis. censoring and treatment |
|---|---|---|---|---|---|---|---|
| Unadjusted KM | RCT & independent censoring | ~ | ~ | ~ | ~ | ~ | ~ |
| ——————- | ——————- | ——————- | ——————- | ——————- | ——————- | ——————- | ——————- |
| IPCW-KM | RCT & dependent censoring | ~ | ⊠ | ~ | ~ | ~ | ~ |
| BJ | — | ⊠ | ~ | ~ | ~ | ~ | ~ |
| ——————- | ——————- | ——————- | ——————- | ——————- | ——————- | ——————- | ——————- |
| IPTW-KM | Obs & independent censoring | ~ | ~ | ⊠ | ~ | ~ | ~ |
| ——————- | ——————- | ——————- | ——————- | ——————- | ——————- | ——————- | ——————- |
| IPTW-IPCW | Obs & dependent censoring | ~ | ⊠ | ⊠ | ~ | ~ | ~ |
| G-formula | — | ⊠ | ~ | ~ | ~ | ~ | ~ |
| IPTW-BJ | — | ⊠ | ~ | ⊠ | ~ | ~ | ~ |
| AIPTW-AIPCW | — | ✓ | ✓ | ✓ | ⊠ | ⊠ | ✓ |

As discussed in Section 4.2.4.3.1, the only estimator that remains robust to mis-specification is the AIPTW-AIPCW.

After having presented the estimators and their theoretical properties, we will now apply them to simulation sets.

# 7 Simulations

## 7.1 Data generating process

The proposed data generation processes simulate both RCT and observational data, with two versions: one featuring independent censoring and the other with conditionally independent censoring. For the observational data, we include a simple version (where nuisance parameters are well estimated by parametric and semi-parametric models like the Cox model and logistic regression), a more complex version with nonlinear relationships between variables, and another version incorporating covariate interactions.

### 7.1.1 RCT

We conduct two simple simulations to simulate RCTs studies, baseline covariates with no time dependency. The first simulation represents a scenario with independent censoring and the second one with conditionally independent censoring.

The time of event and the censoring time (when there is dependency between the censoring time and the covariates) is simulated using the cumulative hazard inversion method for exponential models (details in Annex Section 9).

For the simulation, $n$ samples $(X_i, A_i, C, T_i(0), T_i(1))$ are generated in the following way:

- $X \sim \mathcal{N}\left(\mu = [1, 1, -1, 1]^\top, \Sigma = I_4\right)$.
- $e(X) = 0.5$ (constant) for the propensity score ($P(A = 1|X) = 0.5$).
- $\lambda(0)(X) = 0.01 \cdot \exp\{0.5X_1 + 0.5X_2 - 0.5X_3 + 0.5X_4\}$ hazard for the event time $T(0)$.
- The hazard for the censoring time $C$:
  - For scenario 1: $\lambda_c = 0.03$ does not depend on covariates.
  - For scenario 2: $\lambda_c(X) = 0.03 \cdot \exp\{0.7X_1 + 0.3X_2 - 0.25X_3 - 0.1X_4 - 0.2A\}$.
- $T(1) = T(0) + 10$.
- the event time is $T = AT(1) + (1 - A)T(0)$.
- The observed time is $\widetilde{T} = \min(T, C)$.
- The status is $\Delta = 1(T \leq C)$.
- The threshold time $\tau$ is set to 25.

The observed samples are $(X_i, A_i, \Delta_i, \widetilde{T}_i)$ represented previously in Table 1.

```
############ RCT
# RCT1:  Random treatment assignment + independent censoring
# RCT2:  Random treatment assignment + dependent censoring (conditional on X)

simulate_data_RCT <- function(n, mu = c(1, 1, -1, 1),
                              sigma = diag(4),
                              colnames_cov = c("X1", "X2", "X3", "X4"),
                              tau,
                              coefT0 = 0.01,
                              parsS = c(0.5, 0.5, -0.5, 0.5),
```

```
                                coefC = 0.03,
                                parsC = c(0.7, 0.3, -0.25, -0.1),
                                parsC_A = c(-0.2),
                                scenario = "RCT2",
                                mis_specification="none") {

if (scenario == "RCT1") {
  # Generate X from a multivariate normal distribution
  X <- MASS::mvrnorm(n, mu, sigma)
  X <- as.data.frame(X)
  colnames(X) <- colnames_cov

  # Treatment variable selection: all X
  X_treatment <- as.matrix(X)

  # Propensity score: constant for random assignment
  e <- rep(0.5, n)

  # Random treatment assignment
  A <- sapply(e, FUN = function(p) rbinom(1, 1, p))

  # Outcome variable selection: all X
  X_outcome <- as.matrix(X)

  # Simulate the outcome using the cumulative hazard inversion method
  epsilon <- runif(n, min = 1e-8, max = 1)
  T0 <- -log(epsilon) / (coefT0 * exp(X_outcome %*% parsS))

  # Simulate independent censoring time
  epsilon <- runif(n, min = 1e-8, max = 1)
  C <- -log(epsilon) / coefC

  # T(1) = T(0) + 10
  T1 <- T0 + 10

  # True survival time
  T_true <- A * T1 + (1 - A) * T0

  # Observed time
  T_obs <- pmin(T_true, C)

  # Status indicator
  status <- as.numeric(T_true <= C)
  censor.status <- as.numeric(T_true > C)

  # Restricted survival time
  T_obs_tau <- pmin(T_obs, tau)
  status_tau <- as.numeric((T_obs > tau) | (T_obs <= tau & status == 1))
```

```r
} else if (scenario == "RCT2") {
  # Generate X from a multivariate normal distribution
  X <- MASS::mvrnorm(n, mu, sigma)
  X <- as.data.frame(X)
  colnames(X) <- c("X1", "X2", "X3", "X4")

  # Treatment variable selection: all X
  X_treatment <- as.matrix(X)

  # Propensity score: constant for random assignment
  e <- rep(0.5, n)

  # Random treatment assignment
  A <- sapply(e, FUN = function(p) rbinom(1, 1, p))

  # Outcome variable selection: all X
  X_outcome <- as.matrix(X)

  # Simulate the outcome using the cumulative hazard inversion method
  epsilon <- runif(n, min = 1e-8, max = 1)
  T0 <- -log(epsilon) / (coefT0 * exp(X_outcome %*% parsS))

  # Simulate dependent censoring time
  X_censoring <- as.matrix(cbind(X,A))
  parsC <- c(parsC,parsC_A)

  epsilon <- runif(n, min = 1e-8, max = 1)
  C <- -log(epsilon) / (coefC * exp(rowSums(X_censoring %*% diag(parsC))))

  # T(1) = T(0) + 10
  T1 <- T0 + 10

  # True survival time
  T_true <- A * T1 + (1 - A) * T0

  # Observed time
  T_obs <- pmin(T_true, C)

  # Status indicator
  status <- as.numeric(T_true <= C)
  censor.status <- as.numeric(T_true > C)

  # Restricted survival time
  T_obs_tau <- pmin(T_obs, tau)
  status_tau <- as.numeric((T_obs > tau) | (T_obs <= tau & status == 1))
}


# Combine all data into a single data frame
```

```
  data_target_population <- data.frame(X, tau, A, T0, T1, C, T_obs, T_obs_tau,
                                       status, censor.status, status_tau, e)

  return(data_target_population)
}
```

```
# data_rct1 simulate the data from RCT with independent censoring
data_rct1 <- simulate_data_RCT(n=2000,
                               tau=25,
                               scenario="RCT1",
                               coefC = 0.03)

# data_rct2 simulate the data from RCT with dependent censoring
data_rct2 <- simulate_data_RCT(n=2000,
                               tau=25,
                               scenario="RCT2",
                               coefC = 0.03,
                               parsC = c(0.7,0.3,-0.25,-0.1),
                               parsC_A = c(-0.2))
```

### 7.1.2 Observational study

In the same way as above, we carried out two simulations of an observational study. The only difference lies in the simulation of the propensity score, which is no longer constant.

For the simulation, n samples $(X_i, A_i, C, T_i(0), T_i(1))$ in the same way than Section 7.1.1, except :

- $\text{logit}\{e(X)\} = -1X_1 - 1X_2 - 2.5X_3 - 1X_4$ for the propensity score $(A)$.
- The hazard for the censoring time $C$:
  - For scenario 2: $\lambda_c(X) = 0.03 \cdot \exp\{0.7X_1 + 0.3X_2 - 0.25X_3 - 0.1X_4\}$.

The observed samples are $(X_i, A_i, \Delta_i, \widetilde{T}_i)$ represented in Table Table 1.

```
# Obs1:  Treatment assignment dependent on X + independent censoring
# Obs2:  Treatment assignment dependent on X + dependent censoring (conditional
# on X)

# Function to simulate observational data for two scenarios: Obs1 and Obs2
simulate_data_obs <- function(n,
                              mu = c(1, 1, -1, 1),
                              sigma = diag(4),
                              colnames_cov = c("X1", "X2", "X3", "X4"),
                              tau,
                              coefT0 = 0.01,
                              parsS = c(0.5, 0.5, -0.5, 0.5),
                              parsA = c(-1, -1, -2.5, -1),
                              parsC_A = c(0),
                              coefC = 0.03,
                              parsC = c(0.7, 0.3, -0.25, -0.1),
```

```r
                                scenario = "Obs2") {

# Generate covariates X from a multivariate normal distribution
X <- mvrnorm(n, mu, sigma)
X <- as.data.frame(X)
colnames(X) <- colnames_cov

# Propensity score model based on X
e <- rowSums(as.matrix(X) %*% diag(parsA))
e <- plogis(e)  # Transform to probability scale

# Treatment assignment based on the propensity score
A <- sapply(e, FUN = function(p) rbinom(n = 1, size = 1, prob = p))

# Outcome model based on X
X_outcome <- as.matrix(X)
epsilon <- runif(n, min = 0.00000001, max = 1)
T0 <- -log(epsilon) / (coefT0 * exp(X_outcome %*% parsS))

# Define treatment effect (shift in survival time due to treatment)
T1 <- T0 + 10

if (scenario == "Obs1") {
  # Scenario 1: Independent censoring
  C <- -log(runif(n, min = 0.00000001, max = 1)) / coefC

} else if (scenario == "Obs2") {
  # Scenario 2: Dependent censoring based on X
  X_censoring <- as.matrix(cbind(X,A))
  parsC <- c(parsC,parsC_A)

  C <- -log(runif(n, min = 0.00000001, max = 1)) /
    (coefC * exp(rowSums(X_censoring %*% diag(parsC))))

} else {
  stop("Invalid scenario. Choose 'Obs1' or 'Obs2'.")
}

# Determine the true survival time based on treatment
T_true <- A * T1 + (1 - A) * T0

# Observed time is the minimum of the true survival time and censoring time
T_obs <- pmin(T_true, C)

# Status indicator: 1 if the event (death) occurred, 0 if censored
status <- as.numeric(T_true <= C)

# Restricted survival time (censored at tau)
T_obs_tau <- pmin(T_obs, tau)
```

```
  status_tau <- as.numeric((T_obs > tau) | (T_obs <= tau & status == 1))

  # Compile the simulated data into a data frame
  DATA_target_population <- data.frame(X, tau, A, T0, T1, C, T_obs, T_obs_tau,
                                       status, status_tau, e)

  return(DATA_target_population)
}
```

```
# Observational data with no informative censoring
data_obs1 <- simulate_data_obs(n=2000,tau=25,scenario="Obs1")

# Observational data simulation with dependent censoring
data_obs2 <- simulate_data_obs(n=2000,tau=25,scenario="Obs2",
                               coefC = 0.03, parsC = c(0.7,0.3,-0.25,-0.1))
```

### 7.1.3 Observational study with nonlinear relationships

We refer to a simulation where the effect of treatment and censoring cannot be captured by a simple parametric (or semi-parametric) model but well estimated by probability forest for propensity model or survival forest for conditional survival or censoring model.

For the simulation, $n$ samples $(X_i, A_i, C, T_i(0), T_i(1))$ are generated in the following way (similar to the scenario 4 in Cui et al. (2023)):

- $X \sim \mathcal{N}\left(\mu = [1,1,1]^\top, \Sigma = I_3\right)$.
- T is generated from a Poisson distribution with mean $X_2 + X_3 + max(0; X_1 - 0,3)A$.
- C from a Poisson distribution with mean $1 + log(1 + exp(X_3))$.
- The propensity score is $e(x) = [(1 + exp(-X_1))(1 + exp(-X_2))]^{-1}$

The maximum follow-up time is $h = 3$. Note that for subjects with $X_1 < 0,3$, treatment does not affect survival time. The horizon time $\tau$ is fixed at 2.

```
simulate_data_complex <- function(n = 2000, tau, parsC = c(0,0,1)){
  # Load necessary library
  library(MASS)

  # Generate covariates
  X <- mvrnorm(n, mu = c(1, 1, 1), Sigma = diag(3))
  X <- as.data.frame(X)
  colnames(X) <- c("X1", "X2", "X3")

  # Convert data frame to matrix for matrix operations
  X_treatment <- as.matrix(X)

  # Generate treatment
  e <- 1 / ((1 + exp(-X_treatment[, "X1"])) * (1 + exp(-X_treatment[, "X2"])))
  A <- sapply(e, function(p) rbinom(1, size = 1, prob = p))
```

```r
  # Generate potential outcomes
  lambda_1 <- X_treatment[, "X2"] + X_treatment[, "X3"] +
              pmax(0, X_treatment[, "X1"] - 0.3) * 1
  lambda_0 <- X_treatment[, "X2"] + X_treatment[, "X3"]

  T1 <- rpois(n, lambda_1)
  T0 <- rpois(n, lambda_0)

  T1[is.na(T1)] <- 0
  T0[is.na(T0)] <- 0

  T_true <- T1 * A + T0 * (1 - A)

  # Generate censoring time
  lambda_C <- 1 + log(1 + exp(parsC[1]*X_treatment[, "X1"] +
                              parsC[2]*X_treatment[, "X2"] +
                              parsC[3]*X_treatment[, "X3"]))
  C <- rpois(n, lambda_C)

  # Observed time and status
  T_obs <- pmin(T_true, C)
  status <- as.numeric(T_true <= C)
  censor_status <- as.numeric(T_true > C)

  # Restricted survival time
  T_obs_tau <- pmin(T_obs, tau)
  status_tau <- as.numeric((T_obs > tau) | (T_obs <= tau & status == 1))

  # Create the final data frame
  DATA_target_population <- data.frame(
    X1 = X$X1,
    X2 = X$X2,
    X3 = X$X3,
    tau = tau,
    A = A,
    T1 = T1,
    T0 = T0,
    T_true = T_true,
    C = C,
    T_obs = T_obs,
    T_obs_tau = T_obs_tau,
    status = status,
    censor_status = censor_status,
    status_tau = status_tau,
    e = e
  )

  return(DATA_target_population)
}
```

### 7.1.4 Observational study with covariates interaction

For this simulation with covariates interaction for outcome and censoring, $n$ samples $(X_i, A_i, C, T_i(0), T_i(1))$ are generated in the following way:

- $X \sim \mathcal{N}\left(\mu = [1, 1, -1, 1, -2, -5]^\top, \Sigma = I_6\right)$.
- T is generated from an exponential distribution with mean $X_1^2 + X_2^2 + X_1 * X_2$.
- C from an exponential distribution with mean $0.1 * X_1^2 + 0.1 * X_3^2 + X_1 * X_3$.
- The propensity score is $e(x) = \frac{1}{1+exp(0.05*X_1-0.01*X_2+1*X_3-0.1X_4)}$.

```
# DGP for mis-specification
simulate_data_mis <- function(n,
                              mu = c(0.5, 0.5, 0.7, 0.5, -0.1, -0.1),
                              sigma =  matrix(c(1, 0, 0, 0, 0, 0,
                                                0, 1, 0, 0, 0, 0,
                                                0, 0, 1, 0, 0, 0,
                                                0, 0, 0, 1, 0, 0,
                                                0, 0, 0, 0, 1, 0,
                                                0, 0, 0, 0, 0, 1),
                                              nrow = 6, byrow = TRUE),
                              colnames_cov = c("X1", "X2", "X3", "X4", "X5",
                                               "X6"),
                              parsA =  c(0.05, -0.01, 1, -0.1, 0, 0),
                              tau){

  # Generate X from a multivariate normal distribution
  X <- MASS::mvrnorm(n, mu, sigma)
  X <- as.data.frame(X)
  colnames(X) <- colnames_cov

  # Treatment variable selection: all X
  X_treatment <- as.matrix(X)

  # Propensity score model based on X
  e <- parsA[1]*X_treatment[, "X1"] + parsA[2]*X_treatment[, "X2"] +
    parsA[3]*X_treatment[, "X3"] + parsA[4]*X_treatment[, "X4"] +
    parsA[5]*X_treatment[, "X5"] + parsA[6]*X_treatment[, "X6"]

  # Logistic regression
  e <- plogis(e)

  # Treatment assignment based on the propensity score
  A <- sapply(e, FUN = function(p) rbinom(n = 1, size = 1, prob = p))

  # Outcome variable selection: all X
  X_outcome <- as.matrix(X)

  lambda <- X[,1]2 + X[,2]2 + X[,1] * X[,2]
  # Simulate the outcome using the cumulative hazard inversion method
```

```r
  epsilon <- runif(n, min = 1e-8, max = 1)
  T0 <- -log(epsilon) / exp(lambda)

  # Simulate independent censoring time
  censoring_lambda <- 0.1*X[,1]2 + 0.1*X[,3]2 + X[,3] * X[,1]
  epsilon <- runif(n, min = 1e-8, max = 1)
  C <- -log(epsilon) / exp(censoring_lambda)


  # T(1) = T(0) + 1
  T1 <- T0 + 1

  # True survival time
  T_true <- A * T1 + (1 - A) * T0

  # Observed time
  T_obs <- pmin(T_true, C)

  # Status indicator
  status <- as.numeric(T_true <= C)
  censor.status <- as.numeric(T_true > C)

  # Restricted survival time
  T_obs_tau <- pmin(T_obs, tau)
  status_tau <- as.numeric((T_obs > tau) | (T_obs <= tau & status == 1))
    # Compile the simulated data into a data frame
  DATA_target_population <- data.frame(X, tau, A, T0, T1, C, T_obs, T_obs_tau,
                                       status, status_tau, censor.status, e)

  return(DATA_target_population)
}
```

```r
mis <- simulate_data_mis(n=2000,tau=0.5)
summary(mis)
```

```
      X1                 X2                 X3                   X4
Min.   :-2.7829   Min.   :-2.7185   Min.   :-2.78941   Min.   :-2.6078
1st Qu.:-0.1627   1st Qu.:-0.1966   1st Qu.: 0.03356   1st Qu.:-0.2062
Median : 0.4946   Median : 0.5082   Median : 0.68502   Median : 0.4536
Mean   : 0.5089   Mean   : 0.5062   Mean   : 0.69467   Mean   : 0.4659
3rd Qu.: 1.1593   3rd Qu.: 1.1994   3rd Qu.: 1.38728   3rd Qu.: 1.1688
Max.   : 4.3390   Max.   : 4.0709   Max.   : 3.88018   Max.   : 3.7725
      X5                 X6                 tau             A
Min.   :-3.4051   Min.   :-3.6047   Min.   :0.5   Min.   :0.000
1st Qu.:-0.8206   1st Qu.:-0.7629   1st Qu.:0.5   1st Qu.:0.000
Median :-0.1340   Median :-0.1111   Median :0.5   Median :1.000
Mean   :-0.1447   Mean   :-0.1096   Mean   :0.5   Mean   :0.642
3rd Qu.: 0.5345   3rd Qu.: 0.5729   3rd Qu.:0.5   3rd Qu.:1.000
Max.   : 3.0334   Max.   : 3.1421   Max.   :0.5   Max.   :1.000
```

```
       T0                 T1                C                 T_obs
 Min.   :0.000000   Min.   :1.000   Min.   :  0.0000   Min.   :0.00000
 1st Qu.:0.008749   1st Qu.:1.009   1st Qu.:  0.1048   1st Qu.:0.03349
 Median :0.075322   Median :1.075   Median :  0.3976   Median :0.18761
 Mean   :0.274209   Mean   :1.274   Mean   :  1.1005   Mean   :0.38257
 3rd Qu.:0.337881   3rd Qu.:1.338   3rd Qu.:  1.0558   3rd Qu.:0.61098
 Max.   :5.544541   Max.   :6.545   Max.   :147.5280   Max.   :2.76181
   T_obs_tau           status        status_tau     censor.status
 Min.   :0.00000   Min.   :0.000   Min.   :0.0000   Min.   :0.000
 1st Qu.:0.03349   1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:0.000
 Median :0.18761   Median :0.000   Median :1.0000   Median :1.000
 Mean   :0.23926   Mean   :0.405   Mean   :0.5575   Mean   :0.595
 3rd Qu.:0.50000   3rd Qu.:1.000   3rd Qu.:1.0000   3rd Qu.:1.000
 Max.   :0.50000   Max.   :1.000   Max.   :1.0000   Max.   :1.000
       e
 Min.   :0.05504
 1st Qu.:0.49622
 Median :0.65871
 Mean   :0.63489
 3rd Qu.:0.79973
 Max.   :0.97923
```

## 7.2 Data description

This section will present the data description of the presented simulation in the Section 7.1.1, Section 7.1.2 and Section 7.1.3 to enhance their characteristics.

### 7.2.1 RCT with independent censoring (RCT scenario 1)

The summary by group of treatment of the generated (observed and unobserved) data set RCT with independent censoring (in Section 7.1.1) is displayed below:

```
[1] "Descriptive statistics for group A=0:   1008"

       X1                 X2                 X3                 X4
 Min.   :-2.4924   Min.   :-1.7340   Min.   :-4.3500   Min.   :-2.4140
 1st Qu.: 0.3647   1st Qu.: 0.3624   1st Qu.:-1.6390   1st Qu.: 0.3107
 Median : 1.0029   Median : 1.0362   Median :-0.9609   Median : 1.0115
 Mean   : 1.0151   Mean   : 1.0519   Mean   :-0.9702   Mean   : 0.9972
 3rd Qu.: 1.7617   3rd Qu.: 1.7555   3rd Qu.:-0.2887   3rd Qu.: 1.6981
 Max.   : 4.2458   Max.   : 3.8338   Max.   : 2.3688   Max.   : 3.9505
       C                   T1                 T0                status
 Min.   :  0.09029   Min.   : 10.01   Min.   :  0.012   Min.   :0.0000
 1st Qu.: 11.09214   1st Qu.: 12.96   1st Qu.:  2.959   1st Qu.:0.0000
 Median : 23.20432   Median : 18.31   Median :  8.315   Median :1.0000
 Mean   : 33.67969   Mean   : 32.24   Mean   : 22.245   Mean   :0.7034
 3rd Qu.: 46.59781   3rd Qu.: 31.72   3rd Qu.: 21.721   3rd Qu.:1.0000
 Max.   :219.86798   Max.   :963.41   Max.   :953.411   Max.   :1.0000
     T_tild
 Min.   :  0.01203
 1st Qu.:  2.26619
 Median :  6.03026
```

```
 Mean   : 11.09681
 3rd Qu.: 14.40920
 Max.   :154.64631

[1] "Descriptive statistics for group A=1:   992"

       X1                 X2                 X3                 X4
 Min.   :-2.0683   Min.   :-2.0704   Min.   :-4.8901   Min.   :-2.2487
 1st Qu.: 0.3669   1st Qu.: 0.3254   1st Qu.:-1.6917   1st Qu.: 0.3415
 Median : 1.0846   Median : 1.0461   Median :-1.0387   Median : 1.0288
 Mean   : 1.0534   Mean   : 1.0232   Mean   :-1.0301   Mean   : 1.0037
 3rd Qu.: 1.7411   3rd Qu.: 1.6804   3rd Qu.:-0.3637   3rd Qu.: 1.6592
 Max.   : 4.1499   Max.   : 4.4575   Max.   : 2.2303   Max.   : 4.6691
       C                   T1                 T0                 status
 Min.   :  0.02092   Min.   : 10.03   Min.   :  0.0269   Min.   :0.000
 1st Qu.:  8.88993   1st Qu.: 12.81   1st Qu.:  2.8109   1st Qu.:0.000
 Median : 21.83453   Median : 18.57   Median :  8.5719   Median :1.000
 Mean   : 32.99138   Mean   : 31.10   Mean   : 21.1022   Mean   :0.506
 3rd Qu.: 47.53068   3rd Qu.: 31.67   3rd Qu.: 21.6722   3rd Qu.:1.000
 Max.   :225.63407   Max.   :664.09   Max.   :654.0904   Max.   :1.000
     T_tild
 Min.   : 0.02092
 1st Qu.: 8.88993
 Median :12.79755
 Mean   :16.19660
 3rd Qu.:19.63480
 Max.   :98.45681
```

The tables summarize the covariates $X1$, $X2$, $X3$, and $X4$, the censoring time $C$, the true time T1 when all observations receive treatment $A = 1$ (T(1)), the true time T0 when all observations receive treatment $A = 0$ (T(0)), the event status (1 if the event occurs, 0 if censored), and $\tilde{T}$, the observed time $min(C, T)$.

Covariates are balanced between groups, and censoring times are the same (independent censoring). However, there are more censored observations in the treated group ($A = 1$) than in the control group ($A = 0$). This is due to the higher instantaneous hazard of the event in the treated group (with $T_1 = T_0 + 10$) compared to the constant hazard of censoring.

### 7.2.2 RCT with conditionally independent censoring (RCT scenario 2)

The summary of the generated (observed and unobserved) data set RCT with conditionally independent censoring (in Section 7.1.1) stratified by treatment is displayed below. As a reminder, the difference between the RCT scenario 1 and 2 is that the censoring time is dependent of the covariates.

```
[1] "Descriptive statistics for group A=0:   976"

       X1                 X2                 X3                 X4
 Min.   :-2.8684   Min.   :-2.0500   Min.   :-4.6263   Min.   :-2.1291
 1st Qu.: 0.3601   1st Qu.: 0.2796   1st Qu.:-1.6400   1st Qu.: 0.3386
 Median : 0.9910   Median : 0.9649   Median :-1.0025   Median : 1.0158
 Mean   : 1.0298   Mean   : 0.9668   Mean   :-0.9888   Mean   : 0.9912
 3rd Qu.: 1.7258   3rd Qu.: 1.6305   3rd Qu.:-0.3175   3rd Qu.: 1.6931
 Max.   : 4.0687   Max.   : 4.3052   Max.   : 2.3954   Max.   : 3.9248
```

```
       C                   T1                 T0                 status
 Min.   :  0.0041   Min.   : 10.00   Min.   :  0.0046   Min.   :0.0000
 1st Qu.:  2.3246   1st Qu.: 12.85   1st Qu.:  2.8474   1st Qu.:0.0000
 Median :  6.6486   Median : 19.00   Median :  8.9994   Median :0.0000
 Mean   : 15.0428   Mean   : 32.44   Mean   : 22.4362   Mean   :0.4518
 3rd Qu.: 16.2614   3rd Qu.: 34.31   3rd Qu.: 24.3077   3rd Qu.:1.0000
 Max.   :333.7865   Max.   :957.79   Max.   :947.7940   Max.   :1.0000
   status_tau         T_tild
 Min.   :0.0000   Min.   :  0.00413
 1st Qu.:0.0000   1st Qu.:  1.39015
 Median :0.0000   Median :  3.35093
 Mean   :0.4857   Mean   :  7.62735
 3rd Qu.:1.0000   3rd Qu.:  8.42498
 Max.   :1.0000   Max.   :120.36830
```

```
[1] "Descriptive statistics for group A=1:   1024"
```

```
       X1                 X2                 X3                 X4
 Min.   :-2.3366   Min.   :-2.9880   Min.   :-4.1692   Min.   :-2.2081
 1st Qu.: 0.3508   1st Qu.: 0.2867   1st Qu.:-1.6723   1st Qu.: 0.2493
 Median : 0.9227   Median : 1.0142   Median :-0.9997   Median : 0.9925
 Mean   : 0.9583   Mean   : 1.0184   Mean   :-1.0009   Mean   : 0.9699
 3rd Qu.: 1.6250   3rd Qu.: 1.6958   3rd Qu.:-0.3269   3rd Qu.: 1.6344
 Max.   : 4.0172   Max.   : 4.0904   Max.   : 2.9363   Max.   : 4.4593
       C                   T1                 T0                 status
 Min.   :  0.0004   Min.   : 10.02   Min.   :  0.0218   Min.   :0.0000
 1st Qu.:  3.3060   1st Qu.: 13.08   1st Qu.:  3.0832   1st Qu.:0.0000
 Median :  8.2765   Median : 18.70   Median :  8.6966   Median :0.0000
 Mean   : 18.7392   Mean   : 33.12   Mean   : 23.1230   Mean   :0.2363
 3rd Qu.: 21.4683   3rd Qu.: 34.07   3rd Qu.: 24.0659   3rd Qu.:0.0000
 Max.   :407.5914   Max.   :563.05   Max.   :553.0502   Max.   :1.0000
   status_tau         T_tild
 Min.   :0.0000   Min.   :  0.00044
 1st Qu.:0.0000   1st Qu.:  3.30603
 Median :0.0000   Median :  8.27647
 Mean   :0.2939   Mean   : 12.47166
 3rd Qu.:1.0000   3rd Qu.: 15.35509
 Max.   :1.0000   Max.   :212.81704
```

Covariates are balanced between the two groups. However, censoring times differ between groups due to conditionally independent censoring based on covariates. There are more censored observations in the treated group ($A = 1$) compared to the control group ($A = 0$). The summary statistics do not reveal the difference between independent and dependent censoring. Dependent censoring affects the rate of censoring among sub-groups without necessarily changing the overall level of censoring.

### 7.2.3 Observational study with independent censoring (Obs scenario 1)

The summary of the generated (observed and unobserved) data set observational study with independent censoring (in Section 7.1.2) stratified by treatment is displayed below to enhance the difference with the other scenario.

```
[1] "Descriptive statistics for group A=0:   1118"
```

```
       X1                 X2                 X3                 X4
 Min.   :-1.9302   Min.   :-2.0318   Min.   :-3.02110   Min.   :-2.5969
 1st Qu.: 0.5841   1st Qu.: 0.5514   1st Qu.:-1.06823   1st Qu.: 0.4949
 Median : 1.1557   Median : 1.1850   Median :-0.51273   Median : 1.1541
 Mean   : 1.2032   Mean   : 1.2119   Mean   :-0.49797   Mean   : 1.1851
 3rd Qu.: 1.8489   3rd Qu.: 1.8741   3rd Qu.: 0.05022   3rd Qu.: 1.8439
 Max.   : 4.2043   Max.   : 3.9571   Max.   : 2.32662   Max.   : 4.6769
       C                   T1                 T0                status
 Min.   :  0.01813   Min.   : 10.01   Min.   :  0.0057   Min.   :0.0000
 1st Qu.:  9.75172   1st Qu.: 12.72   1st Qu.:  2.7192   1st Qu.:0.0000
 Median : 24.32198   Median : 18.70   Median :  8.6958   Median :1.0000
 Mean   : 34.25497   Mean   : 29.09   Mean   : 19.0910   Mean   :0.7111
 3rd Qu.: 47.97677   3rd Qu.: 30.97   3rd Qu.: 20.9715   3rd Qu.:1.0000
 Max.   :235.93660   Max.   :360.88   Max.   :350.8849   Max.   :1.0000
     T_tild
 Min.   :  0.00572
 1st Qu.:  1.99600
 Median :  6.15838
 Mean   : 10.57371
 3rd Qu.: 13.94373
 Max.   :107.93013
```

```
[1] "Descriptive statistics for group A=1:   882"
```

```
       X1                 X2                  X3                 X4
 Min.   :-2.6621   Min.   :-2.66667   Min.   :-4.4216   Min.   :-2.8719
 1st Qu.: 0.1565   1st Qu.: 0.09167   1st Qu.:-2.1079   1st Qu.: 0.1110
 Median : 0.7619   Median : 0.68416   Median :-1.5315   Median : 0.7643
 Mean   : 0.7662   Mean   : 0.70584   Mean   :-1.5586   Mean   : 0.7279
 3rd Qu.: 1.3871   3rd Qu.: 1.34194   3rd Qu.:-1.0212   3rd Qu.: 1.3413
 Max.   : 3.6546   Max.   : 3.62573   Max.   : 0.9356   Max.   : 3.5560
       C                   T1                 T0                status
 Min.   :  0.00312   Min.   : 10.01   Min.   :  0.0056   Min.   :0.0000
 1st Qu.: 10.67663   1st Qu.: 12.92   1st Qu.:  2.9164   1st Qu.:0.0000
 Median : 24.18802   Median : 19.07   Median :  9.0729   Median :1.0000
 Mean   : 34.77617   Mean   : 32.54   Mean   : 22.5375   Mean   :0.5204
 3rd Qu.: 48.41797   3rd Qu.: 33.75   3rd Qu.: 23.7548   3rd Qu.:1.0000
 Max.   :260.85573   Max.   :581.45   Max.   :571.4513   Max.   :1.0000
     T_tild
 Min.   :  0.00312
 1st Qu.: 10.23398
 Median : 13.42348
 Mean   : 17.00916
 3rd Qu.: 20.91715
 Max.   :120.15361
```

The covariates between the two groups of treatment are unbalanced because of dependent treatment assignation. The mean of $X1$, $X2$, $X3$ and $X4$ is bigger in the control group than in the treated group. The censoring times have the same distribution (independent censoring). There are more censored observation in the treated group (A=1) than in the control group (A=0) for the same reason than in the RCT scenario.

### 7.2.4 Observational study with conditionally independent censoring (Obs scenario 2)

The summary of the generated (observed and unobserved) data set Observational study with conditionally independent censoring (in Section 7.1.2) stratified by treatment is displayed below. As a reminder, the difference between the observational scenario 1 and 2 is that the censoring time is dependent of the covariates.

```
[1] "Descriptive statistics for group A=0:   1104"

       X1                 X2                 X3                 X4
 Min.   :-1.6751   Min.   :-1.8284   Min.   :-3.1561   Min.   :-2.0100
 1st Qu.: 0.5839   1st Qu.: 0.5436   1st Qu.:-1.0874   1st Qu.: 0.6188
 Median : 1.2820   Median : 1.2405   Median :-0.5452   Median : 1.2621
 Mean   : 1.2647   Mean   : 1.2452   Mean   :-0.5252   Mean   : 1.2524
 3rd Qu.: 1.9205   3rd Qu.: 1.8829   3rd Qu.: 0.0273   3rd Qu.: 1.8932
 Max.   : 3.9466   Max.   : 4.1821   Max.   : 2.3923   Max.   : 4.3920
       C                   T1                 T0                status
 Min.   :  0.00509   Min.   : 10.00   Min.   :  0.0034   Min.   :0.0000
 1st Qu.:  2.13203   1st Qu.: 12.62   1st Qu.:  2.6205   1st Qu.:0.0000
 Median :  6.12865   Median : 17.97   Median :  7.9711   Median :0.0000
 Mean   : 13.15833   Mean   : 30.31   Mean   : 20.3062   Mean   :0.4466
 3rd Qu.: 15.86713   3rd Qu.: 29.91   3rd Qu.: 19.9141   3rd Qu.:1.0000
 Max.   :207.95560   Max.   :569.09   Max.   :559.0948   Max.   :1.0000
   status_tau         T_obs                 e
 Min.   :0.0000   Min.   : 0.00338   Min.   :0.0000095
 1st Qu.:0.0000   1st Qu.: 1.02440   1st Qu.:0.0214057
 Median :0.0000   Median : 3.15482   Median :0.0942990
 Mean   :0.4774   Mean   : 6.68132   Mean   :0.1976924
 3rd Qu.:1.0000   3rd Qu.: 8.74168   3rd Qu.:0.2927670
 Max.   :1.0000   Max.   :86.28431   Max.   :0.9794667

[1] "Descriptive statistics for group A=1:   896"

       X1                  X2                  X3                  X4
 Min.   :-2.08331   Min.   :-1.82901   Min.   :-4.9477   Min.   :-2.6134
 1st Qu.: 0.04506   1st Qu.: 0.09714   1st Qu.:-2.1148   1st Qu.: 0.1030
 Median : 0.73739   Median : 0.69732   Median :-1.6033   Median : 0.7261
 Mean   : 0.74638   Mean   : 0.74217   Mean   :-1.6042   Mean   : 0.7283
 3rd Qu.: 1.42603   3rd Qu.: 1.38210   3rd Qu.:-1.1151   3rd Qu.: 1.3844
 Max.   : 3.84867   Max.   : 3.46314   Max.   : 0.8952   Max.   : 4.0002
       C                   T1                 T0                status
 Min.   :  0.02621   Min.   :  10.02   Min.   :  0.0234   Min.   :0.0000
 1st Qu.:  2.42413   1st Qu.:  13.02   1st Qu.:  3.0195   1st Qu.:0.0000
 Median :  7.09870   Median :  20.03   Median : 10.0332   Median :0.0000
 Mean   : 15.75147   Mean   :  35.47   Mean   : 25.4703   Mean   :0.1797
 3rd Qu.: 18.64573   3rd Qu.:  34.54   3rd Qu.: 24.5362   3rd Qu.:0.0000
 Max.   :274.81376   Max.   :1003.80   Max.   :993.7981   Max.   :1.0000
   status_tau         T_obs                 e
 Min.   :0.0000   Min.   :  0.02621   Min.   :0.008862
 1st Qu.:0.0000   1st Qu.:  2.42413   1st Qu.:0.581531
 Median :0.0000   Median :  7.09870   Median :0.843467
 Mean   :0.2288   Mean   : 11.53579   Mean   :0.744738
```

```
 3rd Qu.:0.0000   3rd Qu.: 14.38595   3rd Qu.:0.957582
 Max.   :1.0000   Max.   :205.27963   Max.   :0.999950
```

The covariates between the two groups are unbalanced. The censoring time is dependent on the covariates also, as the covariates are unbalanced between the two groups, the censoring time is also unbalanced. In particular, the mean of $X1$, $X2$, $X3$ and $X4$ is bigger in the control group than in the treated group. Also, the number of events is bigger in the control than treated group.

### 7.2.5 Observational study with nonlinear relationships and conditionally independent censoring (Non parametric scenario)

The summary of the generated (observed and unobserved) data set observational study with nonlinear relationships and conditionally independent censoring, stratified by treatment is displayed below.

```
[1] "Descriptive statistics for group A=0:   1058"

       X1                 X2                  X3                 C
 Min.   :-2.8577   Min.   :-1.88536   Min.   :-2.3549   Min.   : 0.000
 1st Qu.: 0.1256   1st Qu.: 0.07196   1st Qu.: 0.3135   1st Qu.: 1.000
 Median : 0.8212   Median : 0.68580   Median : 1.0164   Median : 2.000
 Mean   : 0.8322   Mean   : 0.73776   Mean   : 1.0133   Mean   : 2.418
 3rd Qu.: 1.5215   3rd Qu.: 1.37201   3rd Qu.: 1.6951   3rd Qu.: 3.000
 Max.   : 4.3526   Max.   : 3.88309   Max.   : 4.2669   Max.   :10.000
       T1                T0             status            T_obs
 Min.   : 0.000   Min.   :0.000   Min.   :0.0000   Min.   :0.000
 1st Qu.: 1.000   1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:0.000
 Median : 2.000   Median :1.000   Median :1.0000   Median :1.000
 Mean   : 2.467   Mean   :1.737   Mean   :0.7353   Mean   :1.198
 3rd Qu.: 4.000   3rd Qu.:3.000   3rd Qu.:1.0000   3rd Qu.:2.000
 Max.   :13.000   Max.   :8.000   Max.   :1.0000   Max.   :7.000
   status_tau           e
 Min.   :0.0000   Min.   :0.03752
 1st Qu.:1.0000   1st Qu.:0.29898
 Median :1.0000   Median :0.42284
 Mean   :0.7958   Mean   :0.42782
 3rd Qu.:1.0000   3rd Qu.:0.55245
 Max.   :1.0000   Max.   :0.91526

[1] "Descriptive statistics for group A=1:   942"

       X1                 X2                  X3                 C
 Min.   :-1.5067   Min.   :-1.6735   Min.   :-2.8262   Min.   : 0.000
 1st Qu.: 0.6443   1st Qu.: 0.6385   1st Qu.: 0.3146   1st Qu.: 1.000
 Median : 1.2537   Median : 1.2858   Median : 0.9628   Median : 2.000
 Mean   : 1.2784   Mean   : 1.2947   Mean   : 1.0069   Mean   : 2.356
 3rd Qu.: 1.9117   3rd Qu.: 1.8969   3rd Qu.: 1.6788   3rd Qu.: 3.000
 Max.   : 4.3432   Max.   : 4.3559   Max.   : 4.4213   Max.   :10.000
       T1                T0              status            T_obs
 Min.   : 0.000   Min.   : 0.000   Min.   :0.0000   Min.   :0.000
 1st Qu.: 2.000   1st Qu.: 1.000   1st Qu.:0.0000   1st Qu.:1.000
 Median : 3.000   Median : 2.000   Median :0.0000   Median :2.000
 Mean   : 3.402   Mean   : 2.374   Mean   :0.4448   Mean   :1.795
 3rd Qu.: 5.000   3rd Qu.: 4.000   3rd Qu.:1.0000   3rd Qu.:3.000
```

```
 Max.   :13.000   Max.   :13.000   Max.   :1.0000   Max.   :9.000
   status_tau           e
 Min.   :0.0000   Min.   :0.06378
 1st Qu.:0.0000   1st Qu.:0.45264
 Median :1.0000   Median :0.56691
 Mean   :0.5892   Mean   :0.56051
 3rd Qu.:1.0000   3rd Qu.:0.68562
 Max.   :1.0000   Max.   :0.93531
```

The observations are the same than the previous scenario: The covariates and the censoring time between the two groups are unbalanced.

To be able to evaluate the estimators, we need to know the true $\theta_{RMST}$ at time $\tau$.

### 7.2.6 Observational study with interaction and conditionally independent censoring (Mis scenario)

The summary of the generated (observed and unobserved) data set complex observational study (conditionally independent censoring) stratified by treatment is displayed below.

```
[1] "Descriptive statistics for group A=0:   716"

       X1                 X2                 X3                 C
 Min.   :-2.5377   Min.   :-2.7185   Min.   :-2.7894   Min.   : 0.000022
 1st Qu.:-0.1756   1st Qu.:-0.1188   1st Qu.:-0.4791   1st Qu.: 0.156025
 Median : 0.4768   Median : 0.5640   Median : 0.1899   Median : 0.527458
 Mean   : 0.4893   Mean   : 0.5791   Mean   : 0.1747   Mean   : 1.094063
 3rd Qu.: 1.1435   3rd Qu.: 1.2424   3rd Qu.: 0.7864   3rd Qu.: 1.236481
 Max.   : 4.3390   Max.   : 4.0709   Max.   : 2.8860   Max.   :25.018707
       T1               T0                status           T_obs
 Min.   :1.000   Min.   :0.000000   Min.   :0.0000   Min.   :0.000000
 1st Qu.:1.008   1st Qu.:0.007641   1st Qu.:1.0000   1st Qu.:0.006765
 Median :1.077   Median :0.076608   Median :1.0000   Median :0.054797
 Mean   :1.255   Mean   :0.255437   Mean   :0.7821   Mean   :0.153630
 3rd Qu.:1.319   3rd Qu.:0.318746   3rd Qu.:1.0000   3rd Qu.:0.202005
 Max.   :6.545   Max.   :5.544541   Max.   :1.0000   Max.   :2.614387
   status_tau           e
 Min.   :0.0000   Min.   :0.05504
 1st Qu.:1.0000   1st Qu.:0.37151
 Median :1.0000   Median :0.53343
 Mean   :0.8184   Mean   :0.52782
 3rd Qu.:1.0000   3rd Qu.:0.67942
 Max.   :1.0000   Max.   :0.94248

[1] "Descriptive statistics for group A=1:   1284"

       X1                 X2                 X3                 C
 Min.   :-2.7829   Min.   :-2.4426   Min.   :-2.2068   Min.   :  0.00000
 1st Qu.:-0.1618   1st Qu.:-0.2358   1st Qu.: 0.3645   1st Qu.:  0.08509
 Median : 0.5024   Median : 0.4716   Median : 0.9748   Median :  0.34463
 Mean   : 0.5198   Mean   : 0.4655   Mean   : 0.9846   Mean   :  1.10406
 3rd Qu.: 1.1756   3rd Qu.: 1.1676   3rd Qu.: 1.6021   3rd Qu.:  0.92077
 Max.   : 3.8548   Max.   : 3.6265   Max.   : 3.8802   Max.   :147.52796
```

```
      T1              T0                status          T_obs
 Min.   :1.000   Min.   :0.000000   Min.   :0.0000   Min.   :0.0000001
 1st Qu.:1.009   1st Qu.:0.009266   1st Qu.:0.0000   1st Qu.:0.0850867
 Median :1.075   Median :0.075103   Median :0.0000   Median :0.3446251
 Mean   :1.285   Mean   :0.284676   Mean   :0.1947   Mean   :0.5102383
 3rd Qu.:1.350   3rd Qu.:0.349751   3rd Qu.:0.0000   3rd Qu.:0.9207733
 Max.   :5.449   Max.   :4.449298   Max.   :1.0000   Max.   :2.7618115
   status_tau          e
 Min.   :0.000   Min.   :0.09937
 1st Qu.:0.000   1st Qu.:0.58268
 Median :0.000   Median :0.72345
 Mean   :0.412   Mean   :0.69459
 3rd Qu.:1.000   3rd Qu.:0.83255
 Max.   :1.000   Max.   :0.97923
```

The observations are the same than the previous scenario: The covariates and the censoring time between the two groups are unbalanced.

To be able to evaluate the estimators, we need to know the true $\theta_{RMST}$ at time $\tau$.

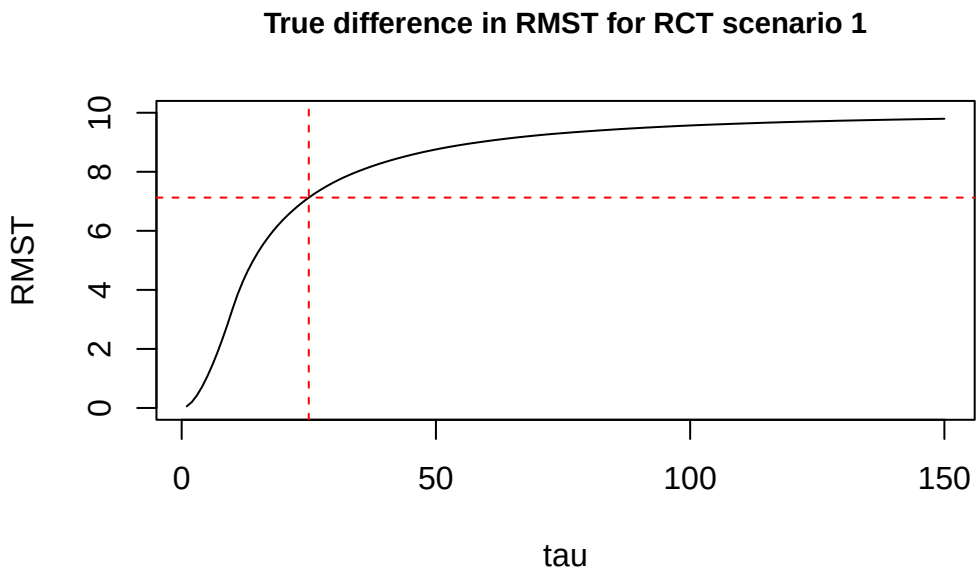
### 7.2.7 True value of RMST

$\theta_{RMST}$ is a time-dependent value. Therefore, the ground truth for $\theta_{RMST}$ must be calculated at the required restricted time.

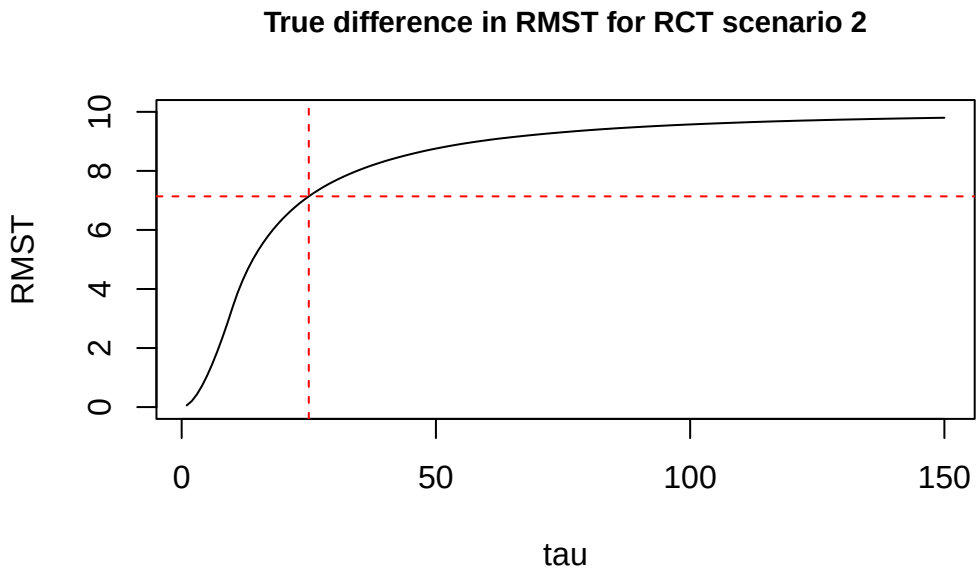
The following implementation computes the true $\theta_{RMST}$ for each of the previous simulations. This is feasible because the simulations produce data that include hypothetical scenarios not observable in real life (within the data frame, we have access to T1, the outcome if the patient had been treated, and T0, the outcome if the patient had not been treated). Thus, calculating of the true $\theta_{RMST}$ becomes straightforward:

```r
# Function to calculate ground truth for RCT and Observational datasets
ground_truth <- function(tau,
                         data) {
  # Compute RMST with the true T1
  data$T1_tau <- ifelse(data$T1 >= tau, tau, data$T1)

  # Compute RMST with the true T0
  data$T0_tau <- ifelse(data$T0 >= tau, tau, data$T0)

  # Compute the difference in RMST if everyone had the treatment
  # and if everyone had the control
  truth <- mean(data$T1_tau) - mean(data$T0_tau)

  return(truth)
}
```

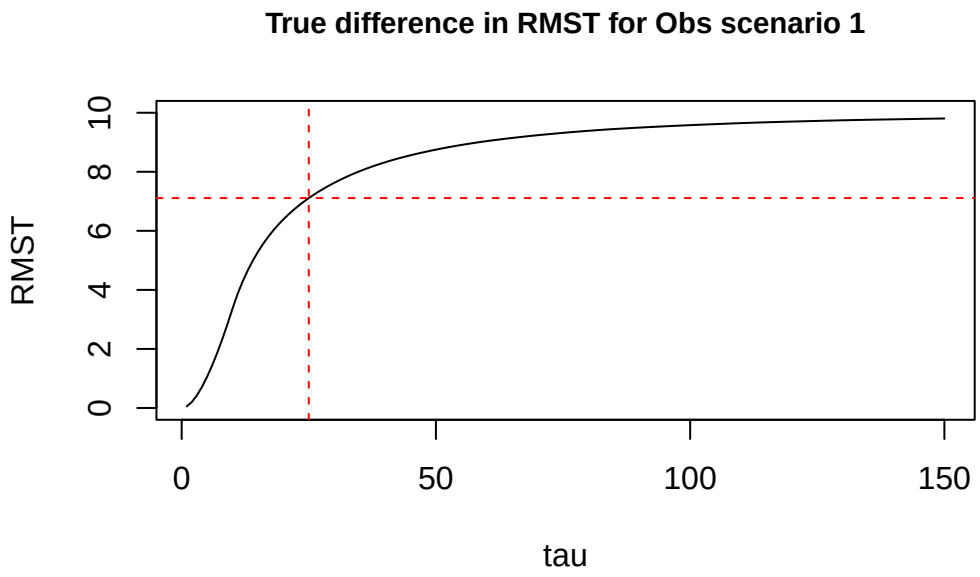
The time-dependent ground truth for all the setting are displayed bellow:



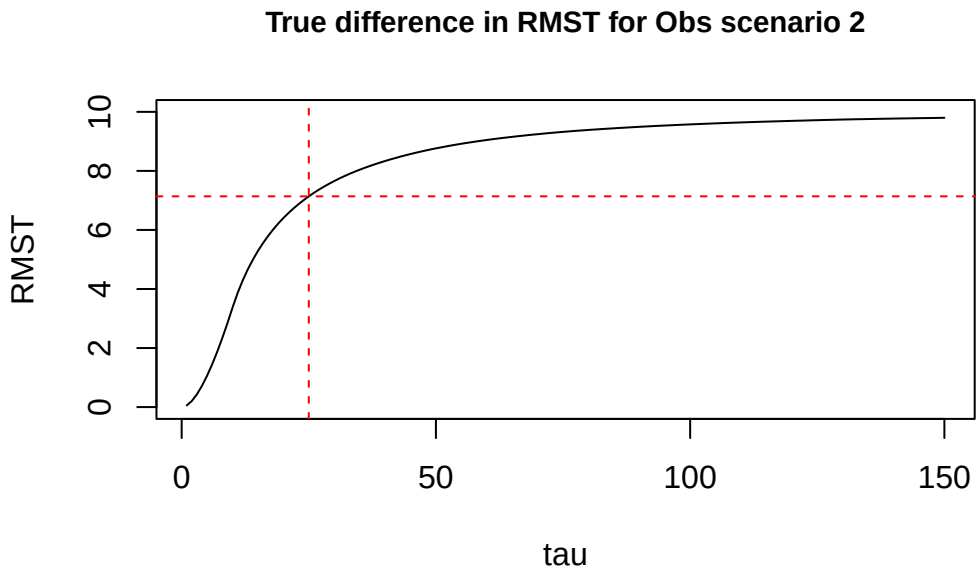

```
[1] "The ground truth for RCT scenario 1 at time 25 is 7.1"
```



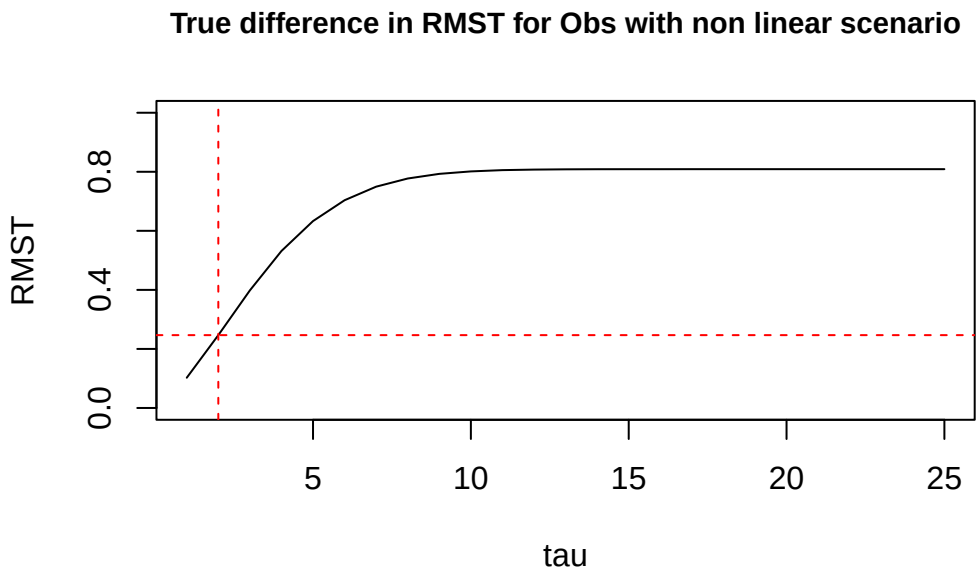

```
[1] "The ground truth for RCT scenario 2 at time 25 is 7.1"
```



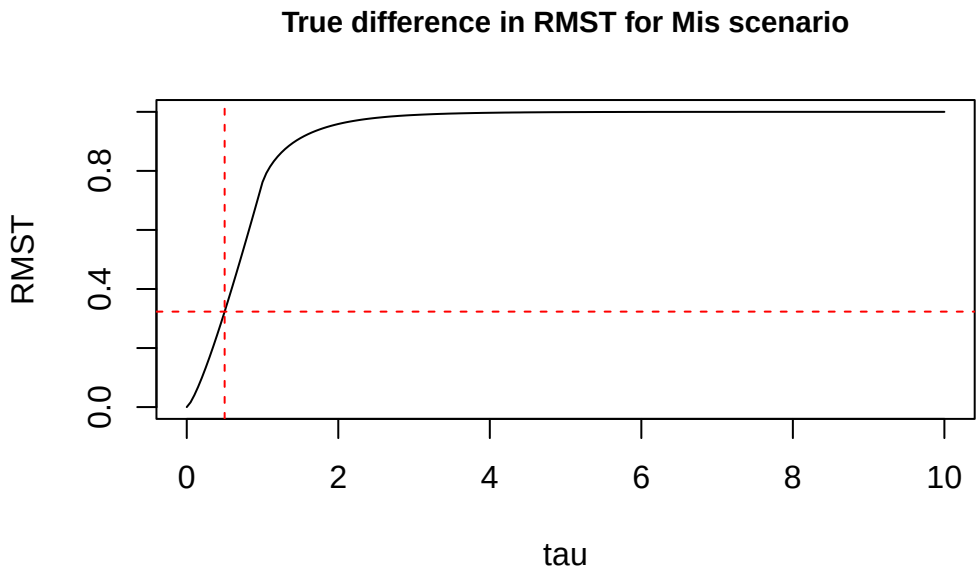

```
[1] "The ground truth for Obs scenario 1 at time 25 is 7.1"
```




```
[1] "The ground truth for Obs scenario 2 at time 25 is 7.1"
```




```
[1] "The ground truth for Observation with non linear scenario at time 2 is 0.25"
```




```
[1] "The ground truth for mis scenario at time 0.45 is 0.32"
```

The following section will apply the previous estimator on the presented simulation to evaluate their performances.

### 7.3 Estimation of the RMST

This section provides an evaluation of the previous estimators when the nuisance parameter specification is correct, as well as when there are mis-specifications of nuisance models or violations of the censoring positivity assumption.

### 7.3.1 Correct specification of the nuisance parameters

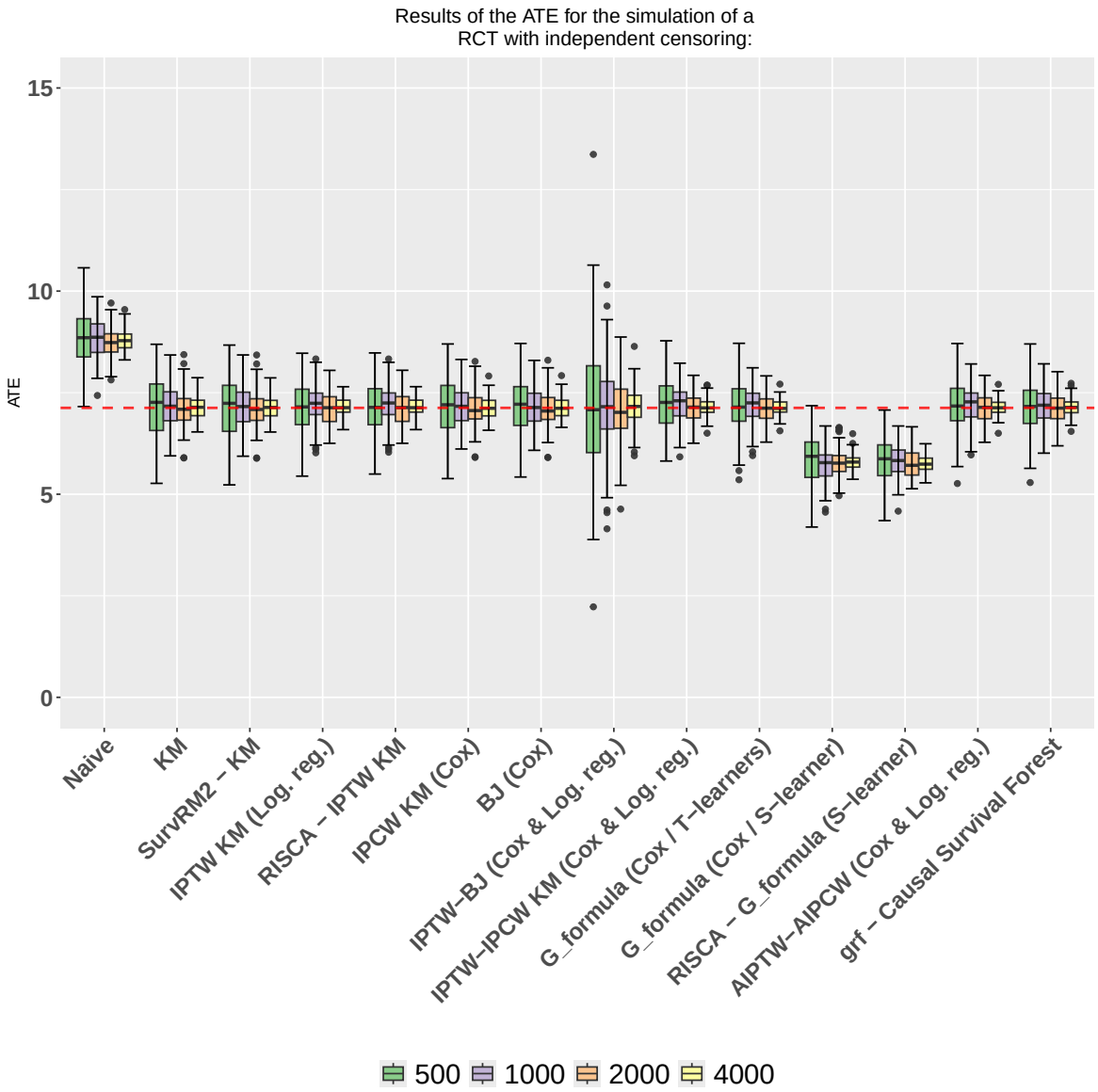
**Est-ce qu'on ajouterais pas le détail des modèles générés ?**

In the case of RCT and observational study presented in Section 7.1.1 and Section 7.1.2, the nuisance models are estimated by Cox model for the conditional censoring model and for the conditional survival and by logistic regression for the propensity model. For the non parametric simulation presented in Section 7.1.3, the nuisance models are estimated by survival forest for conditional survival and conditional censoring, by a probability forest for the propensity model. Default tuning parameters were used for different forest-based methods. Additionally, cross-fitting with five folds is applied to these flexible models.

All the estimators detailed in Section 3 and Section 4 are computed 100 times at each sample size: 500, 1000, 2000, 4000 observations.

The estimations $\theta_{RMST}$ of all the presenting DGP are computed below:

The results are presented in boxplot for each sample size. The true value of $\theta_{RMST}$ is presented as red dotted line for $\tau = 25$. RCT scenario 1 results are displayed bellow:



The boxplot above shows the distribution of the $\theta_{RMST}$ estimates for the RCT scenario 1 (with independent censoring).

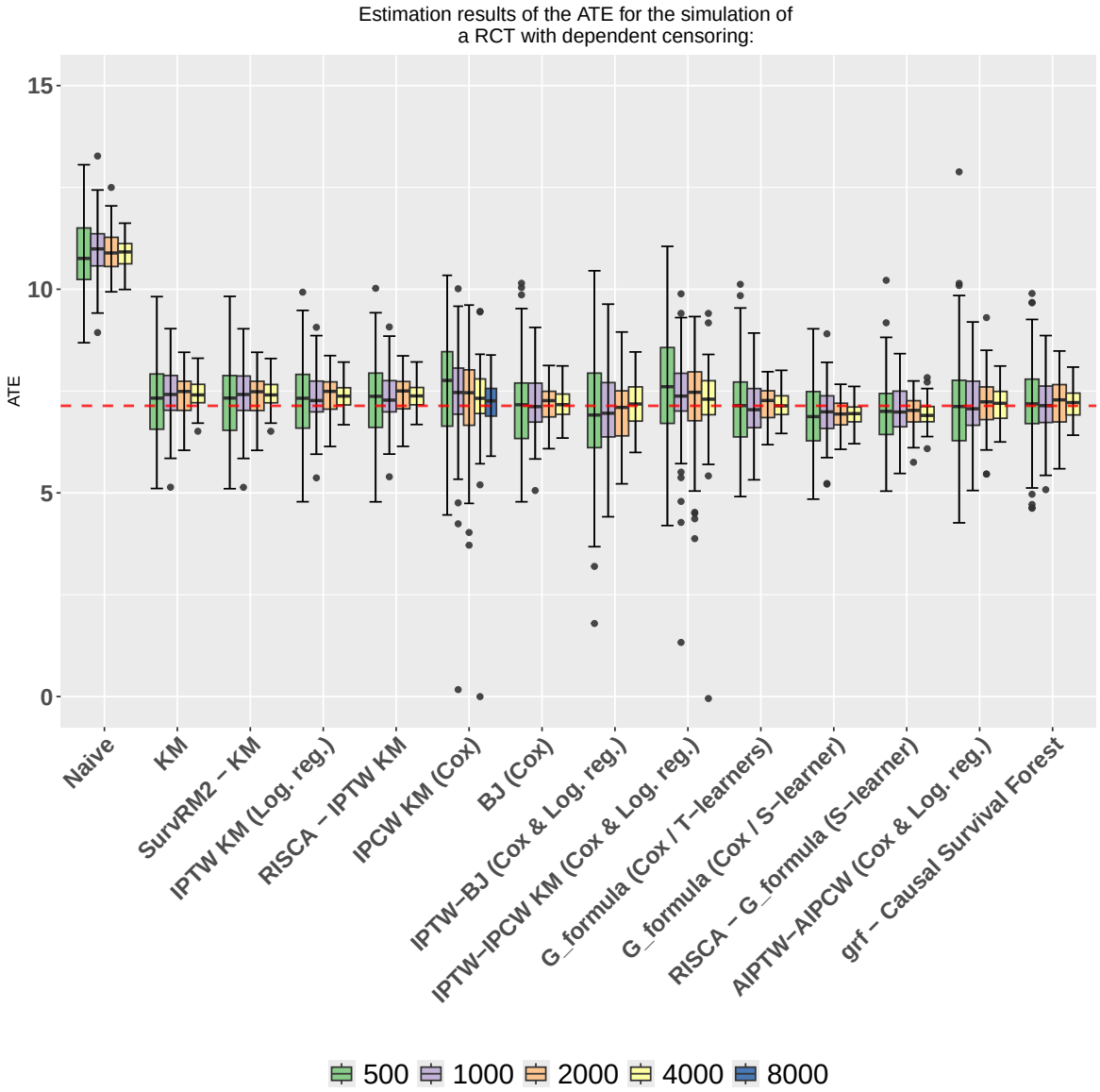
In the case of correct nuisance parameter specification using parametric and semi-parametric models, all estimators converge except for the Naive and G-formula estimators from the RISCA package. The Naive estimator is consistently biased as it excludes censored observations, directly applying the $\theta_{RMST}$ formula on uncensored data. The bias in the G-formula (RISCA) and G-formula S-learner is due to a violation of the proportional hazards assumption, as the treatment effect is additive ($T(1) = T(0) + 10$).

Regarding variance, the G-formula exhibits the lowest, with other estimators maintaining low variance except for the IPTW-BJ estimator, which shows significantly higher variance in small samples. This is likely due to the sensitivity of inverse probability weighting in small samples without correction. Among the converging estimators, AIPCW-AIPTW, Causal Survival Forest from the grf package, and G-formula (Two Learners) are the most efficient, achieving convergence even with 500 observations.

Surprisingly, the Causal Survival Forest converges as quickly as the AIPTW-AIPCW estimator. This is likely due to the simplicity of the RCT design. However, despite convergence, the use of weighted estimators (IPTW KM, IPCW KM, IPTW-IPCW KM, IPTW-BJ, and AIPTW-AIPCW) or the Causal Survival Forest is unnecessary in this RCT and independent censoring scenario. These methods do not improve convergence and increase computational time.

Considering both convergence speed and computational complexity, the G-formula (T-learner) is the most suitable estimator, offering low variance even in small sample sizes.

The results for the RCT with conditionally independent censoring are shown below, with the red dashed line indicating the true $\theta_{RMST}$ for $\tau = 25$.




In the RCT simulation with conditionally independent censoring, as before, the Naive estimator remains biased. Similarly, both the unadjusted Kaplan-Meier (KM) and its SurvRM2 equivalent, as well as the treatment-adjusted IPTW KM and its RISCA equivalent, are biased due to their failure to account for dependent censoring.

Surprisingly, IPCW Kaplan-Meier and IPTW-IPCW Kaplan-Meier estimators exhibit slight bias up to 4,000 observations. This bias stems from the simulation's conditionally independent censoring mechanism, where censoring rates reached nearly 70% in the treatment group and 50% in the control group. Despite maintaining censoring positivity, some subgroups faced near-extreme censoring probabilities (close to 0 or 1). The IPCW correction applies only to uncensored data, representing just 30% of observations, and the instability of weighting—especially double-weighting for IPTW-IPCW—explains their slower convergence and higher variability.

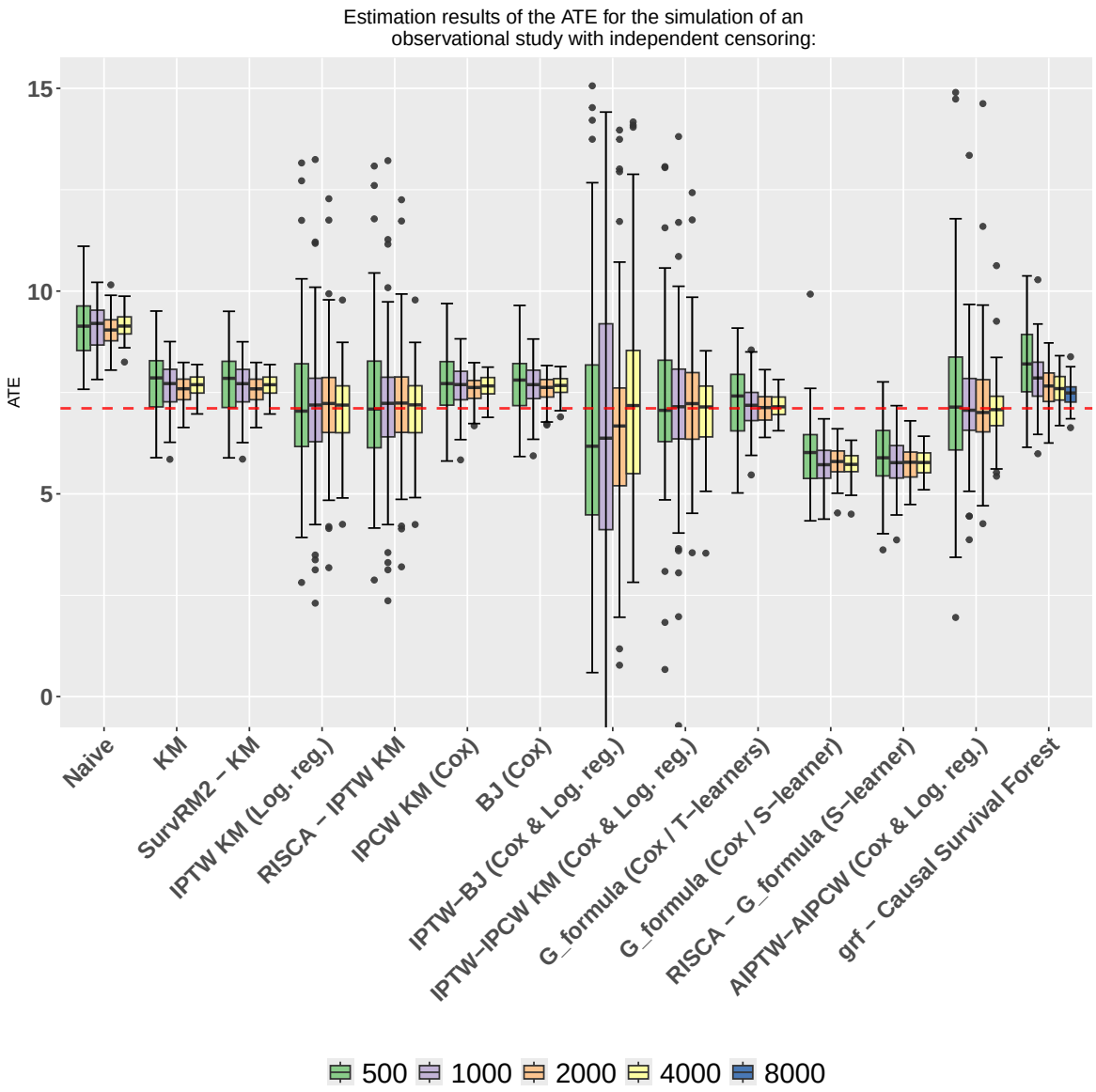
In contrast, the Buckley-James estimators (BJ) and IPTW-BJ, which address censoring using the Buckley-James transformation, converge faster and are unbiased even with as few as 500 observations. The BJ estimator also demonstrates better variance than IPCW methods.

As with RCTs under independent censoring, G-formula (S-learner) and its RISCA equivalent remain

biased. However, the G-formula (T-learners), Causal Survival Forest, and AIPCW-AIPTW estimators show high efficiency, even in small samples, with G-formula (T-learners) and Causal Survival Forest displaying the lowest variability. Notably, all estimators exhibit higher variability compared to the independent censoring scenario.

While AIPCW-AIPTW and Causal Survival Forest perform well, they seem excessive in the context of RCT with conditionally independent censoring, as they involve complex nuisance parameters and flexible regressions. Therefore, when nuisance models are well-specified in RCTs, the G-formula (T-learners) remains the most suitable estimator.

The boxplot below shows the distribution of the $\theta_{RMST}$ estimates for the Observational study with independent censoring. The red dashed line represents the true $\theta_{RMST}$ for $\tau = 25$.



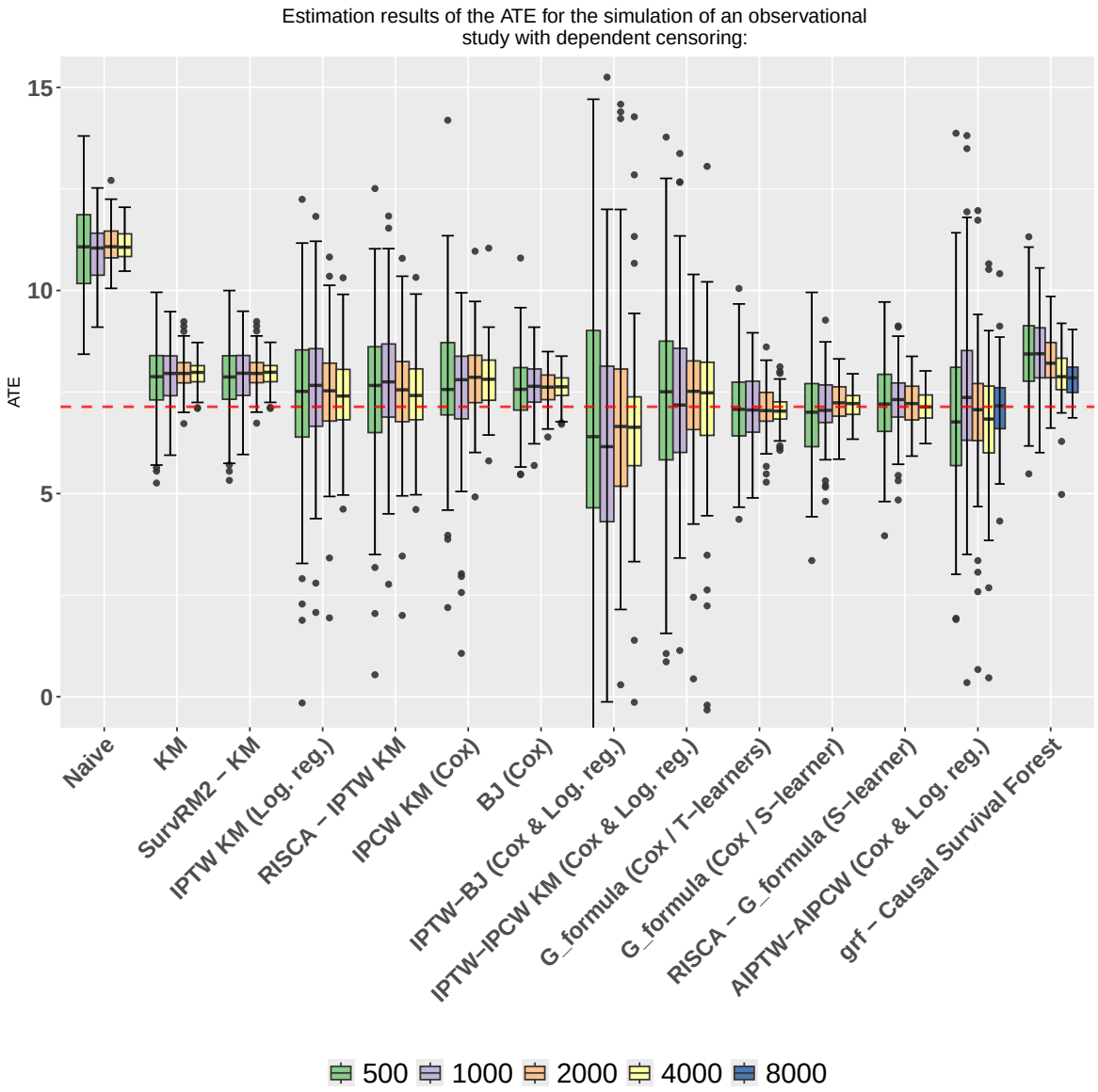

In the simulation of an observational study with independent censoring, confounding bias is introduced, setting it apart from RCT simulations. As expected, estimators that fail to adjust for this bias, such as unadjusted Kaplan-Meier, IPCW Kaplan-Meier, and their equivalents, are biased. However, estimators like IPTW Kaplan-Meier, IPTW-IPCW Kaplan-Meier, IPTW-BJ, G-formula (T-learners), and AIPCW-AIPTW converge successfully.

The IPTW-BJ estimator shows high variability, likely due to inverse probability weighting in observational settings. With small sample sizes, the propensity score model struggles to converge, especially when predicting near-extreme probabilities (close to 0 or 1). IPTW-IPCW Kaplan-Meier also has many outliers, as double-weighting amplifies instability due to extreme weights. Despite this, its overall variability is lower than IPTW-BJ, which applies a single weighting correction.

The Causal Survival Forest from the grf package appears biased, likely due to the presence of confounding and the linear nature of the simulation. With larger samples (8,000 observations), the forest tends to converge asymptotically. The top-performing estimators in this scenario are AIPCW-AIPTW, which converges the fastest (from 500 observations), and G-formula (T-learners), which maintains the lowest variance and converges from 1,000 observations.

The choice between AIPCW-AIPTW and G-formula depends on the confidence in the nuisance models. If the outcome model is well-specified, G-formula is preferred; otherwise, AIPCW-AIPTW offers a more robust alternative.

The boxplot below shows the distribution of the $\theta_{RMST}$ estimates for the Observational study with conditionally independent censoring. The red dashed line represents the true $\theta_{RMST}$ for $\tau = 25$.



In the simulation of an observational study with conditionally independent censoring, estimators that do not account for both censoring and confounding bias, such as KM, IPCW KM, IPTW KM, and their equivalents, are biased. Notably, the IPTW-IPCW Kaplan-Meier and IPTW-BJ estimators remain biased, even with 4,000 observations.

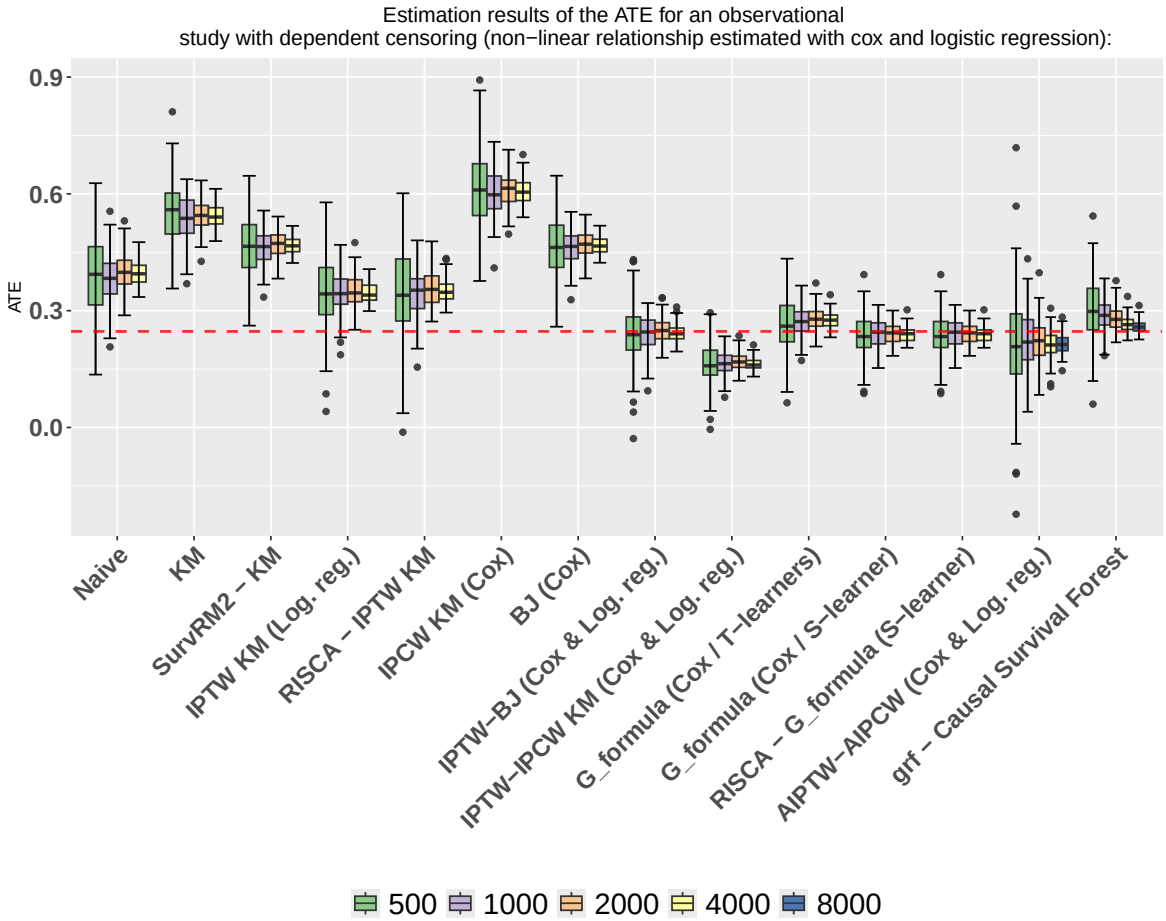
As in prior scenarios, the high censoring rate—approximately 80% in the treatment group and 50% in the control group—complicates convergence, especially since IPCW only accounts for uncensored data. The combination of confounding bias and conditionally independent censoring makes it difficult for the nuisance models in IPTW-IPCW Kaplan-Meier to converge, even when correctly specified. Similarly, the bias in the IPTW-BJ estimator can be attributed to instability from inverse probability weighting on the estimated complete data, a challenge shared with IPTW-IPCW Kaplan-Meier.

The AIPTW-AIPCW estimator fluctuates around the true value with a slight bias but shows convergence at around 8,000 observations, reflecting the complexity of estimating multiple nuisance parameters in this setting. Causal Survival Forest exhibits slow convergence but remains biased, even with 8,000 observations, highlighting its non-parametric convergence rate in this complex scenario.

Interestingly, the G-formula (S-learner) and its equivalent from the RISCA package show a slight bias compared to previous simulations. While the proportional hazards assumption remains violated, the bias is less pronounced, suggesting that conditionally independent censoring might aid Cox model convergence, even when the treatment effect does not strictly adhere to proportional hazards. This demonstrates that bias from violating the outcome model's assumptions can vary depending on the context.

Once again, the top-performing estimator is the G-formula (T-learners), which achieves convergence with lower variance compared to the other estimators.

The boxplot below shows the distribution of the $\theta_{RMST}$ estimates for the observational study with conditionally independent censoring in the context of non-parametric simulation (Section 7.1.3) when the nuisance models are estimated by Cox model and logistic regression. The red dashed line represents the true $\theta_{RMST}$ for $\tau = 2$.

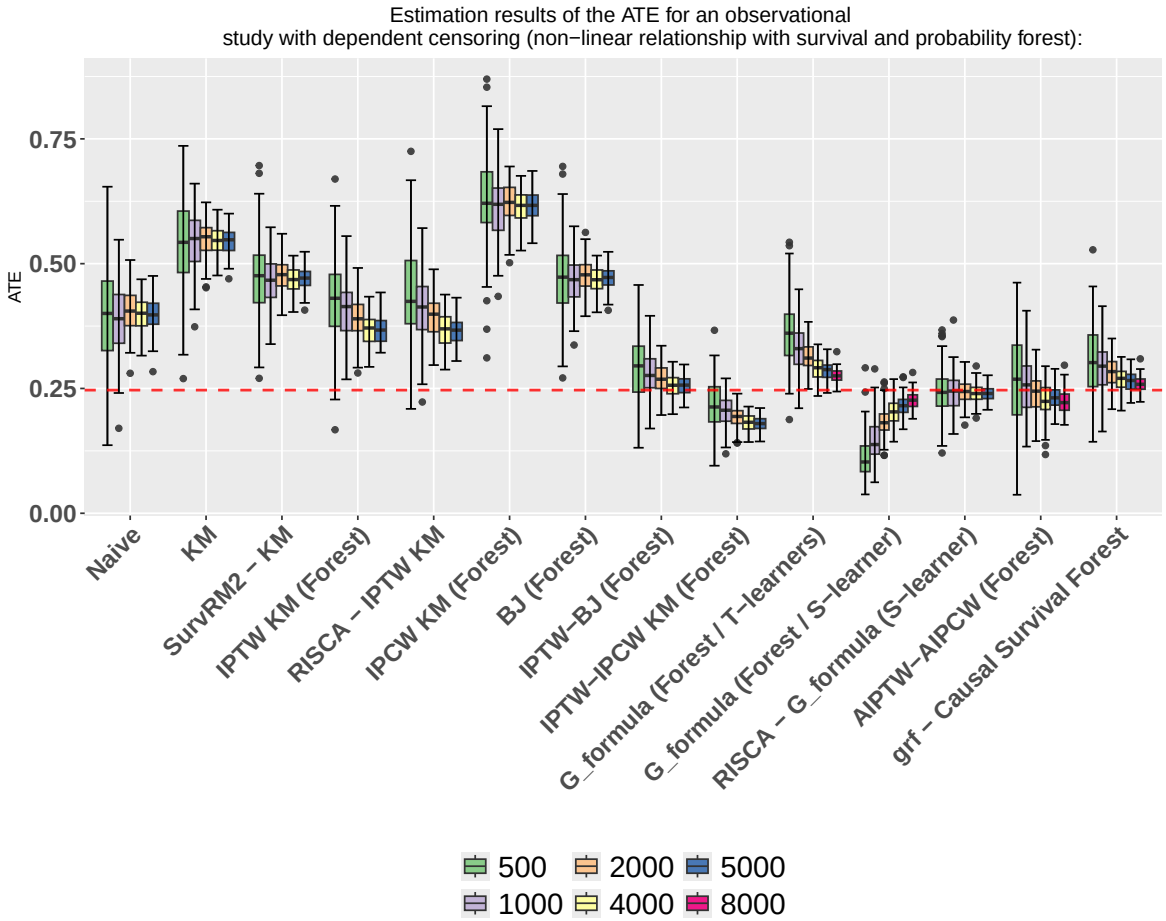



In this non-parametric simulation, most nuisance parameters are estimated using the Cox model and logistic regression, except for the grf function, which employs non-parametric causal survival forests. While the propensity score is well captured by logistic regression, the Cox model struggles to accurately estimate the conditional survival probability due to the violation of the proportional hazards assumption. This issue also affects the conditional censoring probability but to a lesser extent.

The estimators expected to converge under observational and dependent censoring include IPTW-IPCW Kaplan-Meier, IPTW-BJ, G-formula, AIPTW-AIPCW, and Causal Survival Forest. IPTW-IPCW shows the most bias, likely due to the instability of the Cox model in handling conditional censoring. Despite the proportional hazards violation, the G-formula S-learner performs well by chance, unlike the T-learner, which exhibits greater bias.

IPTW-BJ quickly converges to the true $\theta_{RMST}$, benefiting from logistic regression's strong performance in estimating the propensity score and the Buckley-James transformation's correction for censored data. With only 35% of data censored, this estimator converges effectively even with violation of proportional hazard.

Given the results, AIPTW-AIPCW fails to converge as accurately, since the well-estimated propensity score alone is insufficient. Overall, only IPTW-BJ and Causal Survival Forest manage to converge under these conditions, the latter benefiting from its flexible regression approach to nuisance estimation.




This time, flexible models are used for all nuisance parameters. The simulation involves an observational study with conditionally independent censoring. As expected, estimators without corrections for confounding and censoring (Naive, KM, IPTW KM, IPCW KM, BJ, and their equivalents) remain biased. The IPTW-BJ estimator shows some bias but converges after 5,000 observations.

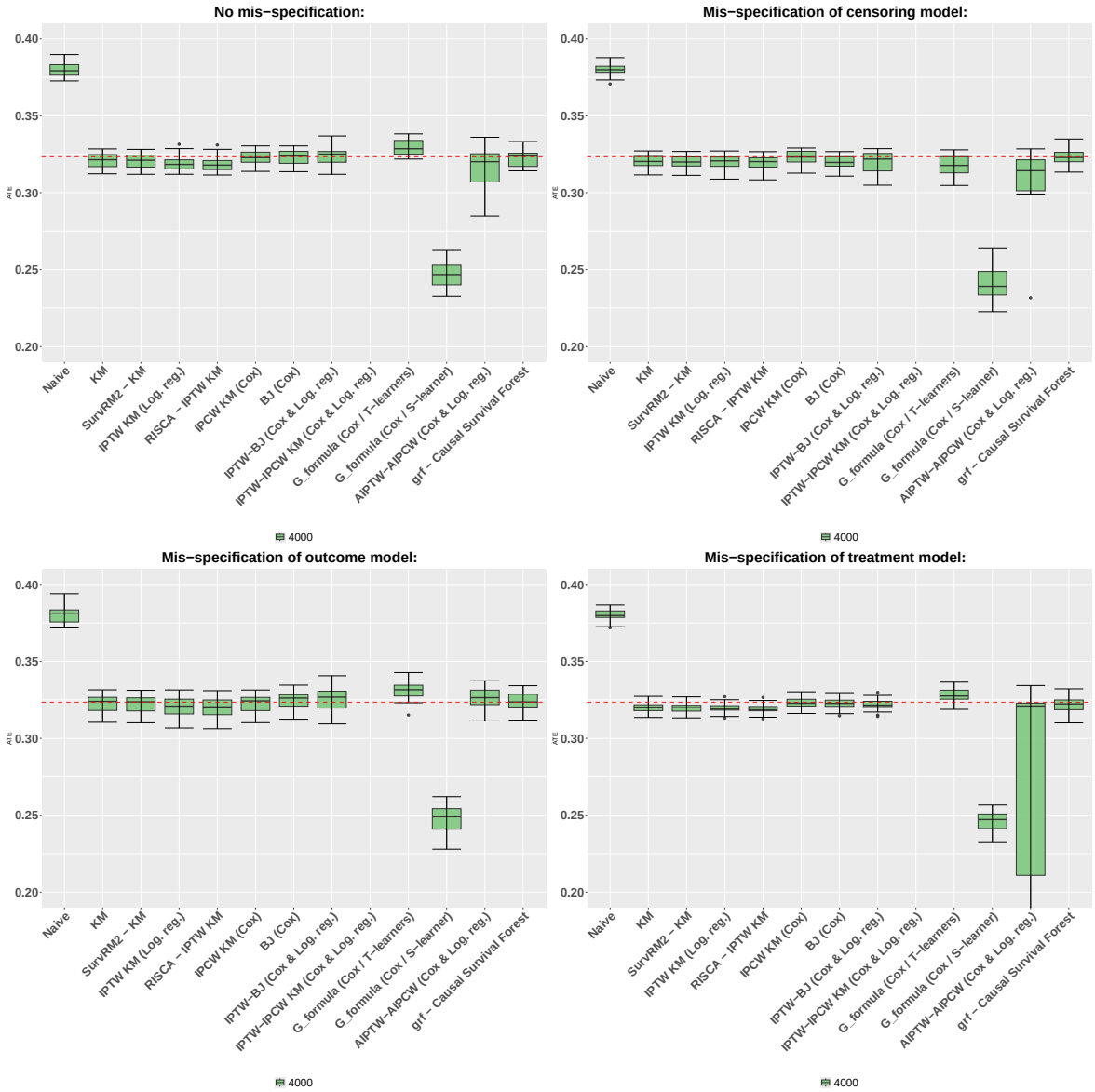
Both G-formula (T-learners and S-learners) and Causal Survival Forest converge after 5,000 observations, with Causal Survival Forest showing the least bias. As noted by Foster, Taylor, and Ruberg (2011), the Virtual Twins (VT) estimator (similar to G-formula with survival random forest) lacks orthogonality (Chernozhukov et al. 2016), which compromises its asymptotic properties and predisposes to bias (Vaart 1998). This is supported by Cui et al. (2023), showing VT's lower efficiency compared to causal survival forests, which is consistent with our findings. Thus, G-formula is not the most efficient estimator in non-parametric settings.

The S-learner performs better than T-learners, as it uses more observations by not stratifying by treatment. The finite sample bias seen across estimators can be attributed to the slower convergence rates of non-parametric models.

AIPTW-AIPCW continues to perform well, with low variability and few outliers, though it fluctuates slightly around the true ATE. Despite this, its bias is smaller compared to other estimators. In this context, IPTW-BJ performs best asymptotically, while AIPTW-AIPCW converges faster with finite samples, despite some fluctuations.

### 7.3.2 Mis-specification of the nuisance parameters

In this section, we challenge the estimators in introducing mis-specification for the nuisance parameters. To do so, the simulation with interaction is used and mis-specification is introduced by selecting only one covariate with the less impact on the model. G-formula from RISCA package is removed of the following graph as it cannot introduce intercation terms . First, we estimate $\theta_{RMST}$ without any model misspecification to ensure the estimator's consistency under a correctly specified nuisance model. Then, we introduce misspecification individually for the treatment model, censoring model, and outcome model. We further explore combined misspecifications for the treatment and censoring models, treatment and outcome models, outcome and censoring models, and finally for all nuisance models simultaneously. The nuisance models are estimated using parametric (logitic regression) and semi-parametric approach (cox model).




We refer to an observational and dependent censoring study, thus the expected consistent estimators are IPTW-IPCW Kaplan-Meier **(completly biased)**, IPTW-BJ estimator, G-formula, AIPTW-AIPCW and Causal survival forest.

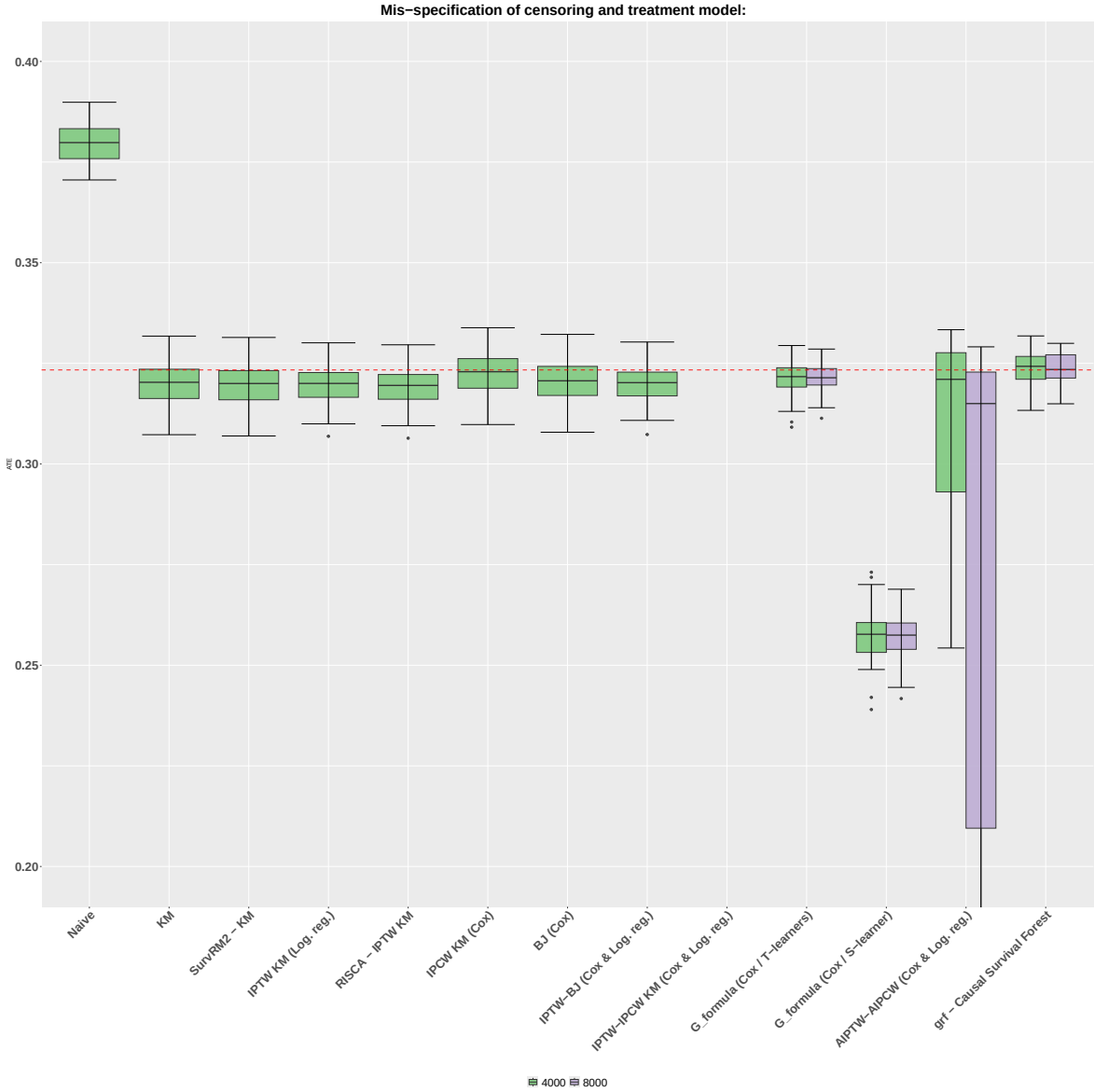
In globality, when one nuisance model is mis-specified, as expected the consistent estimators are

AIPTW-AIPCW and causal survival forest. In all these setup, the two estimators are convergent at 4,000 observations. However, the variance of AIPTW-AIPCW is larger when censoring or treatment is mis-specified.

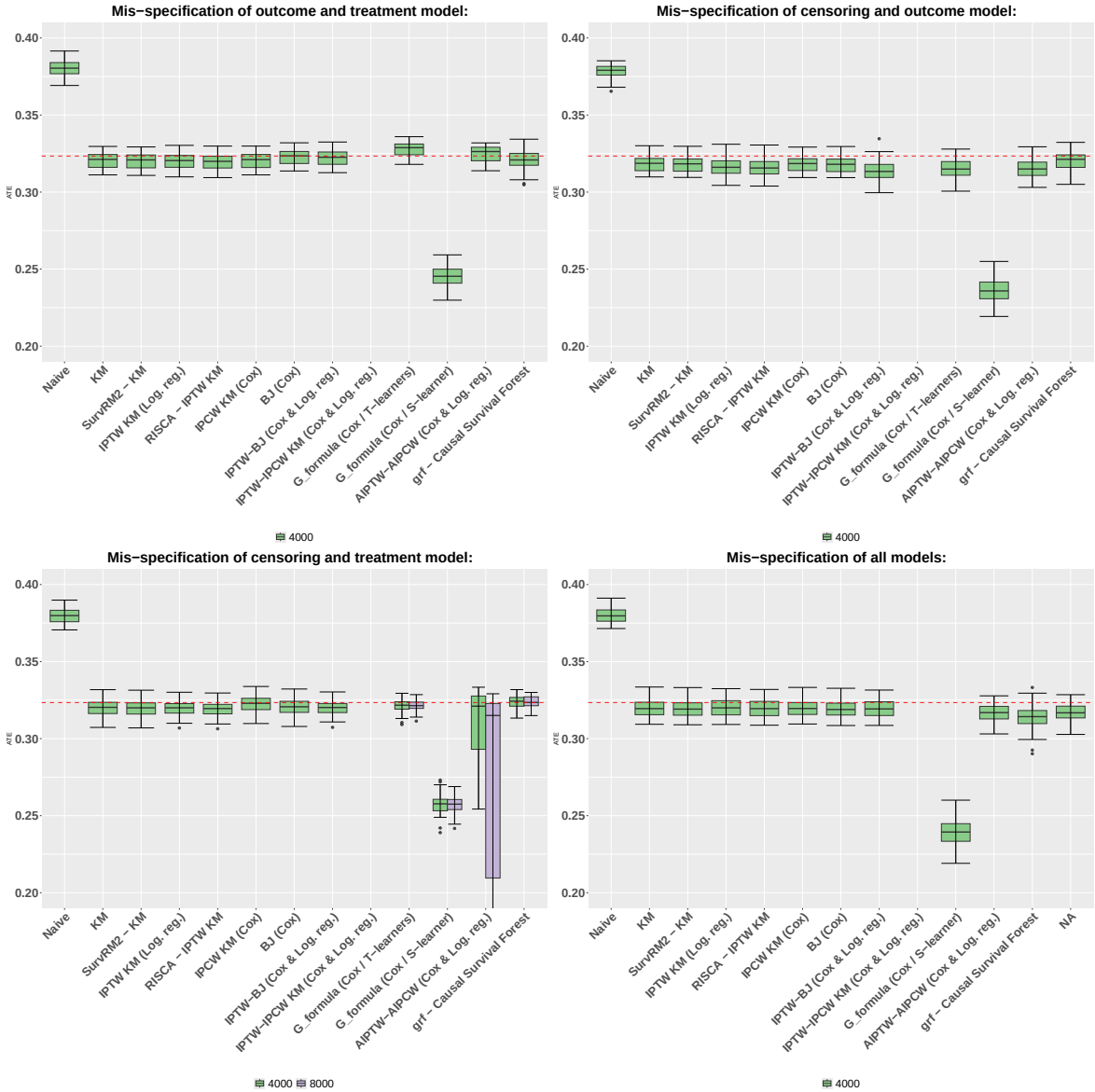
When the censoring model is mis-specified, the estimators with censoring correction are still biased as they already struggle to converge without mis-specification. The other estimators still have the same behavior than without mis-specification.

When the outcome model is mis-specified, IPTW-BJ estimator becomes slightly biased. This estimator is not strongly impacted because it uses the the mis-specified conditional survival function for estimating only the outcome of censored observation which represents 45% of observations. Also, the conditional survival is not strongly impacted by this mis-specification.

As expected, when the treatment model is mis-specified, IPTW-BJ estimator is biased.




The combined graph regroup the $\theta_{RMST}$ value of 4 different setting when there is mis-specification on all nuisance parameters (bottom right) or when two nuisance models are mis-specified either the censoring and the treatment models (top left), the outcome and treatment models (top right) or the outcome and censoring models (bottom left). The dot line represents the true value of the ATE.




As expected, when all nuisance models are misspecified, all estimators exhibit bias. Interestingly, AIPTW-AIPCW seems to converge only in cases where either the outcome and treatment models, or the outcome and censoring models, are misspecified, which deviates from initial expectations. It was anticipated that AIPTW-AIPCW would converge solely when both the censoring and treatment models were misspecified.

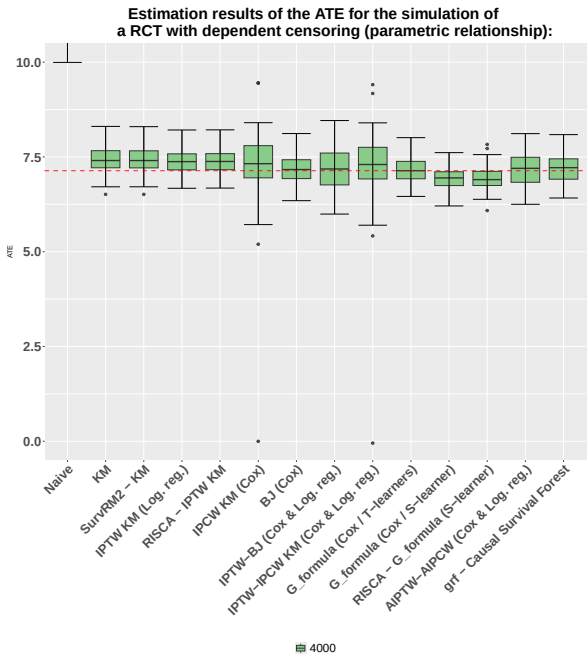
Additionally, and somewhat unexpectedly, the causal survival forest converges in all cases except when all models are misspecified.

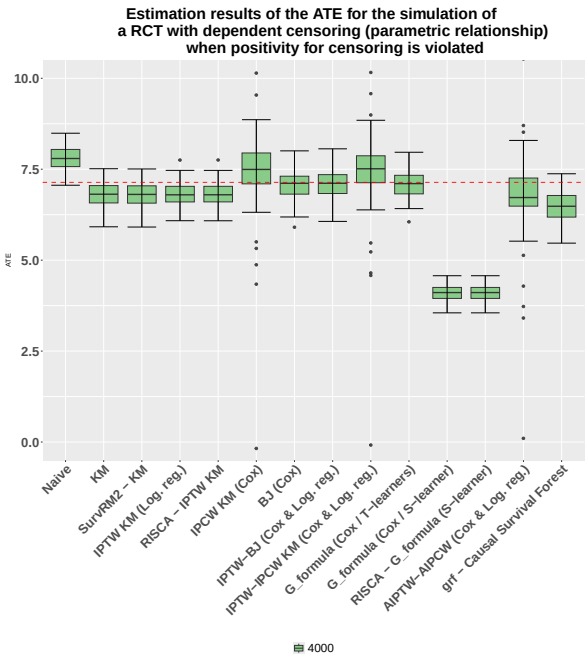
### 7.3.3 Violation of positivity assumption for censoring

In this section, the objectives is to show the impact of the violation of positivity assumption for censoring on the convergence of estimators. To enable this violation, we use the exact same simulation for RCT and conditionally independent censoring and Observational and conditionally independent censoring but with stronger relationship with covariates:

- For our RCT: $\lambda_c(X) = 0.002 \cdot \exp\{2X_1 - 4X_2 - 5X_3 + 0.2X_4 - 0.2A\}$.
- For our observational study: $\lambda_c(X) = 0.002 \cdot \exp\{2X_1 - 4X_2 - 5X_3 + 2X_4\}$.

The violation of positivity censoring can be verified in verifying the probability of remain uncensored for all combination of covariates (with cox model if proportional hazard is verified or with survival forest if not).




When the positivity of censoring is violated in our RCT context (parametric simulation) with conditionally independent censoring, all estimators that adjust for censoring using inverse probability weighting (IPCW) methods, such as IPCW Kaplan-Meier, IPTW-IPCW Kaplan-Meier become biased. The causal survival forest exhibits a small bias while surprisingly the G-formula (T-learner), BJ and IPTW-BJ converge.

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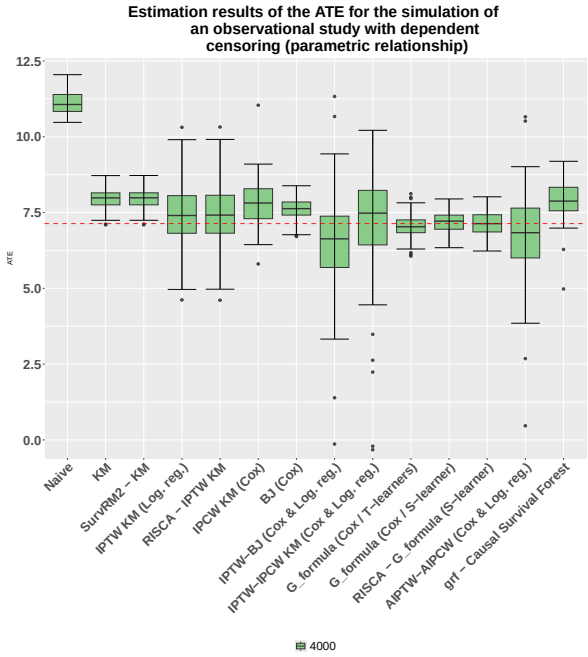
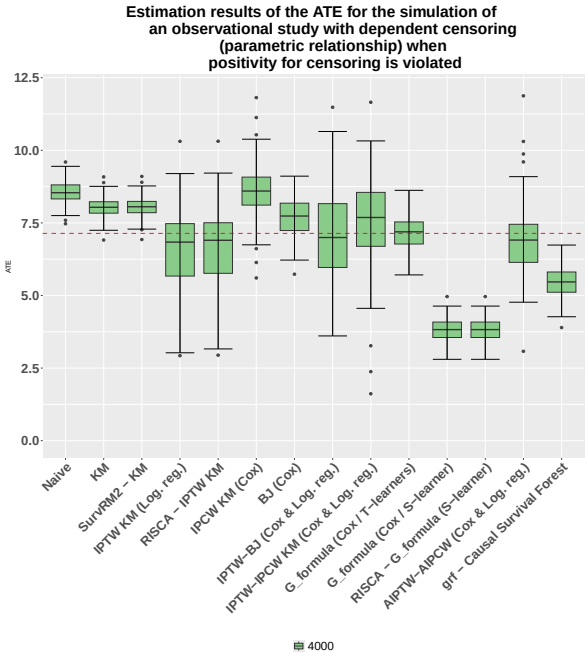
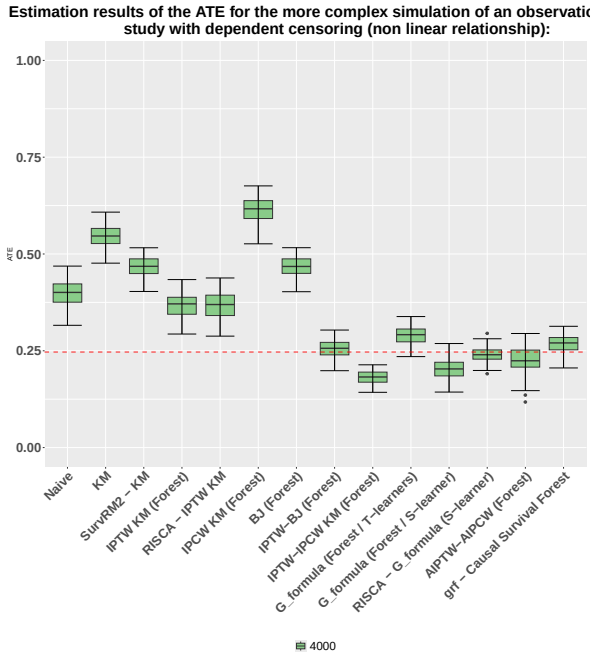
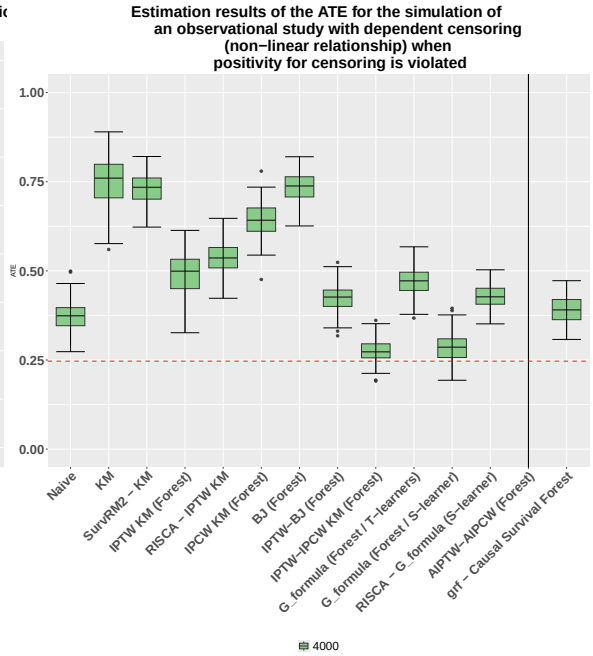



Similarly, estimators that correct for censoring using inverse probability weighting show bias, but IPTW-BJ and the G-formula (T-learner) still converge. The positivity assumption for censoring is implicit in these estimators. However, estimating survival time in a group of individuals who only experience censoring is possible when the relationship between covariates and survival time is linear and can be extrapolated by the model. The below boxplot presents the impact of the violation of positivity in the non-parametric context.




The impact of positivity violation is stronger as all the estimators exhibit bias. It confirms the fact that this violation can be overcome when the relationship between covariates and survival time is linear.

# 8 Conclusion and perspective

In this study on causal survival analysis, we examined a broad range of estimators across multiple simulation settings, particularly focusing on treatment and censoring mechanisms. While the majority of the empirical results align well with the theoretical properties, notable exceptions include estimators based on IPCW transformations, such as the IPTW-IPCW Kaplan-Meier, which failed to converge in non-parametric settings, and the IPCW Kaplan-Meier, which showed some bias under high censoring conditions.

In this study, we made available our code for the estimators tested, addressing a significant gap in current resources. Despite the growing need for causal survival analysis tools, few packages exist that offer robust implementations of various estimators. By providing these codes, we aim to support further research and practical applications of causal inference in survival contexts, where available software is currently limited.

For users, we recommend different estimators based on the context of their data and modeling assumptions. In a parametric setting with correctly specified nuisance parameters, the G-formula is highly efficient due to its low variance and quick convergence, especially in simpler scenarios like randomized controlled trials. In non-parametric settings with no misspecification, estimators like the IPTW-BJ estimator, Causal Survival Forest or AIPTW-AIPCW perform well, though the latter two may require more computing resources. When uncertainty exists around nuisance parameters, robust estimators like AIPTW-AIPCW or Causal Survival Forest are preferable due to their resilience to mis-specification.

Notably, the G-formula and IPTW-BJ estimators implicitly rely on the positivity assumption, although it is not explicitly included in their identifiability conditions. In parametric settings, violations of positivity tend to have minimal impact, as the extrapolation needed to handle these violations is feasible but risky. However, in non-parametric contexts, violating the positivity assumption can lead to erroneous extrapolations, making these estimators unsuitable for use.

In practice, uncertainty around specification of nuisance parameters is real (parametric/non-parametric relationship, variable sélections…) but use always the most complicated estimator is not feasible. Based on all our simulations, we can propose this decision tree to compute $\theta_{RMST}$ (**DRAFT**):

A key limitation of our simulations is the use of larger datasets with relatively simple relationships, which may not reflect the complexity of real-world scenarios. Most survival analysis datasets are smaller and more intricate, so the stability of certain estimators observed here may not fully generalize. It would be valuable to test these methods on real-world datasets to better assess their performance in practical applications.

Looking ahead, one promising avenue for improving these estimators is through optimizing variable selection for the conditional censoring, conditional survival, and treatment models. It has been shown that adding precision variables in causal inference enhances the variance of G-formula-like estimators. A potential area of exploration would be to investigate whether similar improvements can be made in causal survival analysis by refining the selection of covariates that influence censoring and survival outcomes.

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# 9 Annex

## 9.1 Link between RMST and survival probabilities

Survival probabilities and RMST are linked as follows:

$$\begin{aligned}
\theta_{RMST}(\tau) = &= E\left[\int_0^\tau I\{T(1) > t\}dt - \int_0^\tau I\{T(0) > t\}dt\right] \\
&= \int_0^\tau \mathbb{E}[I\{T(1) > t\}]dt - \int_0^\tau \mathbb{E}[I\{T(0) > t\}]dt \\
&= \int_0^\tau S_1(t)dt - \int_0^\tau S_0(t)dt \\
&= \int_0^\tau [S_1(t) - S_0(t)]dt
\end{aligned}$$

with $S_a(t) = P(T(a) > t)$, the probability of surviving at time $t$ when treatment $A = a$.

## 9.2 Cumulative hazard inversion method

When X, a random continuous variable, follow an exponential law (X ~ $\varepsilon(\lambda)$): the corresponding repartition function is: $F_\lambda(x) = P(X \geq x) = 1 - \exp(-\lambda x)$ and the density function is $f_\lambda(x) = \lambda \exp(-\lambda x)$.

$F_\lambda$ is bijective from $\mathcal{R}^+$ to $]0;1[$ thus, $F_\lambda^{-1}$ exists and is also bijective from $]0;1[$ to $\mathcal{R}^+$. The inverse of the repartition function is: $F^{-1}(u) = \frac{-log(1-u)}{\lambda}$ where U ~ $\mathcal{U}(0,1)$ and $\frac{-log(1-u)}{\lambda}$ ~ $\varepsilon(\lambda)$.

In knowing that 1-U ~$\mathcal{U}(0,1)$, we can also simulate X as: $\frac{-log(u)}{\lambda}$ ~ $\varepsilon(\lambda)$.

Following this results, in the case where T, the survival time, follow an exponential distribution (T ~ $\varepsilon(\lambda)$). The variable T can be simulated as: $F_\lambda^{-1}(U) = \frac{-log(U)}{\lambda}$ where U ~ $\mathcal{U}(0,1)$.

## 9.3 Trapezoidal method for integration (TO CHANGE)

The trapezoidal integration method is a numerical technique used to estimate the integral of a function over a given interval by approximating the area under the curve with trapezoids. This method is often employed when the function lacks a simple analytical form or when the integral cannot be computed exactly.

Suppose we want to estimate the integral of a function $f(x) over the interval $[a, b]$. The trapezoidal method divides this interval into $n$ sub intervals of width $h$, where $h = \frac{b-a}{n}$.

Each sub interval is approximated by a trapezoid whose bases are the values of the function $f(x)$ at the endpoints of the sub interval.

The general formula for the area of a trapezoid is: $A = \frac{(b_1+b_2)\times h}{2}$ where $b_1$ and $b_2$ are the lengths of the parallel bases of the trapezoid, and $h$ is its height.

To estimate the integral of $f(x)$ over each sub interval, we calculate the area of each trapezoid and sum them up.

The formula for the trapezoidal integration method for a single pair of trapezoids is:

$$\text{Area of trapezoid} = \frac{(f(x_i) + f(x_{i+1})) \times h}{2}$$

where $ x_i $ and $ x_{i+1} $ are the lower and upper limits of sub interval $ i $ respectively.

To estimate the integral over the entire interval, we sum the areas of all the trapezoids:

$$\int_a^b f(x)dx \approx \sum_{i=1}^{n} \frac{(f(x_i) + f(x_{i+1})) \times h}{2}$$

Below is an example of the trapezoidal integration method used to estimate the integral of $ f(x) = x^2 $ over the interval $[0, 1]$ with $ n = 4 $ sub intervals:

$$\int_0^1 x^2\, dx \approx \frac{(f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)) \times h}{2}$$

Where

$$h = \frac{1-0}{4} = \frac{1}{4}$$

Thus,

$$\int_0^1 x^2 dx \approx \frac{(0 + 2(1/16) + 2(1/4) + 2(9/16) + 1) \times 1/4}{2}$$

## 9.4 Other expression of Hazard ratio

To prove that the hazard ratio is the ratio of the log of survival probabilities:

$$HR = \frac{log(P(T > t|A = 1))}{log(P(T > t|A = 0))}$$

Let's consider $t$ fixed, the survival probabilities can be expressed for each group up to $t$ as follows (based on the fact that $\Lambda(t) = \int_0^t \lambda(u)du = -ln(S(t))$):

$$
\begin{aligned}
log(P(T > t|A = 1)) &= -\int_0^t \lambda_0(s)exp(\beta_1 X_1 + ... + \beta_p X_p + \beta A)ds \\
&= -\int_0^t \lambda_0(s)exp(\beta_1 X_1 + ... + \beta_p X_p + \beta)ds
\end{aligned}
$$

$$
log(P(T > t|A = 0)) = -\int_0^t \lambda_0(s)exp(\beta_1 X_1 + ... + \beta_p X_p)ds
$$

Dividing these two probabilities, we get:

$$
\begin{aligned}
\frac{log(P(T > t|A = 1))}{log(P(T > t|A = 0))} &= \frac{-\int_0^t \lambda_0(s)exp(\beta_1 X_1 + ... + \beta_p X_p + \beta A)ds}{-\int_0^t \lambda_0(s)exp(\beta_1 X_1 + ... + \beta_p X_p)ds} \\
&= \exp(\beta)
\end{aligned}
$$

### 9.5 Identifiability of $\Delta^\tau$

$\Delta^\tau = I\{T \wedge \tau < C|A = 1\}$ cannot be computed directly as we don't have access to $T \wedge \tau$ but to $\widetilde{T} \wedge \tau$. Here's an expression of $\Delta^\tau$ in integrate $\widetilde{T} \wedge \tau$:

$$
\begin{aligned}
\Delta^\tau &= I\{T \wedge \tau < C\} \\
&= I\{min(T, \tau) < C\} \\
&= \underbrace{I\{C > \tau\}.I\{T \geq \tau\}}_{1} + \underbrace{I\{C > T\}.I\{T \leq \tau\}}_{2} \\
&= I\{\widetilde{T} \geq \tau\} + I\{\widetilde{T} \leq \tau\}.I\{C \geq \widetilde{T}\} \\
&= I\{\widetilde{T} \geq \tau\} + I\{\widetilde{T} \leq \tau\}.\Delta
\end{aligned}
$$

As $\widetilde{T} = min(C, T)$ and that $C$ and $T$ is superior to $\tau$, then the first term of the sum is equal to $I\{\widetilde{T} \geq \tau\}$.

The second term $I\{C > T\}.I\{T \leq \tau\}$ is equal to $I\{\widetilde{T} \leq \tau\}.I\{C \geq \widetilde{T}\}$ because when $C$ is superior to $T$, then $\widetilde{T} = T$.

### 9.6 Notion of censoring unbiased transformation

The method consists in transforming the observed data mentioned in Section 2 : $O_i = (X_i, A_i, \Delta_i, \tilde{T}_i)$ into $(X_i, A_i, T_i^*)$ according to:

$$
T^* = \Delta\phi_1(\mathbf{X}, A, \tilde{T}) + (1 - \Delta)\phi_2(\mathbf{X}, A, \tilde{T}) \tag{19}
$$

with $\phi_1(.,.) =$ and $\phi_2(.,.)$ the transformation functions on respectively uncensored and censored observations.

#### 9.6.1 IPC transformation

The IPC transformation can be identified by the following formula:

$$
\begin{aligned}
E\left[T \wedge \tau \mid A, X\right] &= E\left[E\left[1\{T \wedge \tau < C\} \mid A, X, T\right] \cdot \frac{T \wedge \tau}{S_C(\widetilde{T} \wedge \tau \mid A, X)} \middle| A, X\right] \quad (1) \\
&\text{(In color, the terms are equal)} \\
&= E\left[E\left[\Delta^{\tau} \mid A, X, T\right] \cdot \frac{T \wedge \tau}{S_C(\widetilde{T} \wedge \tau \mid A, X)} \middle| A, X\right] \quad (2) \\
&= E\left[E\left[\Delta^{\tau} \mid A, X, T\right] \cdot E\left[\frac{T \wedge \tau}{S_C(\widetilde{T} \wedge \tau \mid A, X)} \middle| A, X, T\right] \middle| A, X\right] \quad (3) \\
&= E\left[E\left[\Delta^{\tau} \cdot \frac{T \wedge \tau}{S_C(\widetilde{T} \wedge \tau \mid A, X)} \middle| A, X, T\right] \middle| A, X\right] \text{(By conditional censoring As.3)} \quad (4) \\
&= E\left[\frac{\Delta^{\tau} \cdot T \wedge \tau}{S_C(\widetilde{T} \wedge \tau \mid A, X)} \middle| A, X\right] \text{(Law of total probability)} \quad (5) \\
&= E\left[\frac{\Delta^{\tau} \cdot \widetilde{T} \wedge \tau}{S_C(\widetilde{T} \wedge \tau \mid A, X)} \middle| A, X\right] \left(\Delta^{\tau} \cdot T \wedge \tau = \Delta^{\tau} \cdot \widetilde{T} \wedge \tau\right) \quad (6) \\
&= \mathbb{E}[T^* | A, X] \quad (7)
\end{aligned}
\tag{20}
$$

The term in color are equal because $E\left[1\{T \wedge \tau < C\} \mid A, X, T\right] = E\left[1\{\widetilde{T} \wedge \tau < C\} \mid A, X, T\right] = S_C(\widetilde{T} \wedge \tau \mid A, X)$. Also, the equality in the line 6 of Equation 20 is easily proven by the fact that $\Delta^{\tau}$, the indicator of censoring, selects observations which declare the event:

$$
\begin{aligned}
(\widetilde{T} \wedge \tau) * \Delta^{\tau} &= (\widetilde{T} \wedge \tau) * I\{T \wedge \tau < C | A = 0\} \\
&= min(T \wedge \tau, C) * I\{T \wedge \tau < C | A = 0\} \\
&= (T \wedge \tau) * I\{T \wedge \tau < C | A = 0\}
\end{aligned}
\tag{21}
$$

### 9.6.2 Buckley-James transformation

Considering Equation 19, this transformation can be expressed as:

$$
T^* = \begin{cases} \phi_1(X, A, \widetilde{T} \wedge \tau) = \tilde{T} \wedge \tau & \text{if} \quad \Delta^{\tau} = 1 \\ \phi_2(X, A, \widetilde{T} \wedge \tau) = \mathbb{E}[T \wedge \tau | X, A, T \wedge \tau \geq \tilde{T} \wedge \tau] & \text{else } \Delta^{\tau} = 0 \end{cases}
$$

$\mathbb{E}[T \wedge \tau | X, A, T \wedge \tau \geq \tilde{T} \wedge \tau]$ can be expressed as:

$$
\begin{aligned}
Q_S(t|x,a) &= E[T \wedge \tau \mid X = x, A = a, T \wedge \tau > t] \\
&= \frac{E[T \wedge \tau . I\{T \wedge \tau > t\} \mid X = x, A = a]}{P(T \wedge \tau > t|X, A)} \quad \text{(by the law of conditional expectation)} \\
&= \int_{-\infty}^{+\infty} \frac{t.I\{T \wedge \tau > t\}.dF(T \wedge \tau|X, A)}{P(T \wedge \tau > t|X, A)} \quad \text{(by def of Riemann–Stieltjes integral)} \\
&= \frac{1}{P(T \wedge \tau > t|X, A)} \int_t^{+\infty} t.dF(T \wedge \tau|X, A) \\
&= \frac{1}{S(T \wedge \tau|X, A)} \int_t^{+\infty} t.dF(T \wedge \tau|X, A) \\
Q_S(t|x,a) &= \frac{1}{S(T \wedge \tau|X = x, A = a)} \int_t^{+\infty} t.d(1 - S(T \wedge \tau|X = x, A = a))
\end{aligned}
$$

The Buckley-James transformation can be identified by the following formula:

$$
\begin{aligned}
\mathbb{E}[T^*|X, A] &= \mathbb{E}[T^* \Delta^\tau|X, A] + \mathbb{E}[T^*(1 - \Delta^\tau)|X, A] \\
&= \mathbb{E}[\Delta^\tau(\tilde{T} \wedge \tau)|X, A] + \\
&\quad \mathbb{E}[(1 - \Delta^\tau)\mathbb{E}[T \wedge \tau|X, A, T \wedge \tau \geq \tilde{T} \wedge \tau]|X, A] \\
&= \mathbb{E}[\Delta^\tau(\tilde{T} \wedge \tau)|X, A] + \\
&\quad \mathbb{E}[(1 - \Delta^\tau)\mathbb{E}[T \wedge \tau|X, A, T \wedge \tau \geq C]|X, A] \\
&= \mathbb{E}[\mathbf{1}\{T \wedge \tau < C\}(\tilde{T} \wedge \tau)|X, A] + \\
&\quad \mathbb{E}[\mathbf{1}\{T \wedge \tau \geq C\}\mathbb{E}[T \wedge \tau|X, A, T \wedge \tau \geq C]|X, A] \\
&= \mathbb{E}[\tilde{T} \wedge \tau|X, A, T \wedge \tau < C]\mathbb{P}[T \wedge \tau < C|X, A] + \\
&\quad \mathbb{E}[\mathbb{E}[T \wedge \tau|X, A, T \wedge \tau \geq C]|X, A, T \wedge \tau \geq C]\mathbb{P}[T \wedge \tau \geq C|X, A] \\
&= \mathbb{E}[T \wedge \tau|X, A, T \wedge \tau < C]\mathbb{P}[T \wedge \tau < C|X, A] + \\
&\quad \mathbb{E}[T \wedge \tau|X, A, T \wedge \tau \geq C]\mathbb{P}[T \wedge \tau \geq C|X, A] \\
&= \mathbb{E}[\mathbf{1}\{T \wedge \tau < C\}(T \wedge \tau)|X, A] + \\
&\quad \mathbb{E}[\mathbf{1}\{T \wedge \tau \geq C\}(T \wedge \tau)|X, A] \\
&= \mathbb{E}[T \wedge \tau|X, A]
\end{aligned}
$$

## 9.7 Influence function

### 9.7.1 Taylor expansion of a function

The Taylor expansion of a function $v$ around a point 1 is given by:

$$
\begin{aligned}
T(P_0) = v(0) &= v(1) + v'(1)(0 - 1) - R_2 \\
&= T(P_1) + \frac{\partial}{\partial \epsilon} T(P_\epsilon)\bigg|_{\epsilon=1} (0 - 1) - R_2,
\end{aligned}
$$

where $R_2$ is the remainder term of the Taylor expansion.

The von Mises expansion is a generalization of the Taylor expansion for functions in an infinite dimension.

### 9.7.2 Derive a corrected estimator with influence function

Based on the notation in Section 4.2.4.1, the direction of the fluctuation is derived by the score function:

$$s_{\epsilon=0}(O) = \frac{d}{d\epsilon} \log \left( \frac{dP_\epsilon}{dP_0}(O) \right) \Bigg|_{\epsilon=0} = \frac{\frac{d}{d\epsilon} dP_\epsilon(O)}{dP_\epsilon(O)} \Bigg|_{\epsilon=0}$$

The goal is to find a quantity that can correct the bias of the estimator $\psi(\bar{P})$ due to the perturbation. For the moment, the estimand of interest is $\psi(P) = \mathbb{E}_P[\mathbb{E}_P(T \wedge \tau | X, A = 1)]$.

Recall the von Mises expansion (can be seen as distributional analog of a Taylor expansion see Annex Section 9):

$$\begin{aligned} \psi(\bar{P}) - \psi(P) &= \int \varphi(z; \bar{P}) d(\bar{P} - P)(z) + R_2(\bar{P}, P) \\ \implies \psi(P) &= \psi(\bar{P}) - \int \varphi(z; \bar{P}) d(\bar{P} - P)(z) - R_2(\bar{P}, P) \end{aligned} \tag{22}$$

where $z$ is the observed data, $\varphi(z; P)$ is the influence function defined as a mean-zero (thus $\int \varphi(z; \bar{P}) d(\bar{P})(z) = 0$), finite variance function and $R_2(\bar{P}, P)$ is a second-order remainder term.

This expansion shows that plug-in estimator such as G-formula (presented in Section 4.2.1.1) has first-order bias and suggests how to correct it by estimating the bias term $-\int \varphi(z; \bar{P}) d(\bar{P} - P)(z)$ (Kennedy 2023) (a straightforward estimator could be the sample average of the influence function on the distribution $P$). Thus, the corresponding bias-corrected estimator is $\hat{\psi} = \psi(\hat{\mathbb{P}}) + \mathbb{P}\{\varphi(Z, \hat{\mathbb{P}})\}$.

It exists several methods to compute influence function, the most general is to compute the following pathwise derivative $\psi'(P_\epsilon)$ derived from Equation 22:

$$\begin{aligned} \frac{\psi(P_\epsilon) - \psi(P)}{\epsilon} \Bigg|_{\epsilon=0} &= \frac{d}{d\epsilon} \psi(P_\epsilon) \Bigg|_{\epsilon=0} \\ &= \int \varphi(O; P) \underbrace{\frac{d}{d\epsilon} dP_\epsilon(O)}_{\equiv s_\epsilon(O) dP_\epsilon(O)} \Bigg|_{\epsilon=0} \\ &= \int \varphi(O; P) s_\epsilon(O) dP_\epsilon(O) \Bigg|_{\epsilon=0} \\ &= \mathbb{E}_{P_\epsilon} [\varphi(O; P) s_\epsilon(O)] |_{\epsilon=0} \end{aligned} \tag{23}$$

One simple method from Kennedy (2023) to compute this quantity Equation 23 is to:

1- Consider that data are discrete.

2- Treat the influence function as derivatives to allow the differentiation rules.

Then derive the correction term.

## 9.8 The augmented inverse probability of censoring transformation

In the context of non parametric regression (from D. Rubin and Laan (2007)):

$$
\begin{aligned}
Y^\star(O) &= Y^\star_{\bar{F},\bar{G}}(O) \\
&= \frac{Y\Delta}{\bar{G}(Y \mid W)} + \frac{Q_{\bar{F}}(W,C)(1-\Delta)}{\bar{G}(C \mid W)} - \int_{-\infty}^{\tilde{Y}} \frac{Q_{\bar{F}}(W,c)}{\bar{G}^2(c \mid W)} dG(c \mid W) \\
&= T_1 + T_2 - T_3.
\end{aligned}
$$

First observe that,

$$
\begin{aligned}
E\left[T_1 \mid W\right] &= E\left[\frac{Y\Delta}{\bar{G}_1(Y \mid W)}\middle| W\right] \\
&= E\left[E\left[\frac{Y\Delta}{\bar{G}_1(Y \mid W)}\middle| W, Y\right]\middle| W\right] \\
&= E\left[\frac{Y}{\bar{G}_1(Y \mid W)} P(\Delta = 1 \mid W, Y)\middle| W\right] \\
&= E\left[\frac{Y}{\bar{G}_1(Y \mid W)} \bar{G}(Y \mid W)\middle| W\right] \\
&= \int_{-\infty}^{\tau} y \frac{\bar{G}(y \mid W)}{\bar{G}_1(y \mid W)} dF(y \mid W).
\end{aligned}
$$

Next note that,

$$
\begin{aligned}
E\left[T_2 \mid W\right] &= E\left[\frac{Q_1(W,C)(1-\Delta)}{\bar{G}_1(C \mid W)}\middle| W\right] \\
&= E\left[E\left[\frac{Q_1(W,C)(1-\Delta)}{\bar{G}_1(C \mid W)}\middle| W, C\right]\middle| W\right] \\
&= E\left[\frac{Q_1(W,C)}{\bar{G}_1(C \mid W)} P(\Delta = 0 \mid W, C)\middle| W\right] \\
&= E\left[\frac{Q_1(W,C)}{\bar{G}_1(C \mid W)} \bar{F}(C \mid W)\middle| W\right] \\
&= E\left[\frac{\bar{F}(C \mid W)}{\bar{F}_1(C \mid W)} \int_C^{\tau} y dF_1(y \mid W) \bar{G}_1^{-1}(C \mid W)\middle| W\right] \\
&= \int_{-\infty}^{\infty} \frac{\bar{F}(c \mid W)}{\bar{F}_1(c \mid W)} \bar{G}_1^{-1}(c \mid W) \left\{\int_c^{\tau} y dF_1(y \mid W)\right\} dG(c \mid W) \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left\{\frac{\bar{F}(c \mid W)}{\bar{F}_1(c \mid W)} \bar{G}_1^{-1}(c \mid W) 1(c < y < \tau) y\right\} dF_1(y \mid W) dG(c \mid W) \\
&= \int_{-\infty}^{\tau} y \left\{\int_{-\infty}^{y} \frac{\bar{F}(c \mid W)}{\bar{F}_1(c \mid W)} \bar{G}_1^{-1}(c \mid W) dG(c \mid W)\right\} dF_1(y \mid W).
\end{aligned}
$$

Finally, observe that,

$$
\begin{aligned}
E\left[T_3 \mid W\right] &= E\left[\int_{-\infty}^{\min(Y,C)} \frac{Q_1(W,c)}{\bar{G}_1^2(c \mid W)} dG_1(c \mid W) \middle| W\right] \\
&= E\left[\int_{-\infty}^{\infty} 1(Y > c)1(C > c) \frac{Q_1(W,c)}{\bar{G}_1^2(c \mid W)} dG_1(c \mid W) \middle| W\right] \\
&= \int_{-\infty}^{\infty} P(Y > c, C > c \mid W) \frac{Q_1(W,c)}{\bar{G}_1^2(c \mid W)} dG_1(c \mid W) \\
&= \int_{-\infty}^{\infty} P(Y > c \mid W) P(C > c \mid W) \frac{Q_1(W,c)}{\bar{G}_1^2(c \mid W)} dG_1(c \mid W) \\
&= \int_{-\infty}^{\infty} \bar{F}(c \mid W) \bar{G}(c \mid W) \frac{Q_1(W,c)}{\bar{G}_1^2(c \mid W)} dG_1(c \mid W) \\
&= \int_{-\infty}^{\infty} \frac{\bar{G}(c \mid W)}{\bar{G}_1^2(c \mid W)} \frac{\bar{F}(c \mid W)}{\bar{F}_1(c \mid W)} \int_c^{\tau} y dF_1(y \mid W) \Big\} dG_1(c \mid W) \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left\{ \frac{\bar{G}(c \mid W)}{\bar{G}_1^2(c \mid W)} \left\{ \frac{\bar{F}(c \mid W)}{\bar{F}_1(c \mid W)} 1(c < y < \tau) y \right\} dF_1(y \mid W) dG_1(c \mid W) \right. \\
&= \int_{-\infty}^{\tau} y \left\{ \int_{-\infty}^{y} \frac{\bar{F}(c \mid W)}{\bar{F}_1(c \mid W)} \frac{\bar{G}(c \mid W)}{\bar{G}_1^2(c \mid W)} dG_1(c \mid W) \right\} dF_1(y \mid W) \\
&= \int_{-\infty}^{\tau} y \int_{-\infty}^{y} \frac{\bar{F}(c \mid W)}{\bar{F}_1(c \mid W)} \frac{\bar{G}(c \mid W)}{\bar{G}_1^2(c \mid W)} \frac{dG_1}{dG}(c \mid W) dG(c \mid W) dF_1(y \mid W)
\end{aligned}
$$

Thus, combining the previous expression, we see that,

$$
\begin{aligned}
E\left[Y^{\star}(O) \mid W\right] &= E\left[T_1 + T_2 - T_3 \mid W\right] = E\left[T_1 \mid W\right] + E\left[T_2 \mid W\right] - E\left[T_3 \mid W\right] \\
&= \int_{-\infty}^{\tau} y \frac{\bar{G}(y \mid W)}{\bar{G}_1(y \mid W)} dF(y \mid W) \\
&+ \int_{-\infty}^{\tau} y \left\{ \int_{-\infty}^{y} \frac{\bar{F}(c \mid W)}{\bar{F}_1(c \mid W)} \bar{G}_1^{-1}(c \mid W) dG(c \mid W) \right\} dF_1(y \mid W) \\
&- \int_{-\infty}^{\tau} y \int_{-\infty}^{y} \frac{\bar{F}(c \mid W)}{\bar{F}_1(c \mid W)} \frac{\bar{G}(c \mid W)}{\bar{G}_1^2(c \mid W)} \frac{dG_1 dG}{dG}(c \mid W) dF_1(y \mid W) \\
&= \int_{-\infty}^{\tau} y \frac{\bar{G}(y \mid W)}{\bar{G}_1(y \mid W)} dF(y \mid W) \\
&+ \int_{-\infty}^{\tau} y \left\{ \int_{-\infty}^{y} \frac{\overline{F}(c|W)}{\overline{F}_1(c|W)} \left[ \frac{1}{\bar{G}_1(c \mid W)} \right. \right. \\
&\left. \left. - \frac{\bar{G}^2(c \mid W)}{\bar{G}_1^2(c \mid W)} \frac{dG_1}{dG}(c \mid W) \right] dG(c \mid W) \right\} dF_1(y \mid W) \\
&= \int_{-\infty}^{\tau} y \frac{\bar{G}(y \mid W)}{\bar{G}_1(y \mid W)} dF(y \mid W) \\
&- \int_{-\infty}^{\tau} y \left\{ \int_{-\infty}^{y} \frac{\bar{F}(c \mid W)}{\bar{F}_1(c \mid W)} \left[ \frac{d}{dc} \frac{\bar{G}(c \mid W)}{\bar{G}_1(c \mid W)} \right] dc \right\} dF_1(y \mid W).
\end{aligned}
$$

If $G = G_1$, then $\frac{d}{dc} \frac{\bar{G}(c|W)}{G_1(c|W)} = 0$, so:

$$
E\left[Y^{\star}(O) \mid W\right] = \int_{-\infty}^{\tau} y \frac{\bar{G}(y \mid W)}{\bar{G}(y \mid W)} dF(y \mid W) = \int_{-\infty}^{\tau} y dF(y \mid W) = m(W).
$$

If $F = F_1$, then it becomes,

$$
\begin{aligned}
E\left[Y^\star(O) \mid W\right] &= \int_{-\infty}^{\tau} y \frac{\bar{G}(y \mid W)}{\bar{G}_1(y \mid W)} dF(y \mid W) \\
&- \int_{-\infty}^{\infty} y \left\{ \int_{-\infty}^{y} \frac{d}{dc} \frac{\bar{G}(c \mid W)}{\bar{G}_1(c \mid W)} dc \right\} dF(y \mid W) \\
&= \int_{-\infty}^{\tau} y \left\{ \frac{\bar{G}(y \mid W)}{\bar{G}_1(y \mid W)} - \int_{-\infty}^{y} \frac{d}{dc} \bar{G}(c \mid W) \right. \\
&= \int_{-\infty}^{\tau} y \left\{ \frac{\bar{G}(y \mid W)}{\bar{G}_1(y \mid W)} - \left[ \frac{\bar{G}(y \mid W)}{\bar{G}_1(y \mid W)} - \frac{\bar{G}(-\infty \mid W)}{\bar{G}_1(-\infty \mid W)} \right] \right\} dF(y \mid W) \\
&= \int_{-\infty}^{\tau} y \left\{ \frac{\bar{G}(y \mid W)}{\bar{G}_1(y \mid W)} - \frac{\bar{G}(y \mid W)}{\bar{G}_1(y \mid W)} + \frac{1}{1} \right\} dF(y \mid W) \\
&= \int_{-\infty}^{\tau} y dF(y \mid W) \\
&= m(W).
\end{aligned}
$$

It proves that $E\left[Y^\star(O) \mid W\right] = m(W) = E[Y|W]$, if $G = G_1$ or $F = F_1$.

#### 9.8.0.1 Simplification for implementation of AIPCW

$$
\begin{aligned}
\int_0^{\bar{T} \wedge T} \frac{Q_s(c|X_i, A_i)}{S_c^2(c|X_i, A_i)} d(1 - S_c(c|X_i, A_i)) &= \int_0^{\bar{T} \wedge T} \frac{Q_s(c|X_i, A_i)}{S_c^2(c|X_i, A_i)} \times -dS_c(c|X_i, A_i) \\
&= -\int_0^{\bar{T} \wedge T} \frac{dS_c(c|X_i, A_i)}{S_c(c|X_i, A_i)} \times \frac{Q_s(c|X_i, A_i)}{S_c(c|X_i, A_i)} \\
&= -\int_0^{\bar{T} \wedge T} d\log(S_c(c|X_i, A_i)) \times \frac{Q_s(c|X_i, A_i)}{S_c(c|X_i, A_i)} ds \\
&= \int_0^{\bar{T} \wedge T} \frac{\lambda_c(s|X_i, A_i) Q_s(c|X_i, A_i)}{S_c(c|X_i, A_i)} ds
\end{aligned}
$$

## 9.9 Classical methods to evaluate survival data

### 9.9.1 Cox model

The Cox proportional hazards estimator is mainly used to evaluate treatment effect in clinical studies (add ref). It is the most used model in survival analysis.

This estimator of the instantaneous hazard function is often referred to as a ***semi-parametric estimator*** due to its mixed nature, combining parametric and nonparametric components.

The Cox survival model, introduced by David R. Cox, is considered semi-parametric because it has two distinct components:

- ***Parametric Component***: The model includes a parametric term that specifies the functional form of the effect of covariates on the relative hazard. However, it does not impose particular restrictions on the form of the baseline survival function.
- ***Nonparametric Component***: The baseline instantaneous hazard function (the hazard function for an individual with all covariates equal to zero) is not specified. Instead, it is estimated in a nonparametric manner, allowing the model to adapt to different shapes of the survival function.

In the case of two states survival models, it can be defined as follows:

$$\lambda_i(t) = \lambda_0(t) exp(\beta_1 X_{1i} + ... + \beta_p X_{pi}) \tag{24}$$

This model is called ***proportional hazards*** because the ratio of instantaneous hazards is constant over time:

$$\frac{\lambda_i(t)}{\lambda_j(t)} = exp(\beta_1 X_{1i} + ... + \beta_p X_{pi} - \beta_1 X_{1j} - ... - \beta_p X_{pj})$$

is independent of time.

The model is also ***log-linear***:

$$\ln(\lambda_i(t)) = \ln(\lambda_0(t)) + \beta_1 X_{1i} + ... + \beta_p X_{pi}$$

.

Basically, cox model provides variables coefficients in maximizing the partial likelihood of Cox model. The underlying principle is that the parameter values that maximize the likelihood of the observed data are the most plausible values, given the statistical model chosen. The maximization of the partial likelihood is calculated instead of the total likelihood to significantly reduce computation time. However, censoring and lifetime must be independent, and censoring must be non-informative.

The exponential of the corresponding coefficients is the hazard ratio:

- For a binary variable coded 0/1: $HR = \exp(\text{coef})$
- For a binary variable coded a/b: $HR = \exp(\text{coef} \cdot (b - a))$
- For a continuous variable, exp(coef) corresponds to the hazard ratio for a one-unit increase in the variable. For example, for "age" variable, each additional year of life multiply the instantaneous risk of death by a factor of 1,01, an increase of 1%.

But even if the variable corresponds to treatment assignment and that the data fit all the model requirement (log linearity, hazard proportional), this quantity is not a causal quantity and can lead to major confounding bias when the data is observational.

**Implementation**

```
# Cox model
library(survival)
Cox_model <- function(data,tau,X.names=c("X1","X2","X3","X4")){
  data$T_obs_tau <- ifelse(data$T_obs>=tau,tau,data$T_obs)
  data$status_tau <- as.numeric((data$T_obs>=tau) | (data$T_obs<tau &  data$status ==
  outcome <- paste(c('Surv(',"T_obs_tau",',',"status_tau",')'),collapse="")
  f <- as.formula(paste(outcome, paste(c(X.names,'A'), collapse = " + "), sep = " ~ "
  fitS <- suppressWarnings(coxph(f, data=data, x=TRUE))
  fitS$coefficients[is.na(fitS$coefficients)] <- 0
  return(fitS)}
```

### 9.9.2 Hazard ratio from cox model are not a causal measure

Analyses of time to event endpoint in RCTs in epidemiological studies focus mainly on HR by the use of Cox model. However, the HR is not known to be a causal measure for different reasons (Hernán 2010; Martinussen, Vansteelandt, and Andersen 2020).

**Explanation**

If we focus on the expression on HR based on Cox model:

In a Cox proportional hazards model, we model the hazard function as a function of time and covariates. Mathematically, this can be expressed as in Equation 24:

$$exp(\beta) = \frac{lim_{h\to 0} P(t \leq T < t+h | T \geq t, A = 1)}{lim_{h\to 0} P(t \leq T < t+h | T \geq t, A = 0)}$$

If we have two treatment groups, say $T = 1$ for the treated group and $T = 0$ for the control group, then the hazard ratio between the two groups at a certain time $t$ is $\exp(\beta)$, where $\beta$ is the coefficient associated with the treatment variable $T$.

HR can be expressed in terms of survival probabilities as follows (explained in annex Section 9.4):

$$exp(\beta) = \frac{log(P(T > t | A = 1))}{log(P(T > t | A = 0)}$$

Under Equation 5, the hazard ratio could be seen as:

$$exp(\beta) = \frac{log(P(T(1) > t))}{log(P(T(0) > t)}$$

This shows that under the proportional hazards assumption, the HR is a function of the survival function of the same population if everyone were treated and if everyone were not treated. But the log function make the interpretation of HR as a causal measure difficult.

Also, under Equation 5, the HR could be expressed as:

$$exp(\beta) = \frac{lim_{h\to 0} P(t \leq T(1) < t+h | T(1) \geq t)}{lim_{h\to 0} P(t \leq T(0) < t+h | T(0) \geq t)}$$

The group with and without treatment will fail to be comparable over the time if the treatment affects the outcome (Hernán 2010). Also, the interpretation of HR is complicated because of its non collapsibility. The HR is not a causal measure because it is not a weighted average of the stratum-specific HRs.

This paradox is due to what is called non-collapsibility of the Hazard Ratio. The average effect on a population could not be written as a weighted sum of effects on sub-populations.

## 9.10 Schoenfeld residuals test

The proportional hazards (PH) assumption can be checked using statistical tests and graphical diagnostics based on the ***scaled Schoenfeld residuals*** to test the proportional hazards assumption for each covariate included in a Cox refression model fit.

In knowing that $\beta$ is the solution of the maximum likelihood estimate (explained in the next part):

$$\sum_{i=1}^{n} (x_i - E(x_i | R(t_i))) = 0$$

Firstly, let's define the Schoenfeld residuals at time $T_i$ :

$$r_i^{\text{schoenfeld}} = x_i - E(x_i | R(t_i))$$

with m covariates in cox model : $r_i = (r_{i,1}, ..., r_{i,m})$

where $x_i$ is the covariate vector of the individual experiencing the i-th event and $R(t_i)$ is the risk set at time $t_i$.

Basically, the Schoenfeld residuals measure the difference between the value of the covariate and the The scaled Schoenfeld residuals are defined as follows :

$$\hat{r}_{i,m}^{\text{scaled}} = \frac{r_{i,m}}{\hat{V}_m}$$

with $\hat{V}_m$ the variance of $r_{i,m}$

To test the time dependent coefficient, for a single covariate $X_m$, the proportional hazards is expanded as :

$$\beta_m(t) = \beta_m + \beta_m * g_m(t)$$

where $g_m(t)$ is a predictable process.

Thus, the scaled Schoenfeld residuals are defined as follows :

$$\hat{r}_{i,m}^{\text{scaled}} = \frac{r_{i,m}(\beta_m)}{\hat{V}(\beta_m, t_i)}$$

with $\hat{V}_m$ the variance of $r_{i,m}$

The esperance of the scaled Schoenfeld residuals is equal to :

$$E(\hat{r}_{i,m}^{\text{scaled}}) = \beta_m(t_i) - \beta_m$$

Grambsch and Therneau have shown that the variance is stable over time, thus in plotting the values of $r_m^{\text{scaled}} + \beta_m$ against time, the plot should be flat if the proportional hazards assumption holds. Then a score test can be used to test the null hypothesis.

Basically, this Schoenfeld residuals measure the scaled difference between ***observed and expected values of covariates under the hypothesis of constant risk*** to assess whether the effect of covariates on risk is constant over time.

All these assumptions can be checked in using the function cox.zph() from survival package.

The statistic of the test is defined as follows :

- Null hypothesis : $H_0$ : ***The scaled schoenfeld residuals are not time dependents*** ($\beta_m(t_i) = \beta_m$)
- Alternative hypothesis : $H_1$ :The scaled schoenfeld residuals are time dependents

The proportional risk hypothesis is confirmed by a non-significant relationship between residuals and time, and refuted by a significant relationship.

Thus a ***low p-value indicates***:

- the Schoenfeld residuals are ***not constant over time***
- there is evidence that the ***variable/predictor may be time-dependent***

Thus proportional-hazards assumption (made when generating the coxph model) may be violated by this variable.

In our example, variable meal.cal and ph.karno have p-value<5% thus the proportional hazards hypothesis is violated. To resolve this problem, we can consider a time dependent covariate :

A violations of proportional hazards assumption can be resolved by:

- Adding covariate*time interaction : The interaction have to be known in analysing the type of interaction
- Stratification : The stratification is a way of splitting the population into subgroups according to the covariate.
- Partition the time axis : The time axis is divided into several intervals and the Cox model is applied to each interval.
- Use a different model such as accelerated failure time model or additive hazard model.

## Session information

```
R version 4.3.3 (2024-02-29 ucrt)
Platform: x86_64-w64-mingw32/x64 (64-bit)
Running under: Windows 11 x64 (build 22631)

Matrix products: default


locale:
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[3] LC_MONETARY=French_France.utf8 LC_NUMERIC=C
[5] LC_TIME=French_France.utf8

time zone: Europe/Paris
tzcode source: internal

attached base packages:
[1] splines   stats     graphics  grDevices datasets  utils     methods
[8] base

other attached packages:
 [1] gridExtra_2.3             forecast_8.23.0
 [3] survRM2_1.0-4             RISCA_1.0.5
 [5] mosaicData_0.20.4         ggformula_0.12.0
 [7] Matrix_1.6-5              lattice_0.22-5
 [9] tune_1.2.1                reticulate_1.39.0
[11] relsurv_2.2-9             date_1.2-42
[13] riskRegression_2023.12.21 grf_2.3.2
[15] rms_6.8-1                 Hmisc_5.1-3
[17] dplyr_1.1.4               survminer_0.4.9
[19] ggpubr_0.6.0              ggplot2_3.5.1
[21] MASS_7.3-60.0.1           survival_3.5-8
```

```
loaded via a namespace (and not attached):
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  [4] hardhat_1.4.0        pROC_1.18.5          xts_0.14.0
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```

# 10 TO DO LIST

- Problème sur mis-spec simulation
  - Complex:
    - Investiguer pourquoi IPTW-IPCW est biaisé

# 11 Question

- Faut-il appeler les packages à chaque fois qu'ils sont utilisés ?